
Minimax-Optimal Policy Regret in Partially Observable Markov Games

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Abstract

We study sequential decision-making in partially observable environments against strategic, adaptive opponents, modeled as partially observable Markov games (POMGs). The central challenge is to learn latent dynamics from partial observations while facing an adversary whose behavior depends on the learner’s strategy, making standard regret notions inadequate. We prove that an epoch-based optimistic maximum-likelihood algorithm achieves $\tilde{O}(\sqrt{T})$ policy regret for fixed problem parameters, with explicit dependence on the horizon, adversary memory, confidence radius, and the aggregate Eluder dimension of the observable-operator error classes. The algorithm deploys one policy per epoch, with geometrically capped epoch lengths, confidence sets built cumulatively from past data, and a statistical termination test that ends an epoch as soon as the data refute the deployed optimistic model. We also prove a matching lower bound, and extend the framework to horizon-adaptive guarantees and geometrically fading adversary memory.

1. Introduction

Sequential decision-making under uncertainty lies at the heart of reinforcement learning (Sutton & Barto, 2018; Bertsekas, 1995). While remarkable progress has been made in understanding single-agent learning (Azar et al., 2017; Jaksch et al., 2010; Jin et al., 2018; Dann et al., 2017), many applications of practical importance involve multiple agents whose interests may conflict. A self-driving car navigates among human drivers who adapt their behavior in response to the autonomous vehicle’s actions (Sadigh et al., 2016). An algorithmic trader operates in markets where other participants learn and respond to observed trading patterns (Hendershott et al., 2011). A cybersecurity system defends against attackers who probe and adapt to defensive mea-

asures (Zhu & Başar, 2013). In each case, the learner faces an environment that is not only partially observable but also strategically responsive to the learner’s own behavior.

These problems go beyond the standard MDP framework (Puterman, 2014; Bertsekas, 1995) in two ways. First, under partial observability the agent cannot directly observe the latent state, requiring belief-based reasoning and history-dependent strategies (Kaelbling et al., 1998; Spaan & Vlassis, 2005). Second, with an adaptive opponent the environment’s response depends on the learner’s past actions (Littman, 1994; Shapley, 1953), introducing counterfactual dependence that standard regret notions do not capture.

We develop a unified framework for learning in partially observable Markov games against adaptive adversaries, with the goal of characterizing when efficient learning is possible and providing algorithms with provable guarantees.

The Challenge of Adaptive Adversaries. Classical online-learning regret benchmarks implicitly assume the environment would behave the same under alternative learner actions (Cesa-Bianchi & Lugosi, 2006; Hazan, 2016), which is appropriate for oblivious adversaries (Auer et al., 2002; Freund & Schapire, 1997) but fails under adaptation. *Policy regret* (Arora et al., 2012) instead evaluates performance against the counterfactual outcome had the learner committed to a fixed policy π^* from the start. Sublinear policy regret is impossible against adversaries with unbounded memory, so attention restricts to memory-bounded adversaries, for which Arora et al. (2012) achieve sublinear policy regret via mini-batching. Their results are limited to bandit settings, and extending them to structured partially observable dynamics requires new ideas.

From POMDPs to POMGs. Recent progress on sample-efficient learning in partially observable MDPs (POMDPs) shows that tractability depends on how informative observations are about latent states (Liu et al., 2022a; 2023). In particular, *weakly revealing* POMDPs exclude degenerate instances and admit optimistic maximum-likelihood methods with polynomial sample complexity. These approaches rely on observable operator models (OOMs) (Jaeger, 2000), which represent observable dynamics via operators indexed by observation-action pairs, enabling tractable likelihood computation even with large latent state spaces; related spectral methods were developed in Boots & Gordon (2011); Hsu et al. (2012). See Appendix A for extended discussion. Our work extends this operator-based viewpoint to the game-

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theoretic setting. In a POMG, transitions depend on both learner and adversary actions, and the adversary's behavior is itself a strategic response to the learner's policy. This coupling creates challenges beyond the single-agent case. The observable operators now entangle world dynamics with the adversary response, and disentangling these components is essential for learning and regret analysis.

1.1. Our Contributions

This paper provides a complete theoretical treatment of policy regret minimization in POMGs.

Formalization and structural assumptions. We formalize learning in POMGs against posterior-Lipschitz adversaries, whose responses vary smoothly with the learner's policy. This captures structured strategic behavior while ruling out discontinuous responses to infinitesimal policy changes. We also introduce a causal decomposition separating world dynamics from adversary behavior, allowing separate likelihood-based estimation of both.

Upper bound via epoch-based optimistic MLE. Our main algorithmic result establishes that policy regret has the desired $\sqrt{T}$ dependence up to problem-dependent and logarithmic factors. In particular, we show that

$$PR(T) = \tilde{O} \left(H \sqrt{\beta_T d_E T} + m H \log T \right),$$

for $\beta_T = H \beta_T$. Here H is the horizon, m is the adversary's memory bound, d_E is the uniform aggregate Eluder dimension, and β_T is the MLE confidence radius after T episodes. The algorithm computes an optimistic MLE confidence set at the start of each epoch and executes a single optimistic policy throughout the epoch; epoch lengths are capped geometrically at $T_e = 2^e$, and an epoch is terminated early as soon as the accumulated data refute the deployed optimistic model. This statistical termination test makes committing to one policy per epoch safe, and its warm-up overhead is dominated by the leading term; a companion proposition shows the test cannot be dropped (Appendix D.7).

Matching lower bound. We prove that any algorithm must incur $\Omega(\sqrt{d_E T})$ policy regret, even in a fully revealing POMG with a fixed memoryless adversary.

Extensions. We show that our algorithm can adapt to unknown time horizons and handle adversaries with unbounded but geometrically fading memory. Finally, we instantiate our bounds for tabular settings.

Paper Organization. Section 2 presents the POMG model, Section 3 the algorithm and upper bound, Section 4 the lower bound, and Section 5 the extensions. Proofs and supplementary results are deferred to the appendices.

2. Problem Setup and Structural Results

We first define the POMG model and learning objective.

2.1. Partially Observable Markov Games

A partially observable Markov game is specified by $\mathcal{M} = (S, A, B, O_A, O_B, T, E^A, E^B, r, H, \rho_0)$. Here, S is the

state space, A and B are the learner's and adversary's action spaces, and O_A and O_B are their observation spaces that reveal partial information about the state. At each step $h \in [H]$, the dynamics are given by transition kernels $T_h : S \times A \times B \rightarrow \Delta(S)$ and emission kernels $E_h^A : S \rightarrow \Delta(O_A)$ and $E_h^B : S \rightarrow \Delta(O_B)$. The learner receives rewards $r_h : O_A \rightarrow [0, 1]$, $\rho_0 \in \Delta(S)$ is the initial state distribution, and all spaces are finite. Throughout, $\log$ denotes the natural logarithm and $\log_2$ the base-2 logarithm.

The game proceeds in episodes. At the beginning of each episode, an initial state s_1 is drawn from ρ_0 . At each step h , both players simultaneously observe signals $o_h^A \sim E_h^A(\cdot | s_h)$ and $o_h^B \sim E_h^B(\cdot | s_h)$, drawn conditionally independently given s_h , then select actions a_h and b_h based on their respective histories. The state transitions according to $s_{h+1} \sim T_h(\cdot | s_h, a_h, b_h)$, and the learner receives reward $r_h(o_h^A)$. The episode ends after H steps.

A learner policy $\pi = (\pi_1, \dots, \pi_H)$ maps observation-action histories to distributions over actions: $\pi_h : \mathcal{T}_{h-1}^A \times O_A \rightarrow \Delta(A)$, where $\mathcal{T}_h^A$ denotes the set of learner post-action histories of length h , that is, sequences $(o_1^A, a_1, \dots, o_h^A, a_h)$, with $\mathcal{T}_0^A$ containing the empty history; the adversary history spaces $\mathcal{T}_h^B$ are defined analogously, and for either player $P \in \{A, B\}$ we write $\mathcal{T}_{\leq H}^P := \bigcup_{h=0}^H \mathcal{T}_h^P$. Similarly, an adversary response $g = (g_1, \dots, g_H)$ maps adversary decision histories to action distributions: $g_h : \mathcal{T}_{h-1}^B \times O_B \rightarrow \Delta(B)$. For notational convenience, we write $\eta_h^B = (\tau_{h-1}^B, o_h^B)$ and denote the response by $g_h(\cdot | \eta_h^B)$. Given a learner policy π and adversary response g , we write $V^\pi(g)$ for the expected cumulative reward over an episode.

2.2. Adaptive Adversaries and Policy Regret

We consider adversaries who adapt their behavior based on the learner's recent policies. Formally, an adversary is *m-memory bounded* if there exists a fixed, time-invariant map $R : \Pi^m \rightarrow \mathcal{G}$, where Π and $\mathcal{G}$ denote the learner's and adversary's policy spaces, respectively, such that the adversary's policy in episode t is $g^t = R(\pi^{t-m+1}, \dots, \pi^t)$. Consequently, once the learner has played a fixed policy π for m consecutive episodes, the response equals the *stationary response* $R_\infty(\pi) := R(\pi, \dots, \pi)$.

The standard notion of external regret compares the learner's performance to the best fixed policy assuming the observed adversary sequence would have been the same. This notion is inadequate for adaptive adversaries. Policy regret instead accounts for counterfactual adversary responses.

Definition 2.1 (Policy Regret). The policy regret over T episodes is

$$PR(T) = \max_{\pi^* \in \Pi} \sum_{t=1}^T \left[V^{\pi^*}(R_\infty(\pi^*)) - V^{\pi^t}(g^t) \right],$$

where $\pi^1, \dots, \pi^T$ are the learner's policies, $g^1, \dots, g^T$ are the adversary's responses, $R_\infty(\pi^*)$ is the stationary response to π^* , and Π is the learner's policy class, a fixed

set of history-dependent policies mapping learner histories to action distributions.

Policy regret compares the learner to the best fixed policy under the adversary response that policy would have induced. It is the natural benchmark for strategic adaptive opponents.

2.3. Model Parameterization and Assumptions

To enable efficient learning, we parameterize the world dynamics by $\theta \in \Theta$ and the adversary's response function by $\Phi \in \Psi$, under which the stationary response to a deployed policy π selects actions according to a response model $g_h^\Phi(\cdot | \eta_h^B, \pi)$. We write $\xi = (\theta, \Phi)$ for the joint parameter and $\Xi = \Theta \times \Psi$ for the joint parameter space. Our algorithm maintains confidence sets over these parameters based on observed data. Throughout, we write τ_B for the adversary's decision history η_h^B defined in Section 2.1, dropping the step index when it is immaterial. The key structural assumption on the adversary is that its response varies smoothly with the learner's policy. To make this precise, we introduce a decoupled formulation that avoids subtle circularity issues.

Definition 2.2 (Reference Posterior-Predictive). Fix a reference adversary policy μ^{ref} independent of the learner's strategy, e.g., uniform over B . For a learner policy π and fixed world model $\theta \in \Theta$, let $P^{\pi, \mu^{\text{ref}}, \theta}$ denote the joint history law under π and μ^{ref} , and $P^{\pi, \mu^{\text{ref}}, \theta}(\cdot | \tau_B)$ its conditional law given the adversary's decision history τ_B . The reference posterior-predictive policy at τ_B is

$$S_{\tau_B, h}^{\text{ref}, \theta}(\pi) := \mathbb{E}_{\tau_A \sim P^{\pi, \mu^{\text{ref}}, \theta}(\cdot | \tau_B)} [\pi_h(\cdot | \tau_A)], \quad (1)$$

the expected step- h learner action distribution given τ_B under the reference dynamics. Note that π enters through the conditioning measure, so $S_{\tau_B, h}^{\text{ref}, \theta}$ depends on the whole policy and not only on π_h .

The reference measure μ^{ref} provides a fixed baseline for computing how the learner's policy appears from the adversary's perspective. This decoupling is essential because the adversary's actual response depends on the learner's policy, but the smoothness condition constraining the adversary must be stated in terms of a quantity that does not itself depend on the adversary. Since $S_{\tau_B, h}^{\text{ref}, \theta}(\pi)$ depends on the world model θ , the condition below must also specify how this dependence on θ is handled.

Assumption 2.3 (Posterior-Lipschitz Adversary). There exists $L \geq 0$ such that, *uniformly over all* $\theta \in \Theta$, for all policy blocks $\pi^{1:m} := (\pi^1, \dots, \pi^m) \in \Pi^m$ and $\nu^{1:m} := (\nu^1, \dots, \nu^m) \in \Pi^m$, where m is the adversary's memory bound, all steps $h \in [H]$, and all adversary histories τ_B :

$$\begin{aligned} & \|g_h(\cdot | \tau_B, \pi^{1:m}) - g_h(\cdot | \tau_B, \nu^{1:m})\|_1 \\ & \leq L \max_{i \in [m]} \|S_{\tau_B, h}^{\text{ref}, \theta}(\pi^i) - S_{\tau_B, h}^{\text{ref}, \theta}(\nu^i)\|_1. \end{aligned} \quad (2)$$

Here $g_h(\cdot | \tau_B, \pi^{1:m})$ denotes the reaction rule's response to the policy block; it determines the response model by stationarity, $g_h^\Phi(\cdot | \eta_h^B, \pi) = g_h(\cdot | \tau_B, \pi, \dots, \pi)$.

Requiring (2) uniformly over $\theta \in \Theta$ keeps the adversary class well-defined without knowledge of the true θ^* and is the form used in the analysis. See Remark D.1 for examples of adversaries satisfying the assumption.

Our second structural assumption concerns the informativeness of observations. It is stated through the *world channel*, which for each step h and learner observation-action pair (o^A, a) is a linear map $W_h^\theta(o^A, a): \mathbb{R}^{|\bar{S}^+|} \rightarrow \mathcal{V} := \mathbb{R}^{|\bar{S}^+| \times B}$ determined by the transition and emission kernels of θ alone; $\bar{S}^+ := S \times \mathcal{T}_{\leq H}^A \times \mathcal{T}_{\leq H}^B$ is the augmented state space pairing a latent state with both players' histories, and a predictive state is a probability vector over $\bar{S}^+$. For a predictive state q , the b -indexed component of $W_h^\theta(o^A, a)q$ is the unnormalized successor of q that results if the adversary plays b . Theorem 2.5 constructs W_h^θ explicitly and shows that observable dynamics factor through it. The weak-revealing condition below requires that discrepancies in this world channel are detectable from short windows of learner observations, after the adversary's private observations and actions are marginalized according to the response model g^Φ .

Assumption 2.4 (Multi-step weak revealing). There exist an integer $\kappa \geq 1$ and a constant $\alpha_\kappa > 0$ such that the following holds. For any two world parameters $\theta, \theta' \in \Theta$, any learner policy π , any adversary response model Φ , any step $h \leq H - \kappa$, any learner observation-action pair (o^A, a) , and any normalized predictive state q , let $\mathcal{O}_{h+1:h+\kappa}^{\theta, \Phi, \pi}(\cdot | q, o^A, a)$ denote the conditional distribution of the next κ learner observations $o_{h+1}^A, \dots, o_{h+\kappa}^A$, generated from q , conditional on observing o^A and taking action a at step h . Then

$$\begin{aligned} & \left\| \mathcal{O}_{h+1:h+\kappa}^{\theta, \Phi, \pi}(\cdot | q, o^A, a) - \mathcal{O}_{h+1:h+\kappa}^{\theta', \Phi, \pi}(\cdot | q, o^A, a) \right\|_1 \\ & \geq \alpha_\kappa \left\| W_h^\theta(o^A, a)q - W_h^{\theta'}(o^A, a)q \right\|_{1, \mathcal{V}}, \end{aligned} \quad (3)$$

where $\|\cdot\|_{1, \mathcal{V}}$ is the ℓ_1 norm on the intermediate space $\mathcal{V}$. Equivalently, world-channel differences affecting the controlled predictive state are detectable from κ -step learner-observation windows with signal strength at least α_κ . For the final steps $h > H - \kappa$ the condition is not imposed; the standard convention pads the episode with κ fully revealing dummy steps (Liu et al., 2022a).

This condition, adapted from Liu et al. (2022a), rules out POMGs where observations are so uninformative that exponentially many samples would be needed to identify the dynamics. The parameter κ represents the window length needed for observations to become revealing.

2.4. OOM and Causal Decomposition

Our analysis builds on the observable operator model (OOM) framework. For a POMG with parameters $\xi = (\theta, \Phi)$ and learner policy π , the dynamics of the learner's observation process can be represented by operators $J_h^{\xi, \pi}(o^A, a) \in \mathbb{R}^{d \times d}$, where $d := |\bar{S}^+|$ is the dimension of the augmented state space (Section 2.3), such that

$\mathbb{P}^{\xi, \pi}(o_1^A, \dots, o_h^A | a_1, \dots, a_h) = \mathbf{1}^\top J_h^{\xi, \pi}(o_h^A, a_h) \dots J_1^{\xi, \pi}(o_1^A, a_1) q_0$. Here q_0 is an initial predictive state. The operators describe the *controlled* conditional process, i.e., the probability of

the learner's observations given its actions. The adversary is not visible in this representation; its private observations and actions are marginalized out according to the response model $g^\Phi(\cdot \mid \eta^B, \pi)$. Learner action probabilities under π are handled separately. A crucial insight is that these operators admit a causal decomposition, a key structural ingredient of our analysis. It separates the world-channel component from the adversary-aggregation component of the observable operator.

Theorem 2.5 (Causal Decomposition). *For any adversary response model $g_h^\Phi(\cdot \mid \eta_h^B, \pi)$ and for each step h , learner observation o^A , and learner action a , there exist linear maps $W_h^\theta(o^A, a) : \mathbb{R}^{|S^+|} \rightarrow \mathcal{V}$, and $G_h^{\Phi, \pi}(o^A, a) : \mathcal{V} \rightarrow \mathbb{R}^{|S^+|}$ such that the controlled observable operator satisfies*

$$J_h^{\xi, \pi}(o^A, a) = G_h^{\Phi, \pi}(o^A, a) \circ W_h^\theta(o^A, a).$$

The map W_h^θ depends only on the world kernels (T_h, E_h^A, E_h^B) , while $G_h^{\Phi, \pi}$ depends only on g_h^Φ and π , not on θ . Both maps are entrywise nonnegative, and $G_h^{\Phi, \pi}$ is ℓ_1 -nonexpansive for arbitrary signed inputs (each output coordinate is a single input coordinate scaled by a response probability, so $\|G_h^{\Phi, \pi} u\|_1 \leq \|u\|_1$).

The decomposition holds for any adversary response function. The world channel W_h^θ maps a predictive state, using only world kernels, to a B -indexed family of hypothetical unnormalized successors, one for each adversary action b . The adversary aggregation $G_h^{\Phi, \pi}$ mixes these successors using $g_h^\Phi(\cdot \mid \eta_h^B, \pi)$. The next augmented state stores the post-action history $\tau_h^B = (\eta_h^B, b_h)$. The proof tracks both players' full observation-action histories, uses $\mathcal{V}$ for type-correctness, and shows that $G_h^{\Phi, \pi}$ is θ -free. Details in Appendix B.

2.5. Uniform Eluder Dimension

We next define the complexity measure used in the regret bound, the aggregate Eluder dimension of the channel error classes introduced below, and record its scaling for several model families. For a joint parameter $\xi = (\theta, \Phi)$ and a learner policy π , write P_ξ^π for the law of the learner's observation-action trajectory when the world follows θ and the adversary plays the stationary response $g^\Phi(\cdot \mid \cdot, \pi)$ to π , and $V^\pi(\xi) := \mathbb{E}_{P_\xi^\pi} [\sum_{h=1}^H r_h(o_h^A)]$ for the corresponding value; for the true parameter ξ^* , $V^\pi(\xi^*) = V^\pi(R_\infty(\pi))$, the value entering the policy-regret benchmark. Write q_{h-1}^ξ for the normalized predictive state under ξ , the conditional distribution over the augmented state given the learner's observation-action history through step $h-1$, so that $\|q_{h-1}^\xi\|_1 = 1$. The classes below are built from these predictive states and the observable operators $J_h^{\xi, \pi}$.

The Eluder dimension, introduced by Russo & Van Roy (2013), measures the sequential complexity of a function class. For a class $\mathcal{F}$ of nonnegative functions and a scale $\alpha > 0$, say a query x is α -dependent on a finite set S of queries if every $f \in \mathcal{F}$ with $\sum_{z \in S} f(z)^2 \leq \alpha^2$ also satisfies $f(x) \leq \alpha$; $\dim_E(\mathcal{F}, \alpha)$ is the length of the longest

sequence of queries such that, for some single $\alpha' \geq \alpha$, every element is α' -independent of its predecessors; under this standard convention (Russo & Van Roy, 2013), $\dim_E(\mathcal{F}, \cdot)$ is nonincreasing in the scale. For our regret analysis, we track the two channels of the causal factorization (Theorem 2.5) separately, through the *channel error classes* $\mathcal{F}_{\text{agg}}^{W, \xi^*} := \{F_\theta^W : \theta \in \Theta\}$ and $\mathcal{F}_{\text{agg}}^{G, \xi^*} := \{F_\Phi^G : \Phi \in \Psi\}$, where (writing $\mathbb{E}^\pi := \mathbb{E}_{\tau \sim P_\xi^\pi}$ for brevity)

$$F_\theta^W(\pi) := \sum_{h=1}^H \mathbb{E}^\pi [\|(W_h^\theta - W_h^{\theta^*}) q_{h-1}^{\xi^*}\|_{1, \mathcal{V}}],$$

$$F_\Phi^G(\pi) := \sum_{h=1}^H \mathbb{E}^\pi [\|(G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi}) W_h^{\theta^*} q_{h-1}^{\xi^*}\|_1].$$

Here and throughout, operators at step h are evaluated at the realized observation-action pair of the trajectory, and both classes depend on the true $\xi^* = (\theta^*, \Phi^*)$ through the reference predictive states $q_{h-1}^{\xi^*}$ and the trajectory distribution $P_{\xi^*}^\pi$. We adopt *uniform* definitions by taking suprema over ξ^* , evaluated at the resolution scale $1/(2T)$:

$$d_{\text{agg}}^W := \sup_{\xi^* \in \Xi} \dim_E(\mathcal{F}_{\text{agg}}^{W, \xi^*}, 1/(2T)),$$

$$d_{\text{agg}}^G := \sup_{\xi^* \in \Xi} \dim_E(\mathcal{F}_{\text{agg}}^{G, \xi^*}, 1/(2T)).$$

We use the channel sum $d_E := d_{\text{agg}}^W + d_{\text{agg}}^G$ as the complexity parameter. The *composed* aggregate error $F_\xi(\pi) := \sum_h \mathbb{E}_{\tau \sim P_{\xi^*}^\pi} [\|(J_h^{\xi, \pi} - J_h^{\xi^*, \pi}) q_{h-1}^{\xi^*}\|_1]$, which is what the regret decomposition measures, is controlled by the two channels. Since $G_h^{\Phi, \pi}$ is ℓ_1 -nonexpansive, the exact identity $(J_h^{\xi, \pi} - J_h^{\xi^*, \pi}) q = G_h^{\Phi, \pi}((W_h^\theta - W_h^{\theta^*}) q) + (G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi})(W_h^{\theta^*} q)$ yields, for every policy π and model $\xi = (\theta, \Phi)$, the pointwise *channel-triangle* inequalities

$$F_\xi \leq F_\theta^W + F_\Phi^G, \quad F_\Phi^G \leq F_\xi + F_\theta^W. \quad (4)$$

The analysis in Appendix D runs the Eluder summability argument once per channel and combines the two via (4). Per-step (stepwise) variants of these classes can be defined analogously; see Remark D.14 in Appendix D.

Proposition 2.6 (Uniform Eluder Dimension Bounds). *Consider model classes satisfying Assumptions 2.3 and 2.4, using the augmented state space $\bar{S}_{\text{red}}^+ = S \times O_A^{\kappa-1} \times O_B^{\kappa-1}$ (dimension $d_W = |S| \cdot |O_A|^{\kappa-1} \cdot |O_B|^{\kappa-1}$). Assume that the learner policy class and the adversary response class depend on histories only through the κ -window statistics retained in $\bar{S}_{\text{red}}^+$, so that the causal factorization descends to the reduced space (see the discussion in Appendix B). Assume the world model class Θ and adversary model class Ψ are parameterized such that the observable operators $\xi \mapsto J_h^{\xi, \pi}$ are Lipschitz in ξ with respect to the ℓ_∞ parameter norm, and that each model class has a finite ϵ -covering number in operator norm, with logarithm polynomial in the stated parameter counts. Then,*

- For **tabular** settings, $d_E = \tilde{O}(H(|S|^2|A||B||O_A|^\kappa|O_B|^\kappa + d_{\text{adv}}|B|))$, where d_{adv} is the feature dimension of the (linear) adversary-response model.

- For **linear world model**, $d_E = \tilde{O}(H(d_{\text{lin}}|O_A|^\kappa|O_B|^\kappa + d_{\text{adv}}|B|))$, where d_{lin} is the linear dimension.
- For **low-rank transitions**, $d_E = \tilde{O}(H(r^2(|A|+|B|)|O_A|^\kappa|O_B|^\kappa + d_{\text{adv}}|B|))$, where r is the rank.

The factor H arises from summing the per-step span dimensions over $h \in [H]$. The factor $|O_B|^\kappa$ arises because the augmented representation must retain enough adversary-side observation information to describe the adversary aggregation operator $G_h^{\Phi, \pi}$ independently of the world parameter θ . For the linear and low-rank families we additionally assume the parameterization is *coordinate-structured*, i.e., each entry of W_h^θ is an affine function of the parameters, so that the world-channel difference family $\{W_h^\theta - W_h^{\theta'}\}$ spans a subspace of the stated dimension. Bilinear parameterizations with unknown factor interactions need not satisfy this (their difference families can span the full ambient space), and the corresponding bounds then fail.

Proof sketch. Since $d_E = d_{\text{agg}}^W + d_{\text{agg}}^G$ by definition, it suffices to bound each channel separately; the channel-triangle inequalities (4) let the regret analysis consume the two bounds directly (Appendix D), so the composed class never needs its own complexity bound. The argument for each channel has two steps. First, each channel error class consists of ℓ_1 norms of linear images of the parameter. For the world channel, the operator difference $W_h^\theta - W_h^{\theta^*}$ is a fixed matrix, independent of the query policy, and each entry is an affine function of the transition and emission parameters; on the reduced window space the difference family therefore lies in a span of dimension $O(|S|^2|A||B| \cdot |O_A|^\kappa|O_B|^\kappa)$ per step (the single-agent analogue of this count is in Liu et al. (2022a)). For the adversary channel, $G_h^{\Phi, \pi}$ is linear in Φ , so $\|(G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi})W_h^{\theta^*}q\|_1 = \|U_h \text{vec}(\Phi - \Phi^*)\|_1$ for a design matrix U_h computable from the query, with $(|B| - 1)d_{\text{adv}}$ free parameters per step.

Second, the sign-witness lemma (Lemma C.1, Appendix C) bounds the Eluder dimension of any such class by its span dimension times logarithm of the evaluation scale. Summing the per-step span dimensions over $h \in [H]$ and adding the two channels gives the tabular bound. Under the coordinate-structure hypothesis, the linear and low-rank families replace the world-channel span count $|S|^2|A||B|$ by d_{lin} and $r^2(|A| + |B|)$ respectively; the adversary channel is unchanged. The full proof is in Appendix C. $\square$

3. Algorithm and Upper Bound

We now present our main algorithm and establish its policy regret guarantee. Algorithm 1 follows the optimistic MLE paradigm, maintaining confidence sets over the joint parameter space and selecting policies that are optimal under the most favorable parameters. The algorithm divides episodes into epochs of geometrically increasing length $T_e = 2^e$. Each epoch uses a single policy selected optimistically from a confidence set built cumulatively from all previous epochs' data. This structure keeps the number of *scheduled* policy

switches logarithmic in T , with data-dependent terminations adding a controlled number more (Lemma D.7); limiting switches matters because the adversary's response changes with the learner's policy and every switch costs warm-up episodes. In epoch e , the algorithm proceeds as follows:

1. **MLE computation:** Compute the maximum likelihood estimate $\hat{\xi}_{e-1} \in \Xi$ of the joint parameter over all trajectories in the cumulative dataset $\mathcal{D}_{e-1}$, which contains data from epochs $1, \dots, e-1$.
2. **Confidence set construction:** Form the confidence set $\mathcal{C}_{e-1}$ consisting of all parameters ξ whose log-likelihood on $\mathcal{D}_{e-1}$ is within β_e of the MLE, where $\beta_e = c(\log \mathcal{N}(\epsilon_e; \Xi) + \log(1/\delta_e) + mB_0)$, with c an absolute constant ($c \geq 16$ suffices), $\mathcal{N}(\epsilon; \Xi)$ the ϵ -covering number of the model class in operator norm (3.1(b)), B_0 the bound on per-trajectory log-likelihood ratios (3.1(c)), discretization scale $\epsilon_e = 2^{-\lceil \log_2(|\mathcal{D}_{e-1}|+2) \rceil} \in [\frac{1}{2} \cdot \frac{1}{|\mathcal{D}_{e-1}|+2}, \frac{1}{|\mathcal{D}_{e-1}|+2}]$ (the dyadic grid keeps the candidate covering nets deterministic; see Lemma D.3), and failure budget $\delta_e = \delta/(2e^2)$. All three quantities are keyed to the data volume and the epoch counter, so the confidence radius requires no knowledge of T ; the mB_0 floor is what prices policy switches (Lemma D.7).
3. **Optimistic planning:** Select the policy-parameter pair (π_e, ξ_e) that maximizes the expected value within the confidence set $\mathcal{C}_{e-1}$. This optimism drives exploration and is the mechanism behind the no-regret guarantee. The selected policy π_e is used throughout the epoch.
4. **Warm-up and execution with statistical termination:** Execute policy π_e for $m-1$ warm-up episodes, discarding data to allow the adversary to stabilize; then execute π_e for at most T_e data-collection episodes. After every data-collection episode t , recompute the maximum-likelihood estimate on all data collected so far; if the deployed model falls more than $\tau_e(t) := 2\beta_e(t) + C_0B_0 \log(2t(t+1)/\delta_e)$ below it in log-likelihood, *terminate the epoch immediately*, retaining the epoch's data. Here $\beta_e(t)$ is the radius of step 2 recomputed at the current dataset size (on the same dyadic grid) and confidence budget $\delta_e/(2t(t+1))$, and C_0 is an absolute constant from the martingale concentration bound behind the test (Appendix D). Let $n_e \leq T_e$ denote the number of data-collection episodes the epoch actually runs. The deviation-aware threshold separates statistical fluctuation from genuine misfit.
5. **Dataset update:** Add the $n_e \leq T_e$ data-collection trajectories (including those of terminated epochs) to the cumulative dataset: $\mathcal{D}_e = \mathcal{D}_{e-1} \cup \{(\pi_e, \tau_1^{(e)}), \dots, (\pi_e, \tau_{n_e}^{(e)})\}$.

The key feature distinguishing this approach is that we use *one policy per epoch*, with epochs ending either at the geometric cap $T_e = 2^e$ or at the statistical termination test. The learner thus deploys $S_T = O(\log T) + \#\{\text{terminations}\}$ policies, one per epoch, over T data-collection episodes, and

we show (Lemma D.7) that terminations are rare enough for their total warm-up cost to be dominated by the leading regret term. This structure serves several purposes:

Cumulative learning: Confidence sets are built from *all historical data*, not just the current epoch. Uniform weighting of these trajectories enables standard concentration and Eluder dimension arguments. This ensures that information accumulates across epochs, with later epochs benefiting from increasingly precise parameter estimates.

Statistical termination: The termination test is what makes committing to a single optimistic policy safe. Optimism deliberately favors directions the data have not yet constrained, and without the test a single ill-informed commitment can run unchecked for $\Theta(T)$ episodes (Prop. D.5); with it, each epoch’s contribution is capped at $\tilde{O}(H\beta_T)$ (Lemma D.6), with β_T the confidence radius at horizon T as in Theorem 3.3. The test is the maximum-likelihood analogue of the count-doubling and determinant-doubling triggers in low-switching-cost RL (Bai et al., 2019; Qiao et al., 2022), though here rare switching is also forced by the adversary, since every policy change costs $m - 1$ warm-up episodes.

Automatic adaptation: Each epoch e sets its length based only on $T_e = 2^e$, requiring no advance knowledge of the final horizon T . The same analysis applies to any final horizon without modifying the algorithm.

Geometric efficiency: The data-collection episodes satisfy $\sum_e n_e \leq T$ by definition of the horizon, and at most $\log_2(2T)$ epochs end by exhausting their caps, since a cap-hit at index e requires $2^e \leq T$; every further epoch ending, except possibly the final horizon-truncated one, is a statistical termination, priced by Lemma D.7. The capped summability bound controls prediction errors over all epochs, and Cauchy–Schwarz yields the $\sqrt{T}$ dependence.

3.1. Main Result: Upper Bound

In addition to Assumptions 2.3 and 2.4, the proof uses the following statistical regularity conditions on the model class. These are standard for the finite/tabular classes considered in Section 5 under positivity, smoothing/discretization, and the weak-revealing conversion described below.

Assumption 3.1 (Statistical Regularity). $\Xi = \Theta \times \Psi$ satisfies:

- a) **Well-specification:** The true parameters $\xi^* = (\theta^*, \Phi^*) \in \Xi$.
- b) **Finite covering:** For every $\epsilon > 0$, the ϵ -covering number $\mathcal{N}(\epsilon; \Xi)$ of Ξ in operator norm is finite; $\log \mathcal{N}(\epsilon; \Xi)$ grows at most polynomially in $1/\epsilon$ and the problem dimensions. Moreover, per-trajectory log-likelihoods $\xi \mapsto \log P_\xi^\pi(\tau)$ are Lipschitz in this norm, uniformly over policies and trajectories (given the positivity in (c)).
- c) **Bounded log-likelihood increments:** Log-likelihood ratios $\log(P_\xi^\pi(\tau)/P_{\xi^*}^\pi(\tau))$ are almost surely bounded by a constant B_0 (which may depend on the smoothing or discretization scale of the model class, and requires all trajectory probabilities to be bounded below, e.g., via smoothing). In particular, all trajectory probabilities are

Algorithm 1 Epoch-Based Optimistic MLE for POMGs

Require: Model class Ξ , policy class Π , confidence level δ , memory bound m , increment bound $B_0 \geq 1$ (Assumption 3.1(c)); absolute constants $c \geq 16$, C_0

- 1: Initialize cumulative dataset $\mathcal{D}_0 \leftarrow \emptyset$
- 2: **for** epoch $e = 1, 2, \dots$ **do**
- 3: Set epoch length $T_e \leftarrow 2^e$
- 4: Set $\delta_e \leftarrow \delta/(2e^2)$, $\epsilon_e \leftarrow 2^{-\lceil \log_2(N_{e-1}+2) \rceil}$ where $N_{e-1} \leftarrow |\mathcal{D}_{e-1}|$, $\beta_e \leftarrow c(\log \mathcal{N}(\epsilon_e; \Xi) + \log(1/\delta_e) + mB_0)$ {keyed to data volume and epoch counter; no knowledge of T required}
- 5: **MLE Estimation:** $\hat{\xi}_{e-1} \leftarrow \arg\max_{\xi \in \Xi} L_{e-1}(\xi)$ where $L_{e-1}(\xi) = \sum_{(\pi, \tau) \in \mathcal{D}_{e-1}} \log P_\xi^\pi(\tau)$
- 6: **Confidence Set:** $\mathcal{C}_{e-1} \leftarrow \{\xi \in \Xi : L_{e-1}(\xi) \geq L_{e-1}(\hat{\xi}_{e-1}) - \beta_e\}$
- 7: **Optimistic Planning:**
 $(\pi_e, \xi_e) \leftarrow \arg\max_{(\pi, \xi) \in \Pi \times \mathcal{C}_{e-1}} V^\pi(\xi)$
- 8: **Warm-up:** Execute π_e for $m - 1$ episodes (discard data)
- 9: **Execution with termination test:** $n_e \leftarrow T_e$; **for** $t = 1, \dots, T_e$: execute π_e , collect $\tau_t^{(e)}$; recompute the MLE $\hat{\xi}_t$ on $\mathcal{D}_{e-1} \cup \{(\pi_e, \tau_1^{(e)}), \dots, (\pi_e, \tau_t^{(e)})\}$; **if** $L_t(\xi_e) < L_t(\hat{\xi}_t) - \tau_e(t)$ **then** set $n_e \leftarrow t$ and end the epoch, where $\tau_e(t) := 2\beta_e(t) + C_0B_0 \log(2t(t+1)/\delta_e)$ and $\beta_e(t)$ is the radius of line 4 recomputed with scale $2^{-\lceil \log_2(|\mathcal{D}_{e-1}|+t+2) \rceil}$ and budget $\delta_e/(2t(t+1))$ {statistical termination}
- 10: **Update Dataset:** $\mathcal{D}_e \leftarrow \mathcal{D}_{e-1} \cup \{(\pi_e, \tau_1^{(e)}), \dots, (\pi_e, \tau_{n_e}^{(e)})\}$ {data-collection trajectories only, terminated epochs included}
- 11: **end for**

uniformly bounded below over Ξ ; we refer to this property as *positivity*. We normalize B_0 so that $B_0 \geq 1$ and B_0 also dominates the Lipschitz constant in (b); this is without loss of generality, since only an upper bound on either quantity enters the analysis.

- d) **Well-defined predictive states:** For every $\xi \in \Xi$, policy π , and trajectory prefix with positive probability under ξ^* , the prefix probability under ξ is positive, so the normalized predictive state q_{h-1}^ξ is well-defined and satisfies $\|q_{h-1}^\xi\|_1 = 1$; this follows from the positivity in (c).
- e) **KL-to-operator conversion (stepwise):** For all $\xi = (\theta, \Phi) \in \Xi$ and policies $\pi \in \Pi$, both

$$\text{KL}(P_\xi^\pi \| P_\xi^\pi) \geq c_{\text{KL}} \sum_{h=1}^H \mathbb{E} \left[\|J_h^{\xi, \pi} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi} q_{h-1}^{\xi^*}\|_1^2 \right]$$
 and the world-channel form

$$\text{KL}(P_\xi^\pi \| P_\xi^\pi) \geq c_{\text{KL}} \sum_{h=1}^H \mathbb{E} \left[\|(W_h^\theta - W_h^{\theta^*}) q_{h-1}^{\xi^*}\|_{1, \mathcal{V}}^2 \right],$$
 where c_{KL} depends on α_κ . The first clause is the standard stepwise conversion proved for POMDPs via weak revealing (Liu et al., 2022a), imposed on the joint-operator class (well-defined by Theorem 2.5); the second is the quantitative content of Assumption 2.4, which makes world-channel discrepancies detectable in κ -step observation windows with strength α_κ , so that a chain-rule/Pinsker argument converts window detectability into trajectory KL. For tabular classes both clauses are verified this way,

with $c_{\text{KL}} = \Omega(\alpha_\kappa^2/\kappa)$, using the positivity in (c). Together the two clauses also control the adversary channel; by the pointwise channel-triangle inequality (4) together with $(a+b)^2 \leq 2a^2 + 2b^2$, $\text{KL}(P_{\xi^*}^\pi \| P_\xi^\pi) \geq (c_{\text{KL}}/4) \sum_h \mathbb{E}[\|(G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi})W_h^{\theta^*} d_{h-1}^{\xi^*}\|_1^2]$. Under this joint conversion no posterior-Lipschitz transport cost enters the analysis; see Remark D.8 (and Lemma D.9 for the weakened, world-channel-only variant).

- f) **Exact warm-up stationarity:** The reaction rule R is time-invariant (as in Section 2.2); consequently, after $m-1$ warm-up episodes of policy π , the m -memory adversary's response is exactly $R_\infty(\pi)$ for all data-collection episodes in that epoch, with no residual transient.

Remark 3.2. For finite/tabular classes, these conditions hold under standard positivity, namely trajectory probabilities are bounded below after smoothing or discretization if needed. Condition (e) is a stepwise KL-to-operator bound; applying Cauchy–Schwarz over h gives $\Delta_e^2 = (\sum_h \mathbb{E}[\|\cdot\|_1])^2 \leq H \sum_h \mathbb{E}[\|\cdot\|_1^2]$, which yields the $H\beta_T$ term in $\beta_T = H\beta_T$. A stronger aggregate version of (e) would remove this factor, giving $\bar{\beta}_T = \beta_T$; see Remark D.13.

Theorem 3.3 (Policy Regret Upper Bound). *Under Assumptions 2.3, 2.4, and 3.1, Algorithm 1 achieves*

$$PR(T) \leq C(H\sqrt{\beta_T} \cdot d_E T \log T + mH S_T)$$

with probability at least $1 - 2\delta$, where $d_E = d_{\text{agg}}^W + d_{\text{agg}}^G$ is the channel-sum aggregate Eluder dimension at scale $1/(2T)$, $\beta_T := H\beta_T$ with $\beta_T = O(\log \mathcal{N}(\epsilon_T; \Xi) + \log T + \log(1/\delta) + mB_0)$ for $\epsilon_T := 1/(2T+4)$, and S_T is the number of policies deployed by the algorithm, which satisfies $S_T \leq 1 + \log_2(2T) + \tilde{O}(\frac{B_0}{1+mB_0} \sqrt{d_E \beta_T T})$ (Lemma D.7).

Consequently $PR(T) = \tilde{O}(H\sqrt{\beta_T d_E T} + mH \log T)$, with no condition relating m , B_0 , and T .

The theorem gives the desired $\sqrt{T}$ dependence on the number of episodes. The effective confidence parameter $\bar{\beta}_T = H\beta_T$ arises from applying Cauchy–Schwarz over H steps to convert the stepwise KL-to-operator bound (Assumption 3.1(e)) into a past-consistency radius for the aggregate class; under the stronger aggregate KL conversion discussed in Remark D.13, the $H\beta_T$ factor improves to β_T . No transport term enters $\bar{\beta}_T$, even though the observable operators depend on the deployed policy; Remark D.8 explains why this dependence costs nothing here. The mB_0 floor in β_e is what buys the unconditional domination of the warm-up term. The constant C depends on α_κ (through c_{KL}) and on B_0 , with additional logarithmic dependence on $|B|$ and H ; the posterior-Lipschitz constant L enters only through d_E and $\mathcal{N}(\cdot; \Xi)$. This rate is optimal in its T and d_E dependence, as shown in Section 4. The full proof is in Appendix D, and an extended sketch in Appendix D.1; here we record the skeleton in four steps.

Optimism reduces regret to simulation error. After its warm-

up episodes, an m -memory adversary plays its stationary response, so $V^{\pi_e}(\xi^*) = V^{\pi_e}(R_\infty(\pi_e))$ throughout epoch e 's data collection. On the confidence event $\xi^* \in \mathcal{C}_{e-1}$, optimism gives $V^{\pi^*}(\xi^*) - V^{\pi_e}(\xi^*) \leq V^{\pi_e}(\xi_e) - V^{\pi_e}(\xi^*)$, and the simulation lemma bounds the right side by $H\Delta_e$, where Δ_e is the aggregate observable-operator prediction error of the optimistic model along the trajectories of π_e .

Likelihood fit controls past errors. Membership $\xi_e \in \mathcal{C}_{e-1}$ means the optimistic model nearly matches the true likelihood on all past data; the KL-to-operator conversion (3.1(e)), available under weak revealing, turns this into small prediction error at the historical policies $\pi_1, \dots, \pi_{e-1}$.

The termination test caps fresh errors. Past consistency says nothing about Δ_e at the newly chosen π_e , and a fixed epoch schedule can commit to an ill-informed policy for $\Theta(T)$ episodes (Proposition D.5). If Δ_e is large, the epoch's own data refute the deployed model at a per-episode rate proportional to the squared stepwise error, so the test fires; hence every epoch, terminated or complete, satisfies $n_e \Delta_e^2 \leq \tilde{O}(H\beta_T/c_{\text{KL}})$ (Lemma D.6).

Eluder summability assembles the bound. Past consistency at radius $\bar{\beta}_T = H\beta_T$ and the per-epoch cap feed the capped summability argument, giving $\sum_e n_e \Delta_e^2 \leq \tilde{O}(d_E \bar{\beta}_T)$ channel by channel (Lemma D.12); Cauchy–Schwarz with $\sum_e n_e \leq T$ then yields $\sum_e n_e H\Delta_e \leq \tilde{O}(H\sqrt{\beta_T d_E T})$, and the warm-up cost of $(m-1)H$ per epoch over S_T epochs is controlled by the switch-count lemma, cap-hit epochs being at most $\log_2(2T)$ and each termination being priced in statistical evidence (Lemma D.7).

4. Lower Bound

We now establish that the upper bound in Theorem 3.3 is optimal in its $\sqrt{T}$ and aggregate-Eluder-dimension dependence, up to problem-dependent and logarithmic factors. Our lower bound demonstrates that even under highly favorable conditions, i.e., fully revealing observations, a memoryless adversary, and a smooth Lipschitz response class, any algorithm must incur $\Omega(\sqrt{d_E T})$ policy regret.

4.1. Hard Instance Construction

We embed the standard d -armed stochastic bandit in the simplest possible POMG, one in which observations fully reveal the state, the adversary is trivial, and the horizon is $H = 2$, so that all remaining difficulty is statistical. The learner's first action routes the game to a latent state, the observation there carries a Bernoulli reward signal, and identifying the best route is exactly the embedded bandit problem. The family is indexed by the location $i^* \in [d]$ of the best route, with gap $\Delta = c_0 \sqrt{d/T}$ for a small constant $c_0 > 0$. The blocks below give the construction.

Dynamics. The POMG has d latent states $s_1, \dots, s_d$ and d learner actions $a_1, \dots, a_d$. The episode starts in a fixed, known state (say s_1 ; step 1 serves only to route the learner's choice). At step $h=1$, the learner observes a fixed observation o_0 (the same in all states, so $h=1$ carries no reward),

chooses action a_i , and the state transitions deterministically to s_i . At step $h=2$, in state s_i the learner observes $o = (i, y)$ where $y \sim \text{Bernoulli}(\mu_i)$ with $\mu_{i^*} = 1/2 + \Delta$ and $\mu_i = 1/2$ for $i \neq i^*$; reward $r_2(i, y) = y$, and $r_1 = 0$. The first coordinate i of o reveals the latent state at step $h=2$; together with the fixed and known initial state at step $h=1$, this makes the POMG fully revealing with $\kappa=1$ and a constant signal parameter independent of Δ .

Adversary. There is a single adversary action b_{coop} , played in every episode, so the adversary is memoryless ($m=1$) and its response $g \equiv b_{\text{coop}}$ satisfies Assumption 2.3 with $L=0$.

Policy regret. The best fixed policy $\pi^* = a_{i^*}$ achieves value $1/2 + \Delta$ per episode. Any algorithm's policy regret equals its bandit regret against arm a_{i^*} .

Model class. Define the ambient class as $\{\mu \in [1/2, 1/2 + \Delta]^d\}$ where μ_i is the mean reward of arm a_i , and take the learner policy class to be the d deterministic routing policies (the learner may still randomize its choices across episodes). The hard instances form the subclass $\mu_{i^*} = 1/2 + \Delta$, $\mu_i = 1/2$ for $i \neq i^*$. With this policy class the ambient class has Eluder dimension $\Theta(d)$ whenever dT exceeds an absolute constant (Appendix E); the lower-bound subclass is contained in the ambient class, and therefore establishes a lower bound for a model class with $d_E = \Theta(d)$.

Theorem 4.1 (Lower Bound). *For any $d, T \in \mathbb{N}$ with $T \geq d \geq 2$ and $dT \geq C_{\text{lb}}$ (an absolute constant determined by c_0 ; $C_{\text{lb}} = \lceil 1/(4c_0^2) \rceil + 1$ suffices), the ambient model class of the construction above, embedded as a POMG, satisfies Assumptions 2.3, 2.4, and 3.1 and has $d_E = \Theta(d)$. For any algorithm, there exists an instance in this class such that $\mathbb{E}[PR(T)] \geq c\sqrt{dT}$ for a universal constant $c > 0$.*

The construction uses a constant adversary response, which is memoryless and trivially satisfies the posterior-Lipschitz condition. Thus the hardness comes not from adversarial memory or partial observability, but from the intrinsic statistical difficulty of identifying the best policy. The detailed proof is in Appendix E.

Since the ambient class has $d_E = \Theta(d)$, Theorem 4.1 establishes the claimed $\Omega(\sqrt{d_E T})$ dependence. We do not claim a complete minimax characterization of all horizon and memory factors. The outer factor H in the upper bound comes from the simulation lemma, and the $mH \log T$ term is the warm-up cost needed to let an m -memory adversary stabilize at the beginning of each epoch. Refined hard instances can force horizon and stabilization costs. Closing the remaining constants and problem-dependent factors in H and m is an interesting direction for future work.

5. Extensions

Next, we study several natural extensions of our framework.

5.1. Adaptive Horizon Property

Algorithm 1 naturally handles unknown time horizons T through its epoch-based structure. Since epochs grow geometrically and each epoch builds its confidence set from all historical data (not just the current epoch), the algorithm requires no prior knowledge of when it will terminate.

Theorem 5.1 (Adaptivity). *Without knowledge of T , and for all horizons T simultaneously, Algorithm 1 achieves $PR(T) = \tilde{O}(H\sqrt{\beta_T d_E T} + mH \log T)$ with probability at least $1 - 2\delta$, where β_T and d_E are as in Theorem 3.3.*

The result follows from Theorem 3.3. Each epoch uses only information about its own length to determine how many episodes to run, requiring no knowledge of the total horizon T . The regret bound holds for all T by a union bound over epochs with confidence parameters $\delta_e = \delta/(2e^2)$, ensuring $\sum_{e=1}^{\infty} \delta_e \leq \delta$; a second, identical budget covers the per-epoch cap and switch-count events, giving the stated $1 - 2\delta$. The key observation is that the confidence and Eluder summability arguments are prefix-stable. They apply to any completed prefix of epochs and to the possibly partial final epoch. Since $\sum_e n_e \leq T$ for every horizon and the deployed-policy count is bounded by Lemma D.7, the same Cauchy-Schwarz step as in Theorem 3.3 gives the desired bound for every final horizon T . This natural adaptivity arises from the epoch-based structure combined with cumulative confidence sets, requiring no modification to the base algorithm. The detailed proof is in Appendix F.

5.2. Fading Memory Adversaries

Our base model assumes the adversary's response depends on the learner's most recent m policies through a hard cutoff window. A natural generalization considers adversaries with exponentially decaying memory. Formally, a γ -fading memory adversary maintains an internal state using the *normalized* recursion $z_t = \gamma z_{t-1} + (1 - \gamma)\phi(\pi^t)$, where $\phi : \Pi \rightarrow \mathbb{R}^{d_z}$ is a feature map with $\|\phi(\pi)\|_2 \leq 1$, and its response is $g^t = g(z_t)$ for some L_g -Lipschitz function g . The normalization $(1 - \gamma)$ ensures $\|z_t\|_2 \leq 1$ uniformly and the stationary state under constant policy π is exactly $z_\infty = \phi(\pi)$, with $\|z_t - z_\infty\|_2 \leq \gamma^k \cdot 2$ after k steps.

Theorem 5.2 (Fading Memory). *Against a γ -fading memory adversary with L_g -Lipschitz response function, run Algorithm 1 with memory parameter $m := m_{\text{warm}} := \lceil \log(2T)/(1 - \gamma) \rceil$. Then, with probability at least $1 - 2\delta$,*

$$PR(T) = \tilde{O}\left(H\sqrt{\beta_T^\gamma d_E T} + \frac{HS_T \log T}{1 - \gamma}\right),$$

where $\beta_T^\gamma := H\beta_T^\gamma$ with $\beta_T^\gamma := O(\log \mathcal{N}(\epsilon_T; \Xi) + \log T + \log(1/\delta) + B_0 \log(2T)/(1 - \gamma) + B_0 H L_g)$, and S_T is the deployed-policy count of Theorem 3.3.

The algorithmic modification is simple. At the beginning of each epoch e , the algorithm selects an optimistic policy π_e and plays it for $m_{\text{warm}} - 1$ warm-up episodes, whose

data are discarded, followed by at most T_e data-collection episodes. The analysis shows that after m_{warm} episodes of playing a fixed policy π , the adversary's state is within $O(\gamma^{m_{\text{warm}}}) \leq O(1/T)$ of the stationary state $z_\infty = \phi(\pi)$.

Additionally, when computing posterior-predictive quantities for the Lipschitz condition, we truncate the memory to the most recent $O(\log T/(1-\gamma))$ policies, incurring approximation error $O(\gamma^{m_{\text{warm}}}) = O(1/T)$ that does not affect the asymptotic regret rate. As stated, the fading variant fixes T through m_{warm} ; an epoch-indexed warm-up restores the horizon-free property at the same rate (Appendix G).

This result shows that fading memory adversaries admit efficient learning with regret depending on the effective memory $1/(1-\gamma)$ rather than requiring a hard cutoff. The extreme cases are instructive. Setting $\gamma = 0$ recovers a memoryless adversary, for which the warm-up may be set to zero, while $\gamma \rightarrow 1$ approaches a long-memory regime in which the effective memory $1/(1-\gamma)$ diverges.

The epoch-based structure with one policy per epoch handles fading memory naturally. Within each epoch, we use the extended warm-up phase once, then execute the policy for the epoch's data-collection episodes. Since the adversary sees the same policy repeatedly within the epoch, its state remains stable after warm-up. With S_T epochs started, the warm-up cost is $O(m_{\text{warm}} \cdot S_T \cdot H) = O(\frac{H S_T \log T}{1-\gamma})$, which is $O(H \log^2 T/(1-\gamma))$ when only scheduled epoch endings occur. The complete proof, including the stability analysis of the adversary's internal state and the truncation argument, is provided in Appendix G.

5.3. Tabular Instantiation

For finite POMGs, we can derive explicit bounds in terms of the state space size $|S|$, action space sizes $|A|$ and $|B|$, and observation space sizes $|O_A|$ and $|O_B|$.

Theorem 5.3 (Tabular Bounds). *Consider a tabular POMG as above, with weak-revealing parameter κ , memory m , and a linear adversary response of dimension d_{adv} . Under the assumptions of Theorem 3.3, Algorithm 1 achieves, with probability at least $1 - 2\delta$,*

$$PR(T) = \tilde{O}(Hm \log T + H\sqrt{\tilde{\beta}_T^{\text{tab}} \cdot d_E^{\text{tab}} \cdot T}),$$

where d_E^{tab} is the tabular Eluder bound of Proposition 2.6 and $\tilde{\beta}_T^{\text{tab}} = \tilde{O}(H^2(|S|^2|A||B| + |S|(|O_A|+|O_B|) + d_{\text{adv}}|B|) + HmB_0)$.

In $\tilde{\beta}_T^{\text{tab}}$, the middle term accounts for the unknown emission kernels, and the $\tilde{O}$ absorbs an additive $H \log(1/\delta)$. Under the standard tabular scaling $|O_A| + |O_B| = O(|S||A||B|)$, the emission term is absorbed leaving $\tilde{\beta}_T^{\text{tab}} = \tilde{O}(H^2(|S|^2|A||B| + d_{\text{adv}}|B|) + HmB_0)$. The outer factor H comes from the simulation lemma (Lemma D.2), while the tabular Eluder dimension itself scales linearly with H because the operator class contains separate time-step-dependent operators for $h \in [H]$. Substituting these

quantities into Theorem 3.3 yields the bound. The detailed proof is in Appendix H.

5.4. Computational Considerations

The optimistic planning step in Algorithm 1 requires solving $\max_{(\pi, \xi) \in \Pi \times \mathcal{C}_{e-1}} V^\pi(\xi)$, which is computationally challenging in general. POMDP planning is PSPACE-complete even with known parameters, and optimization over the confidence set adds another layer of difficulty.

The statistical guarantees above should therefore be interpreted independently of computational tractability in the most general model class. For small tabular instances or structured subclasses, the planning and confidence-set optimization problems may be approachable by dynamic programming, discretization, or problem-specific optimization methods. The epoch structure is still useful computationally because the optimistic planning problem is solved only once per epoch, that is, S_T times in total (Lemma D.7), rather than after every episode, although the termination test does require a per-episode MLE evaluation.

A possible practical alternative is to replace optimistic planning with posterior sampling. At the start of each epoch e , sample parameters $\tilde{\xi}$ from a posterior distribution given the cumulative data $\mathcal{D}_{e-1}$, and then compute the optimal policy for $\tilde{\xi}$. This may be easier to implement in some model classes, although proving an analogous regret guarantee for posterior sampling would require a separate analysis. Posterior sampling has been analyzed for POMGs in the self-play and oblivious-adversary regimes (Qiu et al., 2023); extending such guarantees to policy regret against adaptive adversaries is open.

For additional structured models, such as linear-Gaussian or control-oriented special cases, parts of estimation or planning may admit more efficient implementations. Extending computational tractability guarantees to broad POMG classes remains an important direction for future work.

6. Conclusion

We have developed a comprehensive theoretical framework for policy regret minimization in partially observable Markov games against adaptive adversaries. Our results show that $\sqrt{d_E T}$ policy regret is both achievable and unavoidable, up to problem-dependent and logarithmic factors; along the way, we showed that data-dependent policy switching is necessary for optimistic algorithms that update their policies only rarely. Several directions remain open, including efficient algorithms beyond tabular and linear POMGs, extensions to multi-player games, non-stationary adversaries, and continuous state-action spaces, and sharper domain-specific models of how partial observability interacts with strategic behavior. More broadly, as AI systems increasingly operate in environments with multiple adaptive agents, these foundations are essential for designing learning algorithms with reliable performance guarantees.

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Impact Statement

This paper presents work whose goal is to advance the foundations of machine learning in partially observable multi-agent environments. Potential societal impacts are those broadly associated with reinforcement learning and multi-agent decision-making, including applications in autonomous systems, markets, and security. Our results are primarily mathematical and do not introduce new deployed systems or datasets; we do not identify specific ethical concerns beyond those generally associated with this area.

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Organization of the appendices. Appendix A surveys related work and situates our guarantees (Tables 1 and 2). Appendix B proves the causal decomposition (Theorem 2.5). Appendix C proves the Eluder dimension bounds of Proposition 2.6. Appendix D proves Theorem 3.3, opening with an extended proof sketch (Appendix D.1). Appendices E, F, G, and H prove the lower bound, the adaptivity property, the fading-memory extension, and the tabular instantiation. Appendix I reports the empirical validation, and Appendix J gives a detailed technical comparison with concurrent work.

A. Related Work

The concept of policy regret was introduced by Arora et al. (2012) in the context of multi-armed bandits with adaptive adversaries. They showed that while no algorithm can achieve sublinear regret against adversaries with unbounded memory, bounded-memory adversaries admit efficient learning through mini-batching. Competing against reactive opponents with bounded memory has earlier roots in the information-theoretic literature on sequential prediction with loss functions that have memory (Merhav et al., 2002). Cesa-Bianchi et al. (2013) sharpened the minimax picture for switching costs and other adaptive adversaries, and Dekel et al. (2014) established the matching $\tilde{O}(T^{2/3})$ characterization for bandits with switching costs, while Arora et al. (2019) extended the switching-cost analysis to bandits with feedback graphs, quantifying how side observations change the attainable rates. Rakhlin & Sridharan (2013) studied online learning with predictable sequences, showing how regret improves when the opponent’s moves are partially predictable. We carry this program to partially observable sequential decision-making with structured dynamics.

Arora et al. (2018) studied the game-theoretic implications of policy regret and introduced the notion of policy equilibrium. They showed that when both players use no-policy-regret algorithms, the empirical distribution of play converges to a policy equilibrium. This result complements the earlier policy-regret literature by showing that no-policy-regret learning has an associated equilibrium interpretation in repeated games. Our work is different in focus. We study policy regret for a single learner in a partially observable Markov game against an adaptive adversary, rather than convergence of two no-policy-regret players to an equilibrium. A complementary line of work studies the reverse question of how an opponent can strategize against a no-regret learner (Deng et al., 2019); policy regret is the benchmark a learner adopts to be robust against precisely such strategic responses.

More recently, Nguyen-Tang & Arora (2024) initiated the study of policy regret in Markov games with adaptive adversaries, identifying fundamental barriers under unbounded memory or non-stationarity and giving positive results under structural restrictions on the adversary. Nguyen-Tang & Arora (2025) studied policy regret in Markov games with function approximation, obtaining $\sqrt{T}$ policy regret under Eluder-type complexity conditions. Our work brings these ideas to the more complex setting of partially observable Markov games. The partial-observation setting introduces new difficulties as the learner must reason through observable operator representations, and the adversary response is entangled with hidden world dynamics. Our causal decomposition separates hidden-world estimation from adversary-response estimation. The epoch structure, with geometric caps and statistical termination, then limits scheduled policy changes to $O(\log T)$ and prices each data-driven termination in statistical evidence, so that comparing adversary responses across deployed policies imposes no cost on the final confidence radius.

Learning in POMDPs Sample-efficient learning in POMDPs has been studied extensively. Early work established computational hardness results (Papadimitriou & Tsitsiklis, 1987) and developed approximate planning methods (Cassandra et al., 1994; Kaelbling et al., 1998). More recent work has focused on sample complexity under structural assumptions. Krishnamurthy et al. (2016) established tractability for POMDPs with rich observations, while Azizzadenesheli et al. (2016) and Guo et al. (2016) developed spectral methods for parameter estimation. The breakthrough work of Jin et al. (2020) introduced sample-efficient algorithms for undercomplete POMDPs where the number of observations exceeds the number of latent states.

Liu et al. (2022a) significantly generalized these results by identifying the class of weakly revealing POMDPs, showing that observability over multi-step windows suffices for efficient learning. Liu et al. (2023) developed an optimistic MLE framework that underlies our approach, establishing $\tilde{O}(\sqrt{T})$ regret bounds under weak revealing conditions. Recent work by Golowich et al. (2023) addresses computational aspects of planning and learning in observable POMDPs. Beyond POMDPs, predictive state representations admit sample-efficient learning (Zhan et al., 2023; Uehara et al., 2022), and unified complexity frameworks such as the generalized Eluder coefficient (Zhong et al., 2022) and the decision-estimation coefficient (Foster et al., 2021) cover broad classes of partially observable decision problems. These frameworks, however, treat the environment as fixed. The data-generating process does not change in response to the learner’s strategy, and that response is precisely the coupling that policy regret against an adaptive adversary must address.

Learning in Markov Games Multi-agent reinforcement learning in Markov games has been studied from both computational and statistical perspectives (Littman, 1994; Hu & Wellman, 2003; Greenwald & Hall, 2003). For fully observable games, significant progress has been made on finding Nash equilibria and learning against adaptive opponents. Bai & Jin (2020) and Xie et al. (2020) developed sample-efficient algorithms for finding Nash equilibria in two-player zero-sum games. Jin et al. (2022) introduced the exploiter framework for provably learning non-exploitable policies via self-play, while Tian et al. (2021) studied online learning in unknown Markov games without observing the opponent’s actions, achieving sublinear regret against the minimax value. Liu et al. (2022c) studied Markov games against adversarial opponents, delineating which feedback models permit sublinear external regret, and Foster et al. (2023) characterized the statistical complexity of multi-agent decision making. These works measure external regret against oblivious or worst-case opponents, rather than policy regret against adaptive ones.

The partially observable case is considerably more challenging. Nayyar et al. (2013) developed the common-information approach to decentralized stochastic control with partial history sharing, a structural foundation for games with asymmetric information. Liu et al. (2022b) studied POMGs under weak revealing assumptions, showing polynomial sample complexity for self-play and, against adversarial opponents, for learning the maximin policy. Qiu et al. (2023) obtained sublinear-regret guarantees for POMGs with general function approximation via posterior sampling, in self-play and adversarial regimes with regret measured against the game value, and Liu & Zhang (2023) studied partially observable multi-agent RL under information-sharing structures that restore (quasi-)efficiency. However, the adaptive adversary setting we consider here requires fundamentally different techniques, as the adversary’s response depends on the learner’s policy in ways that create complex feedback loops.

Eluder Dimension and Function Approximation The Eluder dimension, introduced by Russo & Van Roy (2013), provides a complexity measure for function classes that governs the sample complexity of exploration in bandits and MDPs. Osband & Van Roy (2014) applied Eluder-based analysis to model-based reinforcement learning, while Wang et al. (2020) extended these ideas to general value function approximation. Ayoub et al. (2020) developed model-based algorithms using Eluder dimension for structured MDPs, and Zanette et al. (2020) studied exploration under low inherent Bellman error, a structural condition complementary to Eluder-type complexity measures. Our work builds on these ideas, using Eluder dimension to characterize the complexity of the joint world-adversary model class in POMGs.

Observable Operators and Spectral Methods The observable operator model (OOM) framework was introduced by Jaeger (2000) as an alternative to hidden Markov models, representing dynamics through matrix operations on a predictive state space. Boots & Gordon (2011) developed an online spectral algorithm for learning predictive-state representations of partially observable nonlinear dynamical systems. Hsu et al. (2012) gave a spectral, method-of-moments algorithm for learning hidden Markov models with finite-sample guarantees. Song et al. (2010) extended these ideas to continuous state spaces using kernel methods. The OOM framework has proven particularly valuable for POMDPs, enabling likelihood computation and sample-efficient learning analyses under suitable observability conditions.

Low-Switching Reinforcement Learning Our algorithm deploys a single policy per epoch, ending epochs at geometric caps or by a statistical termination test. Deploying few policies is also the goal of low-switching-cost RL (Bai et al., 2019; Qiao et al., 2022), where switches are limited for operational reasons such as deployment or recomputation cost, and data-dependent triggers (count or determinant doubling) achieve logarithmic switching in single-agent settings. Two aspects differentiate our setting: against an adaptive adversary every policy switch additionally costs warm-up episodes for the opponent’s response to re-stabilize, and, as we show in Proposition D.5, data-dependent termination is not merely convenient but *necessary*. With a fixed switching schedule, optimistic planning combined with infrequent updates provably incurs linear policy regret (Appendix D.7).

Concurrent and Related Work on POMGs The most closely related work is a concurrent manuscript by Sang & Nguyen-Tang (2025), which also studies policy regret minimization in POMGs via optimistic MLE under a posterior-Lipschitz response condition. Our paper differs in several technical respects: (i) our decoupled formulation of the posterior-Lipschitz condition, which avoids circular dependence of the smoothness condition on the adversary being constrained; (ii) our causal decomposition tracks both players’ private observation-action histories, so it applies to adversaries that act on private observations, whereas their factorization models the opponent’s response as a function of the learner-side history; (iii) our one-policy-per-epoch structure with cumulative confidence sets and statistical termination, under which no posterior-Lipschitz transport term enters the final confidence radius (Remark D.8), whereas a flat batching schedule lets the transport term grow polynomially (Appendix J); (iv) our matching lower bound, proving the optimality of the $\tilde{O}(\sqrt{T})$ dependence on

T and the aggregate Eluder dimension; and (v) our extensions to fading memory adversaries and horizon-adaptive guarantees. In the other direction, [Sang & Nguyen-Tang \(2025\)](#) give a counterexample showing that without a posterior-Lipschitz-type condition, sublinear policy regret is impossible even under weakly revealing observations and a memory-bounded, stationary opponent, a useful complement establishing that smoothness conditions of this kind are necessary rather than an artifact of the analysis.

Table 1 situates our guarantees among the most closely related settings. Table 2 summarizes the bounds proved in this paper.

Table 1. Comparison with the most closely related work. “ m -memory adaptive” means the opponent reacts to the learner’s recent policies, and policy regret charges the learner for that reaction; PAC and regret guarantees measured against the maximin or Nash value treat the opponent as worst-case within episodes instead. Rates suppress horizon and problem-dependent factors; d_E is the channel-sum aggregate Eluder dimension of Section 2.

Reference	Observability	Opponent	Benchmark	Rate
Arora et al. (2012)	bandit feedback	m -memory adaptive	policy regret	$\tilde{O}(T^{2/3})$
Dekel et al. (2014)	bandit feedback	switching cost	policy regret	$\tilde{\Theta}(T^{2/3})$
Liu et al. (2022b)	POMG, weak revealing	self-play; adversarial	PAC vs. maximin	$\text{poly}(1/\epsilon)$
Qiu et al. (2023)	POMG, general classes	self-play; adversarial	regret vs. Nash	$\tilde{O}(\sqrt{T})$
This paper	POMG, weak revealing	m -memory adaptive	policy regret	$\tilde{O}(\sqrt{d_E T})$, matching lower bound

Table 2. Summary of the guarantees proved in this paper. Upper bounds hold with probability at least $1 - 2\delta$ and lower bounds are on expected policy regret; $\tilde{\beta}_T = H\beta_T$ is the effective confidence radius and S_T the deployed-policy count.

Result	Setting	Bound
Theorem 3.3	m -memory adversary	$PR(T) = \tilde{O}(H\sqrt{\tilde{\beta}_T d_E T} + mH \log T)$, unconditionally
Theorem 4.1	fixed memoryless adversary	$\mathbb{E}[PR(T)] = \Omega(\sqrt{d_E T})$
Proposition D.5	fixed epoch schedule	$\mathbb{E}[PR(T)] \geq T/4$ on an instance satisfying all assumptions
Theorem 5.1	unknown horizon	same rate as Theorem 3.3, for all T simultaneously
Theorem 5.2	γ -fading memory	$\tilde{O}(H\sqrt{\tilde{\beta}_T^\gamma d_E T} + HS_T \log T/(1 - \gamma))$
Theorem 5.3	tabular POMG	explicit $\tilde{O}(H\sqrt{\tilde{\beta}_T^{\text{tab}} d_E^{\text{tab}} T}) + \tilde{O}(Hm \log T)$

B. Proof of Theorem 2.5 (Causal Decomposition)

We construct explicit operators W_h^θ and $G_h^{\Phi, \pi}$ and verify the factorization $J_h^{\xi, \pi} = G_h^{\Phi, \pi} \circ W_h^\theta$. Several issues must be handled simultaneously: (i) the operators must have compatible types, which requires an intermediate B -indexed space $\mathcal{V}$; (ii) the aggregation operator $G_h^{\Phi, \pi}$ must be genuinely independent of θ ; (iii) since the adversary’s private history cannot be inferred from the learner’s history in a general POMG, the augmented state must track both players’ histories; (iv) the construction must also respect the causal timing of the game, where observations come first, actions are then chosen, and the world transitions afterward; and (v) the OOM operator represents the controlled conditional process of observations given learner actions, rather than the joint probability of learner actions and observations.

Proof of Theorem 2.5. The proof proceeds in stages, constructing the two operators explicitly and then verifying the factorization and its structural properties (Figure 1).

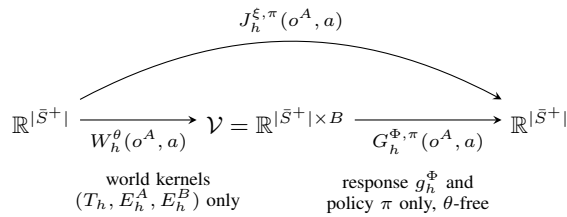


Figure 1. The causal decomposition of Theorem 2.5. The observable operator factors through the B -indexed intermediate space; the world channel carries all dependence on θ , and the adversary aggregation carries all dependence on the response model and the deployed policy.

Timing convention and histories. We begin by fixing the timing convention. At step h , both players observe (o_h^A, o_h^B) , then choose actions (a_h, b_h) , and then the latent state transitions to s_{h+1} . This order matters because the adversary's response at step h is allowed to depend on its current observation o_h^B , but not on the action b_h before it is chosen.

It is useful to distinguish the adversary history available before the action from the history stored after the action. The post-action adversary history after step h is

$$\tau_h^B = (o_1^B, b_1, \dots, o_{h-1}^B, b_{h-1}, o_h^B, b_h),$$

which includes the chosen action b_h . This is the object carried into the next augmented state. The adversary's decision history at step h is

$$\eta_h^B = (\tau_{h-1}^B, o_h^B) = (o_1^B, b_1, \dots, o_{h-1}^B, b_{h-1}, o_h^B),$$

which includes the observation-action history through step $h-1$ plus the new observation o_h^B . This is the information available after observing o_h^B and before choosing b_h , so $g_h^\Phi(\cdot \mid \eta_h^B, \pi)$ is well-defined. The two histories are related by $\tau_h^B = (\eta_h^B, b_h)$, i.e., τ_h^B is η_h^B extended by b_h .

The learner histories are handled analogously. At the beginning of step h , the learner carries τ_{h-1}^A , and after observing o_h^A and taking action a_h , the history becomes

$$\tau_h^A = (\tau_{h-1}^A, o_h^A, a_h).$$

The augmented state space. We now define the augmented state at the beginning of step h as

$$\bar{s}_h = (s_h, \tau_{h-1}^A, \tau_{h-1}^B),$$

where both histories are post-action observation-action histories (consistent with $\mathcal{T}^A$ and $\mathcal{T}^B$). The augmented state space is

$$\bar{S}^+ = S \times \mathcal{T}_{\leq H}^A \times \mathcal{T}_{\leq H}^B.$$

This full-history space is finite for every fixed finite-horizon problem instance, since all component spaces are finite (Section 2). It is intentionally larger than the reduced predictive representations used later for complexity bounds. Here the goal is exact factorization, and exact factorization requires enough information to recover both players' histories. We adopt the following length convention throughout the construction: all step- h operators act as zero on coordinates whose history components are not length-matched to the step. For $J_h^{\xi, \pi}$ and W_h^θ , inputs carry histories of length $h-1$ and outputs of length h ; $G_h^{\Phi, \pi}$ acts on the B -indexed intermediate coordinates that W_h^θ produces, whose learner-history component already has length h . With this convention every operator below is totally defined on its space, and the OOM products only ever traverse length-matched supports.

World channel W_h^θ mapping into the B -indexed intermediate space. We next construct the world channel. Let

$$\mathcal{V} = \mathbb{R}^{|\bar{S}^+| \times B}.$$

An element of $\mathcal{V}$ is a family $(u_b)_{b \in B}$, where each $u_b \in \mathbb{R}^{|\bar{S}^+|}$. The role of this space is to keep the adversary action b as an explicit index before it is mixed according to the adversary response. For each $b \in B$, define the linear map

$$W_{h,b}^\theta(o^A, a) \in \mathcal{L}(\mathbb{R}^{|\bar{S}^+|}, \mathbb{R}^{|\bar{S}^+|})$$

by

$$[W_{h,b}^\theta(o^A, a)v]_{(s', \tau_h^A, \tau_h^B)} = \sum_{\substack{(s, \tau_h^A, \tau_h^B) \\ (\tau_h^A, o^A, a) = \tau_h^A}} \sum_{\substack{o^B \in O_B \\ (\tau_h^B, o^B, b) = \tau_h^B}} E_h^A(o^A \mid s) E_h^B(o^B \mid s) T_h(s' \mid s, a, b) v_{(s, \tau_h^A, \tau_h^B)}. \quad (5)$$

This formula has a simple interpretation; the product $E_h^A(o^A \mid s) E_h^B(o^B \mid s)$ uses the conditional independence of the two observations given s (Section 2). Starting from augmented state (s, τ_h^A, τ_h^B) , the world emits o^A to the learner and o^B to the adversary. The learner action a is fixed because the controlled OOM operator is conditioned on learner actions. The adversary action b is not yet randomized. Instead, $W_{h,b}^\theta$ computes the hypothetical successor mass if the adversary were to choose this particular b . The learner history becomes (τ_h^A, o^A, a) , the adversary history becomes (τ_h^B, o^B, b) , and the latent state moves to s' according to $T_h(\cdot \mid s, a, b)$.

The history-consistency conditions in (5) only select legal predecessors. Once (s', τ_h^A, τ_h^B) , o^A , a , and b are fixed, the equations

$$\tau_h^A = (\tau_h^A, o^A, a), \quad \tau_h^B = (\tau_h^B, o^B, b)$$

determine the predecessor histories and the adversary observation whenever they exist. The summation therefore reduces to a single genuine sum over the latent state $s \in S$ (the histories and the adversary observation are pinned by the output index), which is exactly the belief-propagation sum of the controlled forward kernel on the augmented state space.

The full world channel is the B -indexed collection

$$W_h^\theta(o^A, a)v = (W_{h,b}^\theta(o^A, a)v)_{b \in B} \in \mathcal{V}.$$

By construction, W_h^θ depends only on the world kernels (T_h, E_h^A, E_h^B) . It does not depend on Φ or on the learner policy π . The interpretation is that for each hypothetical adversary action b , $W_{h,b}^\theta$ computes the unnormalized mass flowing to each successor augmented state (s', τ_h^A, τ_h^B) under world dynamics. Crucially, $\tau_h^B = (\tau_h^B, o_h^B, b)$ includes b (the adversary's action at this step), so the next step's augmented state correctly records the full observation-action history of the adversary.

All strategic weighting over adversary actions by response probabilities is deliberately postponed to the next operator, $G_h^{\Phi, \pi}$.

Adversary aggregation $G_h^{\Phi, \pi}$ mapping from the intermediate space. We now define the adversary aggregation operator. Given $(u_b)_{b \in B} \in \mathcal{V}$, define $G_h^{\Phi, \pi}(o^A, a) : \mathcal{V} \rightarrow \mathbb{R}^{|\bar{S}^+|}$ by

$$[G_h^{\Phi, \pi}(o^A, a)(u_b)_{b \in B}]_{(s', \tau_h^A, \tau_h^B)} = g_h^\Phi(b \mid \eta_h^B, \pi) [u_b]_{(s', \tau_h^A, \tau_h^B)}, \quad (6)$$

where the output history is written as $\tau_h^B = (\eta_h^B, b)$, with $\eta_h^B = (\tau_{h-1}^B, o_h^B)$ and $b \in B$. Coordinates whose history components are not length-matched to step h , or do not parse as a post-action history, are mapped to zero, per the length convention above.

This respects causal timing as the adversary forms $\eta_h^B = (\tau_{h-1}^B, o_h^B)$ after observing o_h^B and before choosing b_h , then $g_h^\Phi(b \mid \eta_h^B, \pi)$ gives the probability of choosing b , and then b is appended to form τ_h^B .

The key property is that $G_h^{\Phi, \pi}$ depends only on (g_h^Φ, π) and does not involve θ . The input $(u_b)_{b \in B} \in \mathcal{V}$ already contains the hypothetical successor masses computed by the world channel W_h^θ , one for each possible adversary action. The aggregation operator then reads the adversary response weight $g_h^\Phi(b \mid \eta_h^B, \pi)$, where both b and η_h^B are determined by the output index τ_h^B , and multiplies the corresponding coordinate of u_b by this weight. Thus $G_h^{\Phi, \pi}$ uses the adversary decision history encoded in the augmented-state index and the response model g_h^Φ , but it never refers to the world parameter θ .

It is worth clarifying how the mixture over adversary actions is represented in this coordinatewise definition. For a fixed output index (s', τ_h^A, τ_h^B) , the post-action history τ_h^B already contains the final adversary action. Writing $\tau_h^B = (\eta_h^B, b)$, where $\eta_h^B = (\tau_{h-1}^B, o_h^B)$ is the decision history and b is the action appended to it, only the component u_b contributes to this particular output coordinate. This does not mean that the other adversary actions are ignored. If we fix a decision history η_h^B and sum over all post-action histories (η_h^B, b) with $b \in B$, the aggregation recovers the full mixture

$$\sum_{b \in B} g_h^\Phi(b \mid \eta_h^B, \pi) [u_b]_{(\cdot, \cdot, (\eta_h^B, b))}.$$

Thus $G_h^{\Phi, \pi}$ represents the adversary's randomized action by assigning each possible b to the successor coordinates whose adversary history ends with that b , and weighting those coordinates by $g_h^\Phi(b \mid \eta_h^B, \pi)$.

Verifying the factorization $J_h^{\xi, \pi} = G_h^{\Phi, \pi} \circ W_h^\theta$. The canonical controlled forward operator of the protocol is, coordinate-wise,

$$\begin{aligned} [J_h^{\xi, \pi}(o^A, a)]_{(s', \hat{\tau}^A, \hat{\tau}^B), (s, \tau^A, \tau^B)} &= \mathbf{1}[\hat{\tau}^A = (\tau^A, o^A, a)] \sum_{\substack{o^B, b: \\ \hat{\tau}^B = (\tau^B, o^B, b)}} E_h^A(o^A \mid s) E_h^B(o^B \mid s) g_h^\Phi(b \mid (\tau^B, o^B), \pi) T_h(s' \mid s, a, b), \end{aligned}$$

on length-matched coordinates and zero otherwise; this is the operator whose products yield the controlled OOM identity below, and the theorem asserts that it factorizes.

We can now verify the factorization. Combining (5) and (6), and writing the adversary post-action history as

$$\tau_h^B = (\eta_h^B, b), \quad \eta_h^B = (\tau_{h-1}^B, o_h^B),$$

we have

$$\begin{aligned}
 & [(G_h^{\Phi, \pi} \circ W_h^\theta)(o^A, a)v]_{(s', \tau_h^A, \tau_h^B)} \\
 &= g_h^\Phi(b \mid \eta_h^B, \pi) [W_{h,b}^\theta(o^A, a)v]_{(s', \tau_h^A, \tau_h^B)} \\
 &= g_h^\Phi(b \mid \eta_h^B, \pi) \sum_{\substack{(s, \tau_h^A, \tau_h^B) \\ (\tau_h^A, o^A, a) = \tau_h^A}} \sum_{\substack{o^B \in O_B \\ (\tau_h^B, o^B, b) = \tau_h^B}} E_h^A(o^A \mid s) E_h^B(o^B \mid s) T_h(s' \mid s, a, b) v_{(s, \tau_h^A, \tau_h^B)}. \tag{7}
 \end{aligned}$$

This is exactly the controlled conditional forward dynamics of the POMG. For each current augmented state (s, τ_h^A, τ_h^B) with mass $v_{(s, \tau_h^A, \tau_h^B)}$, the expression in (7) follows exactly one step of the POMG protocol. The learner observation o^A and adversary observation o^B are emitted from the same latent state s , with probabilities $E_h^A(o^A \mid s)$ and $E_h^B(o^B \mid s)$. The adversary observation o^B extends the adversary's pre-step history to the decision history $\eta_h^B = (\tau_h^B, o^B)$. The adversary then chooses action b with probability $g_h^\Phi(b \mid \eta_h^B, \pi)$, and its post-action history becomes $\tau_h^B = (\tau_h^B, o^B, b) = (\eta_h^B, b)$. At the same step, the learner takes the fixed controlled action a , so its post-action history becomes $\tau_h^A = (\tau_h^A, o^A, a)$. Finally, the world transitions to s' with probability $T_h(s' \mid s, a, b)$. Thus the right-hand side of (7) is precisely the (s', τ_h^A, τ_h^B) coordinate of the controlled conditional forward operator $J_h^{\xi, \pi}(o^A, a)v$. Therefore

$$J_h^{\xi, \pi}(o^A, a) = G_h^{\Phi, \pi}(o^A, a) \circ W_h^\theta(o^A, a).$$

The OOM identity and learner action probabilities. It remains to connect the factorization above to the controlled OOM convention used in the paper. The operator $J_h^{\xi, \pi}(o^A, a)$ is indexed by the learner observation o^A and the learner action a , and it propagates unnormalized mass over the augmented state space conditional on that controlled observation-action pair. Therefore, if $q_0 = \rho_0 \otimes \delta_\emptyset \otimes \delta_\emptyset$ is the initial augmented-state distribution (mass $\rho_0(s)$ on $(s, \emptyset, \emptyset)$; it depends on no model parameter), then for any fixed learner observation-action sequence

$$(o_1^A, a_1, \dots, o_h^A, a_h),$$

the vector

$$\tilde{q}_h^\xi = J_h^{\xi, \pi}(o_h^A, a_h) \cdots J_1^{\xi, \pi}(o_1^A, a_1) q_0$$

is the unnormalized augmented-state distribution after realizing that controlled sequence (its normalization is the predictive state q_h^ξ of Section 2). Its total mass is obtained by summing its coordinates over $\bar{s} \in \bar{S}^+$, namely

$$\sum_{\bar{s} \in \bar{S}^+} [J_h^{\xi, \pi}(o_h^A, a_h) \cdots J_1^{\xi, \pi}(o_1^A, a_1) q_0]_{\bar{s}} = \mathbb{P}^{\xi, \pi}(o_{1:h}^A \mid a_{1:h}),$$

which is the controlled conditional OOM identity used in Section 2.4. The learner action probabilities are not included inside $J_h^{\xi, \pi}(o^A, a)$. They enter only when converting this controlled conditional law into the probability of a learner trajectory under policy π :

$$P_\xi^\pi(o_{1:h}^A, a_{1:h}) = \mathbb{P}^{\xi, \pi}(o_{1:h}^A \mid a_{1:h}) \prod_{t=1}^h \pi_t(a_t \mid \tau_{t-1}^A, o_t^A).$$

Thus the OOM operators describe the environment's controlled observation process, while the learner policy contributes the separate action-selection factors in the likelihood.

Non-negativity and ℓ_1 -nonexpansiveness. The nonnegativity properties follow directly from the construction. Each $W_{h,b}^\theta$ has nonnegative entries because its coefficients are products of transition probabilities, emission probabilities, and history-consistency indicators. Thus $v \geq 0$ implies $u_b = W_{h,b}^\theta v \geq 0$ for every b . The operator $G_h^{\Phi, \pi}$ multiplies each successor coordinate whose adversary post-action history is $\tau_h^B = (\eta_h^B, b)$ by the nonnegative probability $g_h^\Phi(b \mid \eta_h^B, \pi)$. For each fixed decision history η_h^B , these weights sum to one over $b \in B$. Consequently, the aggregation preserves nonnegativity and correctly represents the adversary's randomized action. Hence $J_h^{\xi, \pi}$ is nonnegative as well.

The ℓ_1 -nonexpansiveness asserted in the theorem holds in signed form, which is how the operator is used in the channel-triangle inequalities (4) and the transfer arguments of Appendix D. Fix any $u \in \mathcal{V}$ with coordinates $u_{\bar{s}, b}$. By the construction above, the output coordinate of $G_h^{\Phi, \pi} u$ at an augmented state $\bar{s}$ whose adversary post-action history ends with the action $b(\bar{s})$ reads the single input coordinate $u_{\bar{s}, b(\bar{s})}$ and scales it by $g_h^\Phi(b(\bar{s}) \mid \eta_h^B(\bar{s}), \pi) \in [0, 1]$, and distinct output coordinates read distinct input coordinates. Hence

$$\|G_h^{\Phi, \pi} u\|_1 = \sum_{\bar{s}} g_h^\Phi(b(\bar{s}) \mid \eta_h^B(\bar{s}), \pi) |u_{\bar{s}, b(\bar{s})}| \leq \sum_{\bar{s}, b} |u_{\bar{s}, b}| = \|u\|_{1, \mathcal{V}},$$

so $G_h^{\Phi, \pi}$ is ℓ_1 -nonexpansive on all of $\mathcal{V}$, not only on nonnegative inputs. For nonnegative inputs the weights $g_h^\Phi(\cdot \mid \eta_h^B, \pi)$, summing to one over b at each decision history, additionally conserve the total mass routed through that history, which is the probabilistic content of the aggregation.

Dimension and the weak revealing condition. The construction above uses the full-history augmented space $\bar{S}^+$, whose dimension is exponential in H . This is intentional for the exact factorization. Under the κ -step weakly revealing condition, one can often replace this full-history representation by a lower-dimensional predictive-state representation whose dimension depends on κ -window statistics rather than the full history length. In the single-agent POMDP setting, such reductions are standard in OOM analyses of weakly revealing models (Liu et al., 2022a). In the POMG setting considered here, the analogous finite-window representation requires window sufficiency of the retained statistics for both channels; Proposition 2.6 imposes exactly this as a hypothesis (policy and response classes depending on histories only through the κ -window statistics), with Assumption 2.4 supplying detectability of the world channel. In the tabular finite-window case, this gives predictive dimension on the order of

$$O(|S| \cdot |O_A|^{\kappa-1} \cdot |O_B|^{\kappa-1}).$$

This dimension reduction is separate from the exact causal factorization. Naively truncating (τ^A, τ^B) to the last $\kappa - 1$ observations on each side does not in general preserve the factorization unless π and g^Φ also depend only on those truncated histories, or unless the κ -step predictive state is used as a sufficient statistic in place of raw histories. The construction above is always valid on the full-history space. Dimension reduction via the κ -window predictive state is an additional step that relies on Assumption 2.4 and is invoked in the Eluder dimension bound (Proposition 2.6), not in the factorization itself.

Conclusion. We have therefore constructed the desired causal factorization. In particular:

- The world channel

$$W_h^\theta(o^A, a) : \mathbb{R}^{|\bar{S}^+|} \rightarrow \mathcal{V}$$

depends only on the world kernels (T_h, E_h^A, E_h^B) . For each adversary action b , its b -indexed component computes the hypothetical unnormalized successor mass obtained if the adversary were to choose b , and it updates the adversary history by appending b to form $\tau_h^B = (\tau_{h-1}^B, o_h^B, b)$.

- The adversary aggregation operator

$$G_h^{\Phi, \pi}(o^A, a) : \mathcal{V} \rightarrow \mathbb{R}^{|\bar{S}^+|}$$

depends only on the response model g_h^Φ and the learner policy π . For an output history $\tau_h^B = (\eta_h^B, b)$, it reads the decision history $\eta_h^B = (\tau_{h-1}^B, o_h^B)$ from the augmented-state index and applies the response weight $g_h^\Phi(b \mid \eta_h^B, \pi)$.

Thus $G_h^{\Phi, \pi}$ is genuinely independent of the world parameter θ .

- The composition satisfies

$$J_h^{\xi, \pi}(o^A, a) = G_h^{\Phi, \pi}(o^A, a) \circ W_h^\theta(o^A, a),$$

as verified by direct computation. This composition is exactly the controlled conditional forward dynamics of the POMG, with learner action a fixed and the adversary observation and action incorporated according to E_h^B and g_h^Φ .

Both operators preserve non-negativity. The intermediate space $\mathcal{V} = \mathbb{R}^{|\bar{S}^+| \times B}$ makes the types compatible by storing one hypothetical successor family for each adversary action before the adversary response weights are applied. Tracking both players' full observation-action histories ensures that the adversary decision history η_h^B is available to $G_h^{\Phi, \pi}$ without referring to θ , and conditioning g_h^Φ on the post-observation, pre-action history η_h^B respects the causal protocol of the POMG. $\square$

C. Eluder Dimension of ℓ_1 Operator-Error Classes

The complexity bounds of Proposition 2.6 rest on the following general estimate. For classes of ℓ_1 -norms of linear images of a low-dimensional operator family, the scale-sensitive Eluder dimension is controlled by the *span dimension* of the family, up to a factor logarithmic in the scale.

Lemma C.1 (Sign-witness Eluder bound). (a) Fixed design. Let $\mathcal{A} \subset \mathbb{R}^{D_{\text{out}} \times D_{\text{in}}}$ be a family of matrices contained in a linear subspace of dimension p , with $\|A\|_F \leq B_A$ for all $A \in \mathcal{A}$, and let each query x carry a vector v_x with $\|v_x\|_2 \leq B_v$. Consider the function class $\mathcal{F} := \{x \mapsto \|Av_x\|_1 : A \in \mathcal{A}\}$. Then for every $\varepsilon > 0$,

$$\dim_E(\mathcal{F}, \varepsilon) \leq 12p \log \left(1 + \frac{8 D_{\text{out}} B_v^2 B_A^2}{\varepsilon^2} \right).$$

(b) Query-dependent design. Let instead each query x carry a matrix $U(x) \in \mathbb{R}^{D_{\text{out}} \times p}$ with $\|U(x)\|_F \leq B_U$, and let $\mathcal{C} \subset \mathbb{R}^p$ with $\|c\|_2 \leq B_c$ for all $c \in \mathcal{C}$. For $\mathcal{F} := \{x \mapsto \|U(x)c\|_1 : c \in \mathcal{C}\}$ and every $\varepsilon > 0$,

$$\dim_E(\mathcal{F}, \varepsilon) \leq 12p \log\left(1 + \frac{8D_{\text{out}}B_U^2B_c^2}{\varepsilon^2}\right).$$

Proof. By the scale convention of Section 2 it suffices to bound, for each $\varepsilon' \geq \varepsilon$, the length of any sequence whose elements are ε' -independent of their predecessors at the exact scale ε' . The bound we prove is decreasing in the scale, so the bound at scale ε' together with $\varepsilon' \geq \varepsilon$ gives the lemma. To lighten notation we write ε for ε' .

We first prove part (a). Let $x_1, \dots, x_N$ be a sequence in which each x_i is ε -independent of its predecessors, witnessed by $A_i \in \mathcal{A}$; that is,

$$\|A_i v_{x_i}\|_1 > \varepsilon \quad \text{and} \quad \sum_{k < i} \|A_i v_{x_k}\|_1^2 \leq \varepsilon^2. \quad (8)$$

We need to bound N . We begin by defining the *sign-witness matrices*

$$X_k := \sigma_k v_{x_k}^\top, \quad \sigma_k := \text{sign}(A_k v_{x_k}) \in \{\pm 1\}^{D_{\text{out}}}, \quad (9)$$

where the sign vector σ_k is fixed at step k and never changes afterwards. These matrices turn the ℓ_1 quantities of (8) into inner products. At its own query, each witness aligns with its sign-witness matrix,

$$\langle A_i, X_i \rangle = \sigma_i^\top A_i v_{x_i} = \|A_i v_{x_i}\|_1 > \varepsilon. \quad (10)$$

At earlier queries the inner product is dominated by the corresponding ℓ_1 norm, because $|\sigma_k^\top A_i v_{x_k}| \leq \|A_i v_{x_k}\|_1$. Combining this with the second inequality in (8) gives

$$\sum_{k < i} \langle A_i, X_k \rangle^2 \leq \varepsilon^2. \quad (11)$$

The next step is an elliptic-potential argument. Recall that $\mathcal{A}$ lies in a linear subspace of matrices of dimension p . Fix an orthonormal basis of this subspace. Let $\bar{A}_i \in \mathbb{R}^p$ be the coordinate vector of A_i , and let $\bar{X}_k \in \mathbb{R}^p$ be the coordinate vector of the orthogonal projection of X_k onto the subspace. Since A_i lies in the subspace, $\langle A_i, X_k \rangle = \langle \bar{A}_i, \bar{X}_k \rangle$, with the Frobenius inner product on the left and the Euclidean one on the right; the quantities in (10) and (11) are therefore unchanged, and $\|\bar{A}_i\|_2 = \|A_i\|_F$. Define the potential matrices

$$V_i := \lambda I_p + \sum_{k \leq i} \bar{X}_k \bar{X}_k^\top, \quad \lambda := \varepsilon^2 / B_A^2. \quad (12)$$

By Cauchy–Schwarz in the V_{i-1} -geometry, followed by (11) and $\lambda \|A_i\|_F^2 \leq \varepsilon^2$,

$$\varepsilon^2 < \langle A_i, X_i \rangle^2 \leq \left(\sum_{k < i} \langle A_i, X_k \rangle^2 + \lambda \|A_i\|_F^2 \right) \cdot \|\bar{X}_i\|_{V_{i-1}^{-1}}^2 \leq 2\varepsilon^2 \|\bar{X}_i\|_{V_{i-1}^{-1}}^2,$$

where the strict inequality is (10). Hence $\|\bar{X}_i\|_{V_{i-1}^{-1}}^2 > 1/2$ for every i . The elliptic-potential lemma (Lattimore & Szepesvári, 2020, Lemma 19.4) bounds the sum of these quantities,

$$\sum_{i \leq N} \min(1, \|\bar{X}_i\|_{V_{i-1}^{-1}}^2) \leq 2p \log\left(1 + \frac{N \max_k \|X_k\|_F^2}{\lambda p}\right). \quad (13)$$

The left-hand side of (13) is at least $N/2$, and $\|X_k\|_F^2 = \|\sigma_k\|_2^2 \|v_{x_k}\|_2^2 \leq D_{\text{out}} B_v^2$. Substituting both, together with the value of λ , yields

$$N \leq 4p \log\left(1 + \frac{N D_{\text{out}} B_v^2 B_A^2}{p \varepsilon^2}\right). \quad (14)$$

Inequality (14) is self-referential in N , and it remains to solve it. Write $u := N/p$ and $C_0 := D_{\text{out}} B_v^2 B_A^2 / \varepsilon^2$, so that (14) reads $u \leq 4 \log(1 + u C_0)$. If any witness exists then $\varepsilon < \|A_1 v_{x_1}\|_1 \leq \sqrt{D_{\text{out}}} B_A B_v$, that is, $C_0 > 1$; otherwise $\dim_E = 0$ and the bound is trivial. The elementary inequality $1 + u C_0 \leq (1 + u)(1 + C_0)$ then gives

$$u \leq 4 \log(1 + u) + 4 \log(1 + C_0). \quad (15)$$

The function $u \mapsto u - 4 \log(1 + u)$ is increasing for $u \geq 3$, and at $u = 12 \log(1 + 8C_0) \geq 12 \log 9$ it exceeds $4 \log(1 + C_0)$. By (15) we conclude $u \leq 12 \log(1 + 8C_0)$, which is the stated bound.

Part (b) follows by the same argument with witness *vectors* in place of witness matrices. In place of (9), define $w_k := U(x_k)^\top \sigma_k \in \mathbb{R}^p$ with $\sigma_k := \text{sign}(U(x_k) c_k)$. The alignment (10) becomes $\langle c_i, w_i \rangle = \sigma_i^\top U(x_i) c_i = \|U(x_i) c_i\|_1 > \varepsilon$,

and the domination (11) holds because $|\langle c_i, w_k \rangle| = |\sigma_k^\top U(x_k) c_i| \leq \|U(x_k) c_i\|_1$. The potential (12) runs directly in $\mathbb{R}^p$, with $V_i := \lambda I_p + \sum_{k \leq i} w_k w_k^\top$ and $\lambda := \varepsilon^2/B_c^2$, and no projection is needed. The norm bound becomes $\|w_k\|_2^2 \leq \|\sigma_k\|_2^2 \|U(x_k)\|_F^2 \leq D_{\text{out}} B_U^2$, using $\|U^\top \sigma\|_2 \leq \|U\|_{\text{op}} \|\sigma\|_2 \leq \|U\|_F \|\sigma\|_2$. With these substitutions, the argument from (12) to (15) applies verbatim and gives the stated bound. $\square$

We now use the lemma to prove the proposition.

Proof of Proposition 2.6. Fix a reference parameter $\xi^* = (\theta^*, \Phi^*)$. By definition $d_E = d_{\text{agg}}^W + d_{\text{agg}}^G$, so it suffices to bound the Eluder dimension of each channel class at the evaluation scale $\alpha = 1/(2T)$. The proof goes as follows. We first show that every function in the world-channel class is an expectation of ℓ_1 norms of linear images of one operator, and we count the dimension of the subspace in which these operators live. We then verify that the expectation-form queries provide sign-witness matrices satisfying the alignment (10), the domination (11), and a Frobenius norm bound; these are the only properties of the witness matrices that the proof of Lemma C.1(a) uses, so the lemma's conclusion applies and yields the world-channel bound. The adversary channel is handled in the same way through Lemma C.1(b). Finally we add the two channel bounds and specialize the count to the three model families. Throughout, the bounds we obtain do not depend on ξ^* , so the suprema in d_{agg}^W and d_{agg}^G inherit them; and every norm bound entering Lemma C.1 is polynomial in the stated parameter counts and in T , enters only through the logarithm, and is absorbed into $\tilde{O}(\cdot)$.

We begin with the world channel. A function of $\mathcal{F}_{\text{agg}}^{W, \xi^*}$ is indexed by $\theta \in \Theta$ and evaluates at a query policy π to

$$F_\theta^W(\pi) = \sum_{h=1}^H \mathbb{E}^\pi [\|\Delta W_h(o_h^A, a_h) q_{h-1}^{\xi^*}\|_1], \quad \Delta W_h(o^A, a) := W_h^\theta(o^A, a) - W_h^{\theta^*}(o^A, a),$$

where the operators are evaluated at the realized observation–action pair. Splitting the expectation over that pair gives

$$F_\theta^W(\pi) = \sum_{h=1}^H \sum_{o^A, a} \mathbb{E}^\pi [\mathbf{1}\{(o_h^A, a_h) = (o^A, a)\} \|\Delta W_h(o^A, a) q_{h-1}^{\xi^*}\|_1]. \quad (16)$$

Collect the differences into the block-diagonal operator $\bar{A}_\theta := \oplus_{h, o^A, a} \Delta W_h(o^A, a)$. Every summand in (16) is an ℓ_1 norm of a linear image of $\bar{A}_\theta$, so the class has the form treated by Lemma C.1(a), except that the queries produce expectations rather than single vectors v_x . We return to this point after counting dimensions.

The family $\{\bar{A}_\theta : \theta \in \Theta\}$ lies in the span of *pattern matrices*, the coordinate matrices supported on the positions that the window structure allows to be nonzero, and we now count those positions. On the reduced space $\bar{S}_{\text{red}}^+$, an entry of $W_h^\theta(o^A, a)$ is indexed by a source state $\bar{s}$, a hypothetical adversary action b , and a destination state $\bar{s}'$. There are $d_W = |S| |O_A|^{\kappa-1} |O_B|^{\kappa-1}$ source states. The window components of $\bar{s}'$ are determined by those of $\bar{s}$ together with the new observations, so the free destination indices are only the new latent state s' and the new adversary-side observation o_{new}^B . Ranging also over the block indices (o^A, a) , the possibly-nonzero positions at a fixed step h number at most

$$\underbrace{|S| |O_A|^{\kappa-1} |O_B|^{\kappa-1}}_{\bar{s}} \cdot \underbrace{|A|}_{a} \cdot \underbrace{|B|}_{b} \cdot \underbrace{|O_A|}_{o^A} \cdot \underbrace{|O_B|}_{o_{\text{new}}^B} \cdot \underbrace{|S|}_{s'} = |S|^2 |A| |B| |O_A|^\kappa |O_B|^\kappa =: p_W. \quad (17)$$

Distinct steps occupy distinct blocks of $\bar{A}_\theta$, so the family spans at most $p := H p_W$ dimensions.

So far we have shown that the world-channel functions are expectations of ℓ_1 norms of linear images of operators drawn from an $H p_W$ -dimensional subspace. It remains to supply the sign-witness matrices. For a query policy π and a witness θ , define the block query matrix

$$X_{\pi, \theta} := \oplus_{h, o^A, a} \mathbb{E}^\pi [\mathbf{1}\{(o_h^A, a_h) = (o^A, a)\} \sigma_{h, o^A, a}(\tau) (q_{h-1}^{\xi^*})^\top], \quad \sigma_{h, o^A, a}(\tau) := \text{sign}(\Delta W_h(o^A, a) q_{h-1}^{\xi^*}), \quad (18)$$

with the sign vectors chosen per trajectory for the witness θ and frozen thereafter. The alignment (10) holds for these matrices. Indeed, under the witness's own signs $\sigma^\top \Delta W_h q = \|\Delta W_h q\|_1$, so $\langle \bar{A}_\theta, X_{\pi, \theta} \rangle = F_\theta^W(\pi)$ by (16). The domination (11) holds as well. Indeed, $|\sigma^\top v| \leq \|v\|_1$ pointwise, the expectation preserves this inequality, and therefore $|\langle \bar{A}_{\theta'}, X \rangle| \leq F_{\theta'}^W(\pi')$ for every parameter θ' and every previously frozen query matrix X built from a policy π' . For the norm bound, within each step h the indicator blocks partition the trajectory space; using $\|q_{h-1}^{\xi^*}\|_2 \leq \|q_{h-1}^{\xi^*}\|_1 = 1$ and $\|\sigma\|_2^2 \leq D_{\text{out}}$, where $D_{\text{out}} := |B| d_W$ is the block output dimension, we get $\|X_{\pi, \theta}\|_F^2 \leq H D_{\text{out}}$. The conclusion of Lemma C.1(a) therefore holds with $D_{\text{out}} B_v^2$ replaced by $H D_{\text{out}}$ inside the logarithm and with $p = H p_W$, giving, at $\alpha = 1/(2T)$,

$$\dim_E(\mathcal{F}_{\text{agg}}^{W, \xi^*}, \alpha) \leq 12 H p_W \log\left(1 + \frac{8 H D_{\text{out}} B_A^2}{\alpha^2}\right) = \tilde{O}(H |S|^2 |A| |B| |O_A|^\kappa |O_B|^\kappa), \quad (19)$$

where B_A bounds the Frobenius norms of the probability-entry block operators and is polynomial in the counts.

Next we bound the adversary channel. The argument has the same two parts, a linear representation and a parameter count. A function of $\mathcal{F}_{\text{agg}}^{G, \xi^*}$ evaluates to $F_{\Phi}^G(\pi) = \sum_h \mathbb{E}^{\pi} [\|(G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi}) W_h^{\theta^*} q_{h-1}^{\xi^*}\|_1]$. For fixed π the operator $G_h^{\Phi, \pi}$ is linear in Φ (Theorem 2.5), so each summand equals $\|U_h(x) \text{vec}(\Phi - \Phi^*)\|_1$ for a design matrix $U_h(x)$ computable from the query; its entries are built from the components of $W_h^{\theta^*} q_{h-1}^{\xi^*}$ and the response features, both determined by the query and the reference parameter. In the linear response model, Φ_h carries $(|B| - 1) d_{\text{adv}} \leq |B| d_{\text{adv}}$ free parameters per step, so stacking the steps gives a parameter vector of dimension at most $H|B| d_{\text{adv}}$; a step-shared Φ only lowers this. The expectation device of (18), with the per-step designs stacked into a single $U(x)$ and the signs chosen per trajectory and frozen with the witness, again verifies the alignment, the domination, and the norm bound, and Lemma C.1(b) yields

$$\dim_E(\mathcal{F}_{\text{agg}}^{G, \xi^*}, \alpha) = \tilde{O}(H d_{\text{adv}} |B|). \quad (20)$$

Both channels are now bounded, by (19) and (20), and neither bound involves ξ^* . Taking suprema over ξ^* and adding,

$$d_E = d_{\text{agg}}^W + d_{\text{agg}}^G = \tilde{O}(H(|S|^2|A||B||O_A|^{\kappa}|O_B|^{\kappa} + d_{\text{adv}}|B|)),$$

which is the tabular bullet. For the linear and low-rank families only the span count changes. The coordinate-structure hypothesis states that each entry of W_h^{θ} is an affine function of the parameters, so on each of the $|O_A|^{\kappa}|O_B|^{\kappa}$ window contexts the difference family is an affine image of the parameter space. The per-step count p_W of (17) is therefore at most $d_{\text{lin}} |O_A|^{\kappa}|O_B|^{\kappa}$ in the linear case and $r^2(|A| + |B|)|O_A|^{\kappa}|O_B|^{\kappa}$ in the low-rank case. The rest of the argument is unchanged, and the adversary channel is identical in all three cases. $\square$

D. Proof of Theorem 3.3

We prove that the epoch-based algorithm with statistical termination achieves policy regret $C(H\sqrt{\beta_T d_E T} \log T + mH S_T)$ with high probability, where $\bar{\beta}_T = H\beta_T$ and S_T is the number of deployed policies (Lemma D.7).

Figure 2 illustrates the epoch structure that the proof analyzes. We first give an extended proof sketch in Appendix D.1, and then the subsequent subsections contain the full proof.

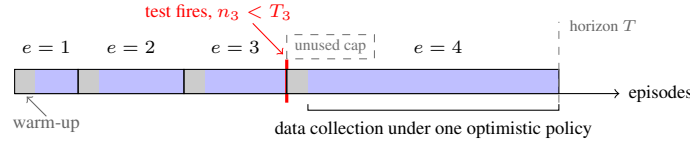


Figure 2. The epoch structure. Each epoch begins with $m - 1$ warm-up episodes (gray), then collects data under a single optimistic policy (blue) for at most $T_e = 2^e$ episodes. The statistical termination test can end an epoch early (red); the unused remainder of the cap (dashed) is forgone, the next epoch begins immediately, and terminated epochs keep their data. The final epoch is truncated by the horizon T .

A remark on policy transport before the proof. Because the adversary’s response changes with the learner’s policy, one might expect a cost for comparing adversary responses across deployed policies. Under Assumption 3.1(e) the likelihood conversion applies directly to the joint operators at each data-collection policy, so no transport term enters the final confidence radius (Remark D.8); posterior-Lipschitz structure instead underpins the well-behavedness of the channel operator classes (Proposition 2.6) and the comparison with flat-batching analyses (Sang & Nguyen-Tang, 2025) in Appendix J.

Remark D.1 (On the choice of θ in S^{ref}). Assumption 2.3 imposes (2) uniformly over $\theta \in \Theta$. This is the cleanest formulation. It makes the adversary class well-defined without knowing the true θ^* , and it is the version needed in the regret analysis, where confidence sets may contain many world models consistent with the data. In parametric models, the resulting Lipschitz constant may depend on the sensitivity of the map $\theta \mapsto S_{\tau_B, h}^{\text{ref}, \theta}(\pi)$, which is controlled when the transition and emission kernels vary continuously in θ and the reference policy μ^{ref} has full support. The assumption is satisfied, for instance, by adversaries who best-respond to a smoothed estimate of the learner’s policy under a fixed world model, or who use gradient-based updates with bounded step sizes.

D.1. Extended Proof Sketch of Theorem 3.3

We outline the main argument, with the technical concentration and Eluder summability lemmas developed in the subsections that follow. The proof follows the same high-level template as optimistic model-based learning, but with one new difficulty. The observable dynamics themselves depend on the learner’s policy, because the adversary’s response depends on that policy.

From policy regret to simulation error. Consider an epoch e , in which the algorithm deploys a single policy π_e for $n_e \leq T_e = 2^e$ data-collection episodes. Before collecting data, it runs $m - 1$ warm-up episodes, so an m -memory adversary sees only π_e during the data-collection phase. Thus, during the n_e data-collection episodes of epoch e , the true model $\xi^* = (\theta^*, \Phi^*)$ induces the stationary adversary response $R_\infty(\pi_e)$, and

$$V^{\pi_e}(\xi^*) = V^{\pi_e}(R_\infty(\pi_e)).$$

The regret in a data-collection episode of epoch e is thus

$$V^{\pi^*}(R_\infty(\pi^*)) - V^{\pi_e}(R_\infty(\pi_e)) = V^{\pi^*}(\xi^*) - V^{\pi_e}(\xi^*).$$

Because $\xi^* \in \mathcal{C}_{e-1}$ on the high-probability confidence event and (π_e, ξ_e) is chosen optimistically,

$$V^{\pi_e}(\xi_e) \geq \max_{\pi} V^{\pi}(\xi^*) \geq V^{\pi^*}(\xi^*).$$

Hence the regret in epoch e is not controlled by how good π_e is in the optimistic model; optimism already handles that. What remains is only the amount by which the optimistic model overestimates the value of the policy it selected:

$$V^{\pi^*}(\xi^*) - V^{\pi_e}(\xi^*) \leq V^{\pi_e}(\xi_e) - V^{\pi_e}(\xi^*). \quad (21)$$

Why observable-operator errors are the right quantity. Inequality (21) reduces the problem to a simulation question, namely how much the value of the same policy π_e can differ between the optimistic model ξ_e and the true model ξ^* . For partially observable models, this difference is naturally measured through the induced observable operators. Define

$$\Delta_e := \sum_{h=1}^H \mathbb{E}_{\tau \sim P^{\pi_e}} \left[\left\| J_h^{\xi_e, \pi_e} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi_e} q_{h-1}^{\xi^*} \right\|_1 \right]. \quad (22)$$

This is the aggregate prediction error of the optimistic model along the trajectory distribution generated by π_e under the true model. The simulation lemma says that an error made at step h can affect rewards over the remaining horizon, giving the horizon prefactor $V^{\pi_e}(\xi_e) - V^{\pi_e}(\xi^*) \leq H \Delta_e$. Thus the regret accumulated in epoch e is bounded by

$$R_e \leq n_e H \Delta_e + (m - 1)H,$$

where the second term is the cost of the warm-up episodes.

How the confidence set controls Δ_e . It remains to bound the cumulative size of the prediction errors Δ_e . The confidence set at the beginning of epoch e is built from data collected under the earlier policies $\pi_1, \dots, \pi_{e-1}$. Since $\xi_e \in \mathcal{C}_{e-1}$, the optimistic model must fit those past data nearly as well as the true model. Consequently (Lemma D.3), the cumulative KL divergence under the historical policies is small:

$$\sum_{k < e} n_k \text{KL}(P_{\xi^*}^{\pi_k} \| P_{\xi_e}^{\pi_k}) \lesssim \beta_T. \quad (23)$$

The KL-to-operator conversion (Assumption 3.1(e)) turns the likelihood statement (23) into a prediction-error statement, and weak revealing (Assumption 2.4) is what makes such a conversion available. If two models induce different observable operators, then a κ -step window of learner observations detects that difference with strength α_κ . Thus likelihood fit under past policies implies small observable-operator error under those past policies.

There is one additional complication that is absent in single-agent POMDPs, and it is where the epoch design earns its keep. Past consistency constrains ξ_e only at the *historical* policies; it says nothing about the error of ξ_e at the freshly chosen π_e . Optimism deliberately steers toward such unconstrained directions, and under a fixed epoch schedule a single ill-informed commitment would run for $\Theta(T)$ episodes; Proposition D.5 shows that the fixed schedule, under a valid tie-breaking of the planning oracle, provably incurs $\Omega(T)$ policy regret. The statistical termination test closes exactly this gap. If the deployed model's prediction error at π_e is large, the epoch's own data refute it at rate $\text{KL}_e := \text{KL}(P_{\xi^*}^{\pi_e} \| P_{\xi_e}^{\pi_e}) \geq c_{\text{KL}} s_e$ per episode, where s_e is the stepwise squared prediction error (written $\mathcal{E}_e^2$ in Definition D.10 below), and the test fires; consequently every epoch, terminated or complete, satisfies the cap

$$n_e \Delta_e^2 \leq H n_e s_e \leq \tilde{O}(H \beta_T / c_{\text{KL}})$$

with high probability (Lemma D.6). The combination of past consistency and the per-epoch cap is precisely what the capped summability lemma (Lemma D.11) consumes; the resulting effective statistical radius is $\bar{\beta}_T := H \beta_T$.

Eluder summability and the final regret bound. Each optimistic model ξ_e is consistent with all past prediction queries up to radius $\bar{\beta}_T$, and every epoch's own contribution is capped at $\tilde{O}(H \beta_T / c_{\text{KL}})$ by the termination test. The aggregate Eluder dimension then limits how often the learner can encounter a large new prediction error. Lemma D.12 gives

$$\sum_{e=1}^E n_e \Delta_e^2 \leq C_1 d_E \bar{\beta}_T \log^2 T. \quad (24)$$

This is the central statistical estimate. It says that although an individual epoch may have nontrivial prediction error, the weighted square-sum of these errors over all epochs is controlled by the aggregate Eluder dimension.

We now return to the regret decomposition. Using Cauchy–Schwarz, $\sum_{e=1}^E n_e \leq O(T)$, and (24), we get

$$\sum_{e=1}^E n_e \Delta_e \leq O\left(\sqrt{C_1 d_E T \bar{\beta}_T} \log T\right).$$

Finally, summing per-epoch regret $R_e \leq n_e H \Delta_e + (m-1)H$ over the S_T epochs gives $PR(T) \leq H \sum_e (n_e \Delta_e + (m-1)) \leq C(H\sqrt{\bar{\beta}_T d_E T} \log T + mHS_T)$, and Lemma D.7 bounds S_T . The design leans on both halves of the termination test. It must fire when the data refute the deployed model, else a fixed schedule suffers $\Omega(T)$ (Proposition D.5), and it must fire rarely, each firing requiring $\Omega(\beta_e)$ bits of evidence, which is what keeps the warm-up cost dominated (Lemma D.7).

D.2. The Epoch-Based Algorithm Structure

Algorithm 1 divides the T episodes into epochs of geometrically increasing cap $T_e = 2^e$ for $e = 1, 2, \dots$; because the termination test can end an epoch early, the number of epochs started by time T is the (data-dependent) deployed-policy count S_T , not $\lceil \log_2 T \rceil$. At the beginning of epoch e , the algorithm:

- Computes the MLE $\hat{\xi}_{e-1}$ from the cumulative dataset $\mathcal{D}_{e-1}$ containing all trajectories collected in epochs $1, \dots, e-1$
- Forms the confidence set $\mathcal{C}_{e-1} = \{\xi : L_{e-1}(\xi) \geq L_{e-1}(\hat{\xi}_{e-1}) - \beta_e\}$ where $\beta_e = c(\log \mathcal{N}(\epsilon_e; \Xi) + \log(1/\delta_e) + mB_0)$, $\delta_e = \delta/(2e^2)$, $\epsilon_e = 2^{-\lceil \log_2(|\mathcal{D}_{e-1}|+2) \rceil}$
- Selects one policy-parameter pair (π_e, ξ_e) that maximizes value within $\mathcal{C}_{e-1}$
- Executes π_e for $m-1$ warm-up episodes (discarding data), then for at most T_e data-collection episodes, recomputing the MLE after each episode and *terminating the epoch* as soon as $L_t(\xi_e) < L_t(\hat{\xi}_t) - \tau_e(t)$, where $\tau_e(t) = 2\beta_e(t) + C_0 B_0 \log(2t(t+1)/\delta_e)$ and $\beta_e(t)$ is the confidence radius recomputed at the current dataset size (Appendix D.8); the realized number of data-collection episodes is $n_e \leq T_e$
- Updates the cumulative dataset: $\mathcal{D}_e = \mathcal{D}_{e-1} \cup \{\text{new trajectories from epoch } e\}$ (terminated epochs' data included)

The total number of data-collection episodes satisfies $\sum_e n_e \leq T$ by definition of the horizon (the final epoch is truncated at time T). Scheduled completions, meaning epochs that exhaust their cap $T_e = 2^e$, number at most $\log_2(2T)$, since a completion at index e requires $2^e \leq T$. The algorithm deploys S_T policies in total (one per epoch), where S_T counts scheduled completions, early terminations, and the final truncated epoch, and is bounded in Lemma D.7. We define the horizon-level radius

$$\beta_T := c(\log \mathcal{N}(\epsilon_T; \Xi) + 5 \log(2T) + \log(1/\delta) + mB_0), \quad \epsilon_T := 1/(2T+4),$$

where the scale ϵ_T lower-bounds every dyadic scale the algorithm can use (all realized scales are at least $2^{-\lceil \log_2(T+2) \rceil} \geq 1/(2T+4)$, since dataset sizes never exceed T). Every realized radius is dominated by it. Since $|\mathcal{D}_{e-1}| \leq T$ and $e \leq S_T \leq T+1 \leq 2T$ (every started epoch except possibly the final horizon-truncated one collects at least one data-collection episode), we have $\log(1/\delta_e) = \log(2e^2/\delta) \leq 3 \log(2T) + \log(1/\delta)$, so $\beta_e \leq \beta_T$; likewise, for the running radii of Appendix D.8, $\log(2t(t+1)/\delta_e) \leq \log(32T^4/\delta) \leq 5 \log(2T) + \log(1/\delta)$ for $t \leq T$, so $\beta_e(t) \leq \beta_T$ for every reachable pair (e, t) . By construction $\log T \leq \beta_T/c$, and $\beta_T = O(\log \mathcal{N}(\epsilon_T; \Xi) + \log T + \log(1/\delta) + mB_0)$.

D.3. Auxiliary Lemmas

We state the simulation lemma and likelihood concentration lemma before the main proof.

Lemma D.2 (Simulation Lemma). *For any policy π and model $\xi \in \Xi$,*

$$|V^\pi(\xi) - V^\pi(\xi^*)| \leq H \sum_{h=1}^H \mathbb{E}_{\tau_{h-1} \sim P_{\xi^*}^\pi} \left[\left\| J_h^{\xi, \pi} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi} q_{h-1}^{\xi^*} \right\|_1 \right].$$

Proof. The proof is the standard telescoping argument for controlled observable operators. Fix π , and let $q_{h-1}^{\xi^*}$ be the predictive state under the true model after the learner observation-action history up to step $h-1$. Replacing the true step- h operator $J_h^{\xi^*, \pi}$ by $J_h^{\xi, \pi}$, while keeping the same reference predictive state, changes the one-step predictive distribution by

$$\left\| J_h^{\xi, \pi} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi} q_{h-1}^{\xi^*} \right\|_1.$$

By telescoping over $h = 1, \dots, H$, the ℓ_1 distance between the trajectory laws induced by ξ and ξ^* under the fixed policy π , which equals twice their total variation distance, is bounded by the sum of these one-step prediction errors, averaged over

the true-model trajectory distribution. Since rewards are in $[0, 1]$, an error in the predictive distribution at any step can affect at most H units of remaining cumulative reward. Therefore,

$$|V^\pi(\xi) - V^\pi(\xi^*)| \leq H \sum_{h=1}^H \mathbb{E}_{\tau_{h-1} \sim P_{\xi^*}^\pi} \left[\left\| J_h^{\xi, \pi} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi} q_{h-1}^{\xi^*} \right\|_1 \right].$$

For POMGs, the controlled observable operators are well-defined by the causal decomposition in Theorem 2.5; the telescoping argument is otherwise the same as in the POMDP/OOM setting (Liu et al., 2023). $\square$

Lemma D.3 (Likelihood Concentration). *Let*

$$K_{e-1}(\xi) := \sum_{k < e} n_k \text{KL}(P_{\xi^*}^{\pi_k} \| P_\xi^{\pi_k}).$$

Under Assumption 3.1(a)–(c) and (f), with probability at least $1 - \delta_e/2$, uniformly over $\xi \in \Xi$, the following two bounds hold simultaneously:

- (i) (validity; absolute constants) $L_{e-1}(\xi) - L_{e-1}(\xi^*) \leq \beta_e/2$;
- (ii) (evidence accumulation) $L_{e-1}(\xi) - L_{e-1}(\xi^*) \leq -\frac{1}{2}K_{e-1}(\xi) + C_B \beta_e/2$ for a constant $C_B \geq 1$ depending only on B_0 .

Consequently, if $\xi \in \mathcal{C}_{e-1}$, then $K_{e-1}(\xi) \leq C\beta_e$ for a constant $C > 0$ depending only on B_0 .

Proof. Fix an epoch e and a model $\xi \in \Xi$. For a trajectory τ collected under policy π_k , define

$$Z_k(\xi; \tau) := \log \frac{P_\xi^{\pi_k}(\tau)}{P_{\xi^*}^{\pi_k}(\tau)} + \text{KL}(P_{\xi^*}^{\pi_k} \| P_\xi^{\pi_k}). \quad (25)$$

Conditioned on the filtration before this trajectory is sampled, the policy π_k is fixed and the trajectory is distributed according to $P_{\xi^*}^{\pi_k}$. Hence

$$\mathbb{E}_{\xi^*}[Z_k(\xi; \tau) \mid \mathcal{F}_{k-1}] = 0, \quad (26)$$

so the sequence of Z_k 's over all trajectories collected before epoch e is a martingale difference sequence.

By Assumption 3.1(c), log-likelihood ratios are uniformly bounded:

$$\left| \log \frac{P_\xi^{\pi_k}(\tau)}{P_{\xi^*}^{\pi_k}(\tau)} \right| \leq B_0. \quad (27)$$

Moreover, the KL term is also bounded by B_0 , so

$$|Z_k(\xi; \tau)| \leq 2B_0. \quad (28)$$

For $K_{e-1}(\xi) := \sum_{k < e} n_k \text{KL}(P_{\xi^*}^{\pi_k} \| P_\xi^{\pi_k})$, the conditional variance satisfies

$$\sum_{k < e} \sum_{i=1}^{n_k} \mathbb{E}_{\xi^*}[Z_{k,i}(\xi)^2 \mid \mathcal{F}_{k,i-1}] \leq C_{\text{var}} K_{e-1}(\xi), \quad (29)$$

for a constant $C_{\text{var}} = O(B_0)$ depending only on the likelihood-ratio bound (recall $B_0 \geq 1$). This is the standard Bernstein/Freedman variance control for bounded likelihood ratios.

Freedman's inequality in its fixed-parameter exponential-supermartingale form (Freedman, 1975), which applies to the adapted policy sequence and the stopping-time epoch lengths, together with (28) and (29), gives that for any $u > 0$, with probability at least $1 - e^{-u}$,

$$\sum_{k < e} \sum_{i=1}^{n_k} Z_{k,i}(\xi) \leq \frac{1}{2} K_{e-1}(\xi) + Cu, \quad (30)$$

where $C = O(B_0)$ depends only on B_0 and C_{var} . By the definition of Z_k in (25), this is equivalent to

$$L_{e-1}(\xi) - L_{e-1}(\xi^*) + K_{e-1}(\xi) \leq \frac{1}{2} K_{e-1}(\xi) + Cu, \quad (31)$$

or equivalently,

$$L_{e-1}(\xi) - L_{e-1}(\xi^*) \leq -\frac{1}{2} K_{e-1}(\xi) + Cu. \quad (32)$$

To make the bound uniform over Ξ , we must account for the fact that the scale $\epsilon_e = 2^{-\ell_e}$, $\ell_e := \lceil \log_2(|\mathcal{D}_{e-1}| + 2) \rceil$, is data-dependent, so the realized net is not known in advance. The dyadic grid makes the candidate nets a deterministic family. Since $|\mathcal{D}_{e-1}| \leq \sum_{k < e} 2^k \leq 2^e$, the realized level satisfies $\ell_e \leq e + 1$. For each candidate level $\ell \in \{1, \dots, e + 1\}$, let $\mathcal{N}^{(\ell)}$ be a fixed $2^{-\ell}$ -net of Ξ in the operator norm from Assumption 3.1, with

$$|\mathcal{N}^{(\ell)}| \leq \mathcal{N}(2^{-\ell}; \Xi, \|\cdot\|_{\text{op}}). \quad (33)$$

Apply (32) to every $\xi \in \mathcal{N}^{(\ell)}$ and every level ℓ with

$$u_\ell = \log |\mathcal{N}^{(\ell)}| + \log(4(\ell+1)(\ell+2)/\delta_e). \quad (34)$$

A union bound over the net at each level, then over levels ($\sum_{\ell \geq 1} \frac{1}{(\ell+1)(\ell+2)} = \frac{1}{2}$), gives (32) simultaneously at every level with probability at least $1 - \delta_e/8$; on this event the bound holds in particular at the realized level ℓ_e . The level overhead is affordable: $2 \log(4(\ell_e+1)(\ell_e+2)) \leq 2 \log(1/\delta_e) + 7$ for every $\ell_e \leq e + 1$, since $(\ell_e+1)(\ell_e+2) \leq (e+2)(e+3) \leq 12e^2$ for $e \geq 1$ and $\log(1/\delta_e) \geq \log(2e^2)$. Hence $Cu_{\ell_e} \leq C_B \beta_e/2$ for a constant $C_B \geq 1$ depending only on B_0 , since $\beta_e \geq c(\log |\mathcal{N}^{(\ell_e)}| + \log(1/\delta_e) + mB_0)$ with $mB_0 \geq 1$. This yields part (ii) at the realized net.

Part (i) does not require Freedman and carries absolute constants. For a fixed ξ , conditionally on the filtration each factor $\mathbb{E}_{\xi^*}[\exp(\frac{1}{2} \log(P_\xi/P_{\xi^*})(\tau)) \mid \mathcal{F}]$ equals the Hellinger affinity of the two trajectory laws, which is at most 1; hence $\exp(\frac{1}{2}(L_{e-1}(\xi) - L_{e-1}(\xi^*)))$ is a supermartingale with initial value 1, and Markov's inequality gives $\mathbb{P}[L_{e-1}(\xi) - L_{e-1}(\xi^*) \geq 2u] \leq e^{-u}$. A union bound over $\mathcal{N}^{(\ell)}$ and the levels $\ell \leq e + 1$, with u_ℓ as in (34), gives part (i) at the realized net with probability at least $1 - \delta_e/8$ and deviation $2u_{\ell_e} \leq \beta_e/2$, using the level-overhead estimate above together with $c \geq 16$ and the floor $mB_0 \geq 1$.

The Lipschitz regularity in Assumption 3.1(b) extends both inequalities from the net to all $\xi \in \Xi$. Replacing a model by its nearest net point changes each of the $|\mathcal{D}_{e-1}|$ per-trajectory log-likelihoods by at most $\epsilon_e L_{\text{lik}}$, where L_{lik} denotes the Lipschitz constant of Assumption 3.1(b). Since the realized scale satisfies $\epsilon_e \leq 1/(|\mathcal{D}_{e-1}| + 2)$, the total discretization error in L_{e-1} and K_{e-1} is at most L_{lik} , which the normalization in Assumption 3.1(c) bounds by B_0 , hence by the mB_0 floor; it is absorbed by the same $c \geq 16$ slack, keeping c an absolute constant. Uniformly over $\xi \in \Xi$, with probability at least $1 - \delta_e/2$, both bounds hold:

$$L_{e-1}(\xi) - L_{e-1}(\xi^*) \leq \min\left(\frac{\beta_e}{2}, -\frac{1}{2}K_{e-1}(\xi) + C_B \frac{\beta_e}{2}\right). \quad (35)$$

It remains to verify the stated consequence. If $\xi \in \mathcal{C}_{e-1}$, then by definition of $\mathcal{C}_{e-1}$ and the optimality of $\hat{\xi}_{e-1}$,

$$L_{e-1}(\xi) \geq L_{e-1}(\hat{\xi}_{e-1}) - \beta_e \geq L_{e-1}(\xi^*) - \beta_e.$$

Combining this with part (ii) of (35) gives

$$-\beta_e \leq L_{e-1}(\xi) - L_{e-1}(\xi^*) \leq -\frac{1}{2}K_{e-1}(\xi) + C_B \frac{\beta_e}{2}.$$

Therefore $K_{e-1}(\xi) \leq (2 + C_B)\beta_e$, and the claimed bound $K_{e-1}(\xi) \leq C\beta_e$ holds with $C := 2 + C_B$, a constant depending only on B_0 . $\square$

D.4. Cumulative Confidence Sets Contain True Parameters

Lemma D.4 (Confidence Set Validity). *With probability at least $1 - \delta$, we have $\xi^* \in \mathcal{C}_{e-1}$ for all epochs e simultaneously.*

Proof. Consider epoch e . The confidence set uses threshold β_e :

$$\mathcal{C}_{e-1} = \left\{ \xi \in \Xi : L_{e-1}(\xi) \geq L_{e-1}(\hat{\xi}_{e-1}) - \beta_e \right\}.$$

On the event of Lemma D.3, part (i) holds uniformly over $\xi \in \Xi$, and hence also for the data-dependent MLE $\hat{\xi}_{e-1}$:

$$L_{e-1}(\hat{\xi}_{e-1}) - L_{e-1}(\xi^*) \leq \frac{\beta_e}{2}.$$

Thus $L_{e-1}(\xi^*) \geq L_{e-1}(\hat{\xi}_{e-1}) - \beta_e$, so $\xi^* \in \mathcal{C}_{e-1}$ with probability at least $1 - \delta_e/2$. Equivalently,

$$\mathbb{P}_{\xi^*} \left[L_{e-1}(\hat{\xi}_{e-1}) - L_{e-1}(\xi^*) > \beta_e \right] \leq \frac{\delta_e}{2}.$$

Setting $\delta_e = \delta/(2e^2)$ and applying a union bound over all epochs gives

$$\mathbb{P}[\xi^* \in \mathcal{C}_{e-1} \text{ for all epochs } e] \geq 1 - \sum_{e=1}^{\infty} \frac{\delta_e}{2} = 1 - \frac{\delta}{2} \sum_{e=1}^{\infty} \frac{1}{2e^2} \geq 1 - \delta. \quad \square$$

For the remainder of this proof, we condition on the high-probability event that $\xi^* \in \mathcal{C}_{e-1}$ for all epochs e .

D.5. Per-Epoch Regret Decomposition

Fix an epoch e where the algorithm executes policy π_e for $n_e \leq T_e = 2^e$ data-collection episodes (after warm-up). Let $\xi_e = (\theta_e, \Phi_e)$ be the optimistically selected model. For each episode t in this epoch (after the warm-up phase), the instantaneous regret can be decomposed as:

$$\begin{aligned} V^{\pi^*}(R_\infty(\pi^*)) - V^{\pi_e}(R_\infty(\pi_e)) \\ = \underbrace{[V^{\pi^*}(\xi^*) - V^{\pi_e}(\xi_e)]}_{\text{optimism gap}} + \underbrace{[V^{\pi_e}(\xi_e) - V^{\pi_e}(\xi^*)]}_{\text{simulation error}}, \end{aligned} \quad (36)$$

where we use the key equality $V^{\pi^*}(\xi^*) = V^{\pi^*}(R_\infty(\pi^*))$ (the model ξ^* captures the adversary's stationary response to any fixed policy). This equality holds because, by Assumption 3.1(f), after the $m - 1$ warm-up episodes, every data-collection episode in epoch e is generated under the stationary response $R_\infty(\pi_e)$. The value $V^{\pi_e}(\xi^*)$ is defined as the expected cumulative reward when the learner plays π_e and the adversary responds according to Φ^* ; after warm-up, this equals $V^{\pi_e}(R_\infty(\pi_e))$ since the adversary is in its stationary regime for all n_e data episodes.

Optimism gap: By the optimistic selection procedure, since $\xi^* \in \mathcal{C}_{e-1}$ (by Lemma D.4) and we maximize over all policies within the confidence set:

$$V^{\pi_e}(\xi_e) \geq \max_{\pi \in \Pi} V^{\pi}(\xi^*) \geq V^{\pi^*}(\xi^*) = V^{\pi^*}(R_\infty(\pi^*)).$$

Therefore, the optimism gap is non-positive and contributes zero regret.

Warm-up cost: The $m - 1$ warm-up episodes allow the adversary to stabilize to $R_\infty(\pi_e)$. Each warm-up episode incurs at most H regret, contributing $(m - 1)H$ per epoch.

Simulation error: This measures how much the optimistic model ξ_e overestimates the value of π_e compared to the true model ξ^* . This is the key term we must bound carefully.

Define the *operator error* for epoch e as

$$\Delta_e := \sum_{h=1}^H \mathbb{E}_{\tau_{h-1} \sim P_{\xi^*}^{\pi_e}} \left[\|J_h^{\xi_e, \pi_e} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi_e} q_{h-1}^{\xi^*}\|_1 \right]. \quad (37)$$

By the simulation lemma (Lemma D.2):

$$V^{\pi_e}(\xi_e) - V^{\pi_e}(\xi^*) \leq H \Delta_e.$$

The total regret in epoch e is thus:

$$R_e \leq n_e \cdot H \Delta_e + (m - 1)H,$$

where $(m - 1)H$ accounts for warm-up.

D.6. Bounding Operator Errors via Eluder Dimension

It remains to control the cumulative prediction errors Δ_e of the optimistic models selected across epochs. This is the main statistical step of the proof, and it rests on three ingredients. First, the confidence-set construction ensures that each optimistic model ξ_e remains consistent with the data collected under earlier epoch policies (*past consistency*). Second, and this is where the statistical termination test is essential, past consistency alone cannot control the error of ξ_e at the freshly selected policy π_e . Proposition D.5 shows that without the termination test the algorithm, under a valid tie-breaking of its planning oracle, provably suffers linear policy regret. The termination test supplies the missing ingredient, a *per-epoch cap* on each epoch's contribution (Lemma D.6). Third, a capped weighted summability argument (Lemma D.11) combines the two constraints with the aggregate Eluder dimension to give

$$\sum_e n_e \Delta_e^2 \leq \tilde{O}(d_E \bar{\beta}_T), \quad \bar{\beta}_T = H \beta_T. \quad (38)$$

D.7. Why Data-Dependent Termination Is Necessary

The following proposition shows that the statistical termination test cannot be dispensed with. Under the fixed schedule $T_e = 2^e$ alone, optimism can commit to a direction the data have never constrained for an entire epoch, and the final epoch has length $\Theta(T)$.

Proposition D.5 (Failure of the fixed epoch schedule). *For every $T \geq 2$ there is a POMG instance and model class satisfying Assumptions 2.3, 2.4, and 3.1, namely the lower-bound family of Appendix E with a constant adversary ($m = 1$, $L = 0$, $\kappa = 1$, $H = 2$) and $d = \lceil \log_2 T \rceil + 1$ arms, so that $d_E = \Theta(\log T)$, on which Algorithm 1 without the termination test (executing every epoch for its full scheduled length $T_e = 2^e$, with any maximizer selection that prefers unplayed arms under exact ties, breaking remaining ties by largest index; the algorithm specification leaves such ties unresolved, so this is a valid instantiation of the planning oracle) incurs*

$$\mathbb{E}[PR(T)] \geq T/4.$$

Proof. Take the construction of Appendix E with d arms and a constant adversary, true means $\mu_1^* = 3/4$ and $\mu_i^* = 1/2$ for $i \geq 2$, and model class $\Theta = [1/8, 7/8]^d$. All outcome probabilities are bounded away from 0 and 1, giving Assumption 3.1(c); Assumptions 3.1(a)–(f), 2.3 (with $L = 0$), and 2.4 (with $\kappa = 1$) hold with absolute constants exactly as verified in Appendix E (the policy class is again the d routing policies). The adversary channel is trivial, so $d_E = d_{\text{agg}}^W$; repeated queries are dependent at every scale, so $d_E \leq d$, and the d single-arm queries are $1/(2T)$ -independent (each arm’s mean varies over an interval of length $3/4$, so witness values are of constant order, above the evaluation scale $1/(2T)$ for every $T \geq 2$), giving $d_E = \Theta(d) = \Theta(\log T)$. The likelihood factorizes over arms, and an arm that has never been played contributes no likelihood terms; hence for any unplayed arm i , every value $\mu_i \in [1/8, 7/8]$ attains the maximum likelihood, and the optimistic value of the policy “commit to arm i ” is $7/8$. Any arm’s optimistic value is at most $7/8$, and arm 1’s true value is $3/4 < 7/8$. Consequently a maximizer always exists among the unplayed arms, and under the stated tie rule optimistic planning selects one whenever an unplayed arm exists. Without the termination test the algorithm runs at most $\lceil \log_2 T \rceil$ epochs, deploying one arm per epoch; since $d > \lceil \log_2 T \rceil$, an unplayed arm exists at every epoch start, and the largest-index tie rule never reaches arm 1. Every episode is therefore spent on an arm $i \geq 2$ and incurs instantaneous policy regret $\mu_1^* - \mu_i^* = 1/4$, so $\mathbb{E}[PR(T)] \geq T/4$. $\square$

The defect this exposes is not specific to our setting. Holding one function fixed across a block of 2^e identically weighted queries invalidates weighted analogues of the standard Eluder summability bound (the arm class above satisfies the past-consistency hypothesis with any radius, yet $\sum_e T_e f_e(x_e)^2 = \Theta(T)$). It also explains why data-dependent update triggers are ubiquitous in low-switching reinforcement learning (Bai et al., 2019; Qiao et al., 2022). Fixed schedules do not mix with optimistic planning that is updated only infrequently.

D.8. Per-Epoch Cap and Switch Count

Throughout this subsection, condition on the event of Lemma D.3 at every epoch boundary (which contains the event of Lemma D.4 and, through part (i), bounds the MLE-to-truth gap by $\beta_e/2$), strengthened as follows. For an episode boundary t of epoch e , let $N_{e,t} := |\mathcal{D}_{e-1}| + t$ denote the current dataset size and define the running radius

$$\beta_e(t) := c \left(\log \mathcal{N}(2^{-\lceil \log_2(N_{e,t}+2) \rceil}; \Xi) + \log(2t(t+1)/\delta_e) + mB_0 \right),$$

which is nondecreasing in t and satisfies $\beta_e(t) \geq \beta_e$ (the dyadic scale is finer and the per-boundary budget smaller) as well as $\beta_e(t) \leq \beta_T$ for every reachable t , by the definition of β_T in Appendix D.2. Within epoch e , condition on the epoch-start σ -field. The dataset $\mathcal{D}_{e-1}$ is then fixed, so the dyadic covering net at scale $2^{-\lceil \log_2(N_{e,t}+2) \rceil}$ is deterministic for every t , and the epoch’s data-collection episodes are the only remaining randomness. For each ξ in the boundary- t net, the conditional Hellinger-affinity supermartingale over the epoch’s episodes (as in part (i) of Lemma D.3) bounds the epoch-local gap $G_t(\xi) := L_t(\xi) - L_t(\xi^*) - (L_{e-1}(\xi) - L_{e-1}(\xi^*))$ by $2u_t$ with conditional failure probability e^{-u_t} ; a union bound over the net with budget $\delta_e/(2t(t+1))$ at boundary t , integrated over the conditioning, sums to at most $\delta_e/2$ per epoch. Combining with the uniform epoch-boundary bound $L_{e-1}(\xi) - L_{e-1}(\xi^*) \leq \beta_e/2$ of Lemma D.3(i) and the net-extension step, on the resulting event, for every epoch e and every episode t within it, $L_t(\xi_t) - L_t(\xi^*) \leq \beta_e/2 + 2u_t + 2B_0 \leq \beta_e/2 + \beta_e(t)/8$, the last inequality because $2u_t + 2B_0 \leq 2(\log |\mathcal{N}_t| + \log(2t(t+1)/\delta_e) + mB_0) \leq (2/c)\beta_e(t) \leq \beta_e(t)/8$ for $c \geq 16$ (the net size is counted by the covering term of $\beta_e(t)$, and $B_0 \leq mB_0$). The termination test of Algorithm 1 uses the threshold $\tau_e(t) := 2\beta_e(t) + C_0 B_0 \log(2t(t+1)/\delta_e)$, where C_0 is the absolute constant from (39) below; we fix $c \geq 16$ throughout (as in the proof of Lemma D.3). Write $\text{KL}_e := \text{KL}(P_{\xi^*}^{\pi_e} \| P_{\xi_e}^{\pi_e})$ and $s_e := \sum_{h=1}^H \mathbb{E}_{\pi_e, \xi^*} [\|J_h^{\xi_e, \pi_e} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi_e} q_{h-1}^{\xi^*}\|_1^2]$ (this is exactly the stepwise error $\mathcal{E}_e^2$ of Definition D.10), so that $\text{KL}_e \geq c_{\text{KL}} s_e$ by the first clause of Assumption 3.1(e), and $\Delta_e^2 \leq H s_e$ by Cauchy–Schwarz over h .

Lemma D.6 (Per-epoch error cap). *With probability at least $1 - \delta_e$ over epoch e ’s data, the number of data-collection episodes n_e of epoch e , when $n_e \geq 1$, satisfies*

$$n_e \text{KL}_e \leq C(\beta_e(n_e) + B_0 \log(2n_e(n_e+1)/\delta_e)) =: \tilde{\beta}_e = O(B_0 \beta_T),$$

for an absolute constant C , however the epoch ends (by the test, the cap, or the horizon). The last estimate uses $n_e \leq T$, $\beta_e(n_e) \leq \beta_T$, and $\log T \leq \beta_T/c$ (both by the definition of β_T), so that the additive logarithm is absorbed at the cost of a factor depending on B_0 . Consequently $n_e s_e \leq \beta_e/c_{\text{KL}}$ and $n_e \Delta_e^2 \leq H \tilde{\beta}_e/c_{\text{KL}}$.

Proof. The pair (π_e, ξ_e) is fixed at the start of the epoch (measurable with respect to the pre-epoch σ -field), and after warm-up the data-collection episodes are i.i.d. with law $P_{\xi^*}^{\pi_e}$ (Assumption 3.1(f)). Let $Z_s := \log(P_{\xi_e}/P_{\xi^*})(\tau_s) + \text{KL}_e$ over the epoch's data-collection episodes s ; as in Lemma D.3, (Z_s) is a martingale difference sequence with $|Z_s| \leq 2B_0$ and conditional variance at most $O(B_0) \text{KL}_e$ (the variance bound (29), using $B_0 \geq 1$). Freedman's inequality with a union bound over the episodes s of the epoch (budget $\delta_e/(2s(s+1))$ at episode s , summing to $\delta_e/2$) gives, with probability $\geq 1 - \delta_e/2$, simultaneously for all s ,

$$G_s := \sum_{s' \leq s} \log \frac{P_{\xi_e}(\tau_{s'})}{P_{\xi^*}(\tau_{s'})} \leq -\frac{1}{2} s \text{KL}_e + C_0 B_0 \log(2s(s+1)/\delta_e), \quad (39)$$

where C_0 is the absolute constant used in the termination threshold $\tau_e(t)$ (taken large enough to cover both tails). Let $g_{\text{pre}} := L_{\text{pre}}(\xi_e) - L_{\text{pre}}(\xi^*)$ denote the pre-epoch cumulative gap. At selection, $\xi_e \in \mathcal{C}_{e-1}$ and $L_{\text{pre}}(\hat{\xi}) \geq L_{\text{pre}}(\xi^*)$ give $g_{\text{pre}} \geq -\beta_e$; in the other direction, $L_{\text{pre}}(\xi_e) \leq L_{\text{pre}}(\hat{\xi})$ and Lemma D.3(i) at the epoch boundary give $g_{\text{pre}} \leq \beta_e/2 \leq \beta_e(n)/2$ for every $n \geq 1$. Consider the last episode n at which the termination test had not yet fired ($n = n_e$ if the test never fired, whether the epoch exhausted its cap or was truncated by the horizon; $n = n_e - 1 \geq n_e/2$ if the test fired with $n_e \geq 2$; for $n_e = 1$ the claim is trivial since $\text{KL}_e \leq 2B_0 \leq \beta_e$, using $c \geq 16$). Not having fired means $L_n(\xi_e) \geq L_n(\hat{\xi}_n) - \tau_e(n) \geq L_n(\xi^*) - \tau_e(n)$, i.e. $g_{\text{pre}} + G_n \geq -\tau_e(n)$, whence $G_n \geq -\tau_e(n) - \beta_e(n)/2$. Combining with (39): $\frac{1}{2} n \text{KL}_e \leq \tau_e(n) + \frac{1}{2} \beta_e(n) + C_0 B_0 \log(2n(n+1)/\delta_e) \leq \frac{5}{2} \beta_e(n) + 2C_0 B_0 \log(2n(n+1)/\delta_e)$, and $n_e \text{KL}_e \leq 2n \text{KL}_e + 2B_0$ yields the claim after adjusting the absolute constant. The consequences follow from Assumption 3.1(e) and $\Delta_e^2 \leq H s_e$. $\square$

Lemma D.7 (Switch count). *Let S_T denote the number of epochs started by time T , and let N_{term} be the number ended by the termination test. On the same event, augmented by the lower-tail analogue of (39) (same per-episode union budget, remaining $\delta_e/2$),*

$$S_T \leq \log_2(2T) + N_{\text{term}} + 1,$$

and

$$N_{\text{term}} \leq \frac{32}{3c(1+mB_0)} \sum_e n_e \text{KL}_e \leq \frac{32B_0}{3c(1+mB_0)} \sqrt{T \sum_e n_e \Delta_e^2} = \tilde{O}\left(\frac{B_0}{1+mB_0} \sqrt{d_E \tilde{\beta}_T T}\right).$$

Consequently the total warm-up cost satisfies, with no condition relating m , B_0 , and T , $(m-1)H N_{\text{term}} \leq \tilde{O}(H \sqrt{d_E \tilde{\beta}_T T})$ and $(m-1)H S_T \leq \tilde{O}(mH \log T + H \sqrt{d_E \tilde{\beta}_T T})$.

Proof. Scheduled completions number at most $\log_2(2T)$ (Appendix D.2), and at most one epoch is truncated by the horizon. Consider an epoch e ended by the termination test at episode $n = n_e$. At the firing time, $L_n(\xi_e) < L_n(\hat{\xi}_n) - \tau_e(n) \leq L_n(\xi^*) + \beta_e/2 + \beta_e(n)/8 - \tau_e(n) \leq L_n(\xi^*) + \frac{5}{8} \beta_e(n) - \tau_e(n)$ on the strengthened confidence event, using $\beta_e \leq \beta_e(n)$. Subtracting the pre-epoch gap, which satisfies $g_{\text{pre}} \geq -\beta_e \geq -\beta_e(n)$, the epoch-local gap obeys $G_n < -\tau_e(n) + \frac{13}{8} \beta_e(n)$. The lower-tail Freedman bound for the same martingale (same union budget) gives $G_n \geq -2n \text{KL}_e - C_0 B_0 \log(2n(n+1)/\delta_e)$ for all n simultaneously with probability $\geq 1 - \delta_e/2$; therefore

$$2n_e \text{KL}_e \geq \tau_e(n) - \frac{13}{8} \beta_e(n) - C_0 B_0 \log(2n(n+1)/\delta_e) = \frac{3}{8} \beta_e(n),$$

by the definition of τ_e , so $n_e \text{KL}_e \geq \frac{3}{16} \beta_e(n)$. Since $\log(1/\delta_e) \geq \log 2 \geq \frac{1}{2}$ and $\beta_e(n)$ carries the mB_0 floor, $\beta_e(n) \geq (c/2)(1+mB_0)$, so each terminated epoch contributes $n_e \text{KL}_e \geq \frac{3c}{32}(1+mB_0)$; summing over terminated epochs gives the first inequality. For the second, bounded log-ratios give $\text{KL}_e \leq 2B_0 \text{TV}(P_{\xi^*}^{\pi_e}, P_{\xi_e}^{\pi_e}) \leq B_0 \Delta_e$ (total variation of the trajectory laws is at most half the sum of one-step prediction errors, by the telescoping in the proof of Lemma D.2), so by Cauchy-Schwarz $\sum_e n_e \text{KL}_e \leq B_0 \sqrt{\sum_e n_e} \sqrt{\sum_e n_e \Delta_e^2} \leq B_0 \sqrt{T \sum_e n_e \Delta_e^2}$, using $\sum_e n_e \leq T$. The final estimate substitutes $\sum_e n_e \Delta_e^2 = \tilde{O}(d_E \tilde{\beta}_T)$ from Lemma D.12, whose proof does not use this lemma. The warm-up consequences follow by multiplying by $(m-1)H$ and using $(m-1)B_0/(1+mB_0) \leq 1$. $\square$

D.9. The Role of the Posterior-Lipschitz Transport Term

Remark D.8 (Role of the transport term). Because the adversary aggregation operator $G_h^{\Phi, \pi}$ depends on the deployed policy, one might expect the analysis to pay a cost for transporting adversary-aggregation errors from historical policies to

the current one. Under Assumption 3.1(e), which imposes the KL-to-operator conversion on the *joint* operators, no such cost arises. Past consistency is established directly at the data-collection policies, and the query shift to π_e is handled by the capped summability argument, so the radius $\bar{\beta}_T = H\beta_T$ carries no transport term. The transport viewpoint remains instructive for two reasons. First, it quantifies what changes if Assumption 3.1(e) is weakened to a world-channel-only conversion, in which case adversary-aggregation errors must be compared across deployed policies and each comparison costs $O(H^2)$ at the aggregate squared-error scale (Lemma D.9 below). Second, it is the lens through which the flat-batching analysis of Sang & Nguyen-Tang (2025) is best understood (Appendix J). With $K = \Theta(\sqrt{d_E T})$ deployed policies, the accumulated transport term grows polynomially in T and obstructs a direct $\tilde{O}(\sqrt{T})$ guarantee by that route.

Lemma D.9 (Posterior-Lipschitz transport). *Suppose Assumption 2.3 holds. Fix two learner policies π, ν , an adversary model Φ , and a step h .*

Let $q_h^{\text{mid}} \in \mathcal{V}$ denote any nonnegative intermediate predictive vector of the B -indexed form produced by the world-channel update W_h^θ , before the adversary action is aggregated by $G_h^{\Phi, \pi}$. Then

$$\left\| (G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi})q_h^{\text{mid}} - (G_h^{\Phi, \nu} - G_h^{\Phi^*, \nu})q_h^{\text{mid}} \right\|_1 \leq C_{\text{tr}} \|q_h^{\text{mid}}\|_1,$$

where C_{tr} depends on the posterior-Lipschitz constant L but not on T . Consequently, at the aggregate squared-error scale,

$$\left(\sum_{h=1}^H \left\| (G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi})q_h^{\text{mid}} - (G_h^{\Phi, \nu} - G_h^{\Phi^*, \nu})q_h^{\text{mid}} \right\|_1 \right)^2 \leq C_{\text{tr}}^2 H \sum_{h=1}^H \|q_h^{\text{mid}}\|_1^2.$$

Proof. The aggregation operator $G_h^{\Phi, \pi}$ differs across policies only through the adversary response probabilities $g_h^\Phi(\cdot | \eta^B, \pi)$. For any intermediate vector q_h^{mid} , linearity of $G_h^{\Phi, \pi}$ and the definition of the ℓ_1 operator norm give

$$\left\| (G_h^{\Phi, \pi} - G_h^{\Phi, \nu})q_h^{\text{mid}} \right\|_1 \leq \sup_{\eta^B} \|g_h^\Phi(\cdot | \eta^B, \pi) - g_h^\Phi(\cdot | \eta^B, \nu)\|_1 \cdot \|q_h^{\text{mid}}\|_1.$$

By Assumption 2.3, uniformly over $\theta \in \Theta$,

$$\|g_h^\Phi(\cdot | \eta^B, \pi) - g_h^\Phi(\cdot | \eta^B, \nu)\|_1 \leq L \|S_{\eta^B, h}^{\text{ref}, \theta}(\pi) - S_{\eta^B, h}^{\text{ref}, \theta}(\nu)\|_1.$$

The two posterior-predictive learner action distributions are probability distributions, so their ℓ_1 distance is at most 2. Hence

$$\left\| (G_h^{\Phi, \pi} - G_h^{\Phi, \nu})q_h^{\text{mid}} \right\|_1 \leq 2L \|q_h^{\text{mid}}\|_1.$$

The same bound holds with Φ^* in place of Φ . Applying the triangle inequality,

$$\begin{aligned} & \left\| (G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi})q_h^{\text{mid}} - (G_h^{\Phi, \nu} - G_h^{\Phi^*, \nu})q_h^{\text{mid}} \right\|_1 \\ & \leq \left\| (G_h^{\Phi, \pi} - G_h^{\Phi, \nu})q_h^{\text{mid}} \right\|_1 + \left\| (G_h^{\Phi^*, \pi} - G_h^{\Phi^*, \nu})q_h^{\text{mid}} \right\|_1 \leq 4L \|q_h^{\text{mid}}\|_1. \end{aligned}$$

Absorbing constants into C_{tr} proves the first claim. The aggregate squared bound follows from Cauchy–Schwarz:

$$\left(\sum_{h=1}^H a_h \right)^2 \leq H \sum_{h=1}^H a_h^2,$$

with

$$a_h = \left\| (G_h^{\Phi, \pi} - G_h^{\Phi^*, \pi})q_h^{\text{mid}} - (G_h^{\Phi, \nu} - G_h^{\Phi^*, \nu})q_h^{\text{mid}} \right\|_1.$$

□

Applied with $\pi = \pi_e$ and $\nu = \pi_k$, and using $\|q_h^{\text{mid}}\|_1 \leq |B|$ along trajectories of the true model (the B -indexed components carry the unit predictive mass once per hypothetical adversary action), each historical policy contributes at most

$$C_{\text{tr}}^2 H \sum_{h=1}^H \mathbb{E} \|q_h^{\text{mid}}\|_1^2 = O(H^2) \quad (40)$$

at the aggregate squared-error scale (the constant absorbs $C_{\text{tr}}^2 |B|^2$). After S deployed policies the accumulated transport term is therefore

$$\Gamma_S = O(SH^2), \quad (41)$$

which is $O(H^2 \log T)$ for $S = O(\log T)$ scheduled epochs, but $\tilde{O}(H^2 \sqrt{d_E T})$ under a flat $\sqrt{d_E T}$ -batch schedule; this is the quantitative content of the comparison in Appendix J.2. We emphasize that none of this enters the proof of Theorem 3.3.

D.10. Eluder Dimension Bound

We first state the weighted Eluder summability result, then derive the main bound. The key issue is that Δ_e is an aggregate over H steps and a trajectory distribution, not a single scalar function value. We handle this by working with the *stepwise squared error* $\mathcal{E}_e^2$ as the primary object for the Eluder argument, then converting back to Δ_e via Jensen's inequality.

Definition D.10 (Stepwise and channel prediction errors). For epoch e , define the *stepwise squared prediction error*

$$\mathcal{E}_e^2 := \sum_{h=1}^H \mathbb{E}_{\tau \sim P_{\xi^*}^{\pi_e}} \left[\|J_h^{\xi_e, \pi_e} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi_e} q_{h-1}^{\xi^*}\|_1^2 \right],$$

and recall the *aggregate prediction error* Δ_e of (37). By Jensen's inequality (or Cauchy–Schwarz over h), $\Delta_e^2 \leq H \mathcal{E}_e^2$. Define also the *channel errors*

$$\Delta_e^W := \sum_{h=1}^H \mathbb{E}_{\tau \sim P_{\xi^*}^{\pi_e}} \left[\|(W_h^{\theta_e} - W_h^{\theta^*}) q_{h-1}^{\xi^*}\|_1 \right], \quad \Delta_e^G := \sum_{h=1}^H \mathbb{E}_{\tau \sim P_{\xi^*}^{\pi_e}} \left[\|(G_h^{\Phi_e, \pi_e} - G_h^{\Phi^*, \pi_e}) W_h^{\theta^*} q_{h-1}^{\xi^*}\|_1 \right],$$

which are the values $F_{\theta_e}^W(\pi_e)$ and $F_{\Phi_e}^G(\pi_e)$ of the channel classes of Section 2.5; by the channel-triangle inequalities (4), $\Delta_e \leq \Delta_e^W + \Delta_e^G$ and $\Delta_e^G \leq \Delta_e + \Delta_e^W$.

Lemma D.11 (Capped Weighted Eluder Summability). *Let $\mathcal{F}$ be a class of functions taking values in $[0, b]$, and suppose epochs $e = 1, \dots, E$ produce triples $(f_e \in \mathcal{F}, x_e, n_e \in \mathbb{N})$ with $\sum_e n_e \leq 2T$ such that (i) past consistency: $\sum_{k < e} n_k f_e(x_k)^2 \leq R$ for every e ; and (ii) per-epoch cap: $n_e f_e(x_e)^2 \leq \tilde{\beta}$ for every e . Then, with $\alpha := \sqrt{\tilde{\beta}/(2T)}$,*

$$\sum_{e=1}^E n_e f_e(x_e)^2 \leq 40 \dim_E(\mathcal{F}, \alpha/2) (R + \tilde{\beta}) \left\lceil \log_2 \frac{2b}{\alpha} \right\rceil \left\lceil \log_2 4T \right\rceil + 2\tilde{\beta}.$$

Without hypothesis (ii), no bound of this form is possible, since a function that vanishes on all past queries but is large at its own query satisfies (i) with any $R \geq 0$, yet contributes up to $n_e b^2$ (cf. Proposition D.5).

Proof. Epochs with $f_e(x_e)^2 \leq \tilde{\beta}/(2T)$ contribute at most $\sum_e n_e \tilde{\beta}/(2T) \leq \tilde{\beta}$ in total; set them aside. Bucket the remaining epochs by the dyadic scales of value and weight. Bucket (j, ℓ) contains the epochs with $f_e(x_e) \in (\varepsilon_{j+1}, \varepsilon_j]$, where $\varepsilon_j := b2^{-j}$ and j ranges over at most $\lceil \log_2(b/\alpha) \rceil$ values before reaching the floor scale α , and $n_e \in [2^\ell, 2^{\ell+1})$ with $\ell \leq \log_2 2T$. Fix a bucket, list its epochs in time order, and write N for the bucket size.

Claim: $N \leq d(R/(2^\ell \varepsilon_{j+1}^2) + 2)$, where $d := \dim_E(\mathcal{F}, \varepsilon_{j+1})$ and $\varepsilon_{j+1} \geq \alpha/2$ for every bucket in range. Process the bucket's queries in order, maintaining a collection of disjoint subsequences, as follows. Append the arriving query to some maintained subsequence on whose current content it is ε_{j+1} -independent if one exists, and otherwise start a new subsequence. Every query is placed, and by the definition of the Eluder dimension no subsequence receives more than d elements, so at the end there are $M \geq \lceil N/d \rceil$ subsequences. Consider the query x_e that opened the M -th subsequence. At its arrival it was ε_{j+1} -dependent on each of the $M - 1$ previously opened, disjoint, nonempty subsequences. Let $f := f_e$ be that epoch's function. Since $f(x_e) > \varepsilon_{j+1}$, dependence on each such set S forces $\sum_{z \in S} f(z)^2 > \varepsilon_{j+1}^2$, and since every bucket predecessor carries weight at least 2^ℓ , $\sum_{k \in S} n_k f(x_k)^2 > 2^\ell \varepsilon_{j+1}^2$. Summing over the $M - 1$ disjoint sets and invoking hypothesis (i) gives $R \geq (M - 1) 2^\ell \varepsilon_{j+1}^2 \geq (N/d - 1) 2^\ell \varepsilon_{j+1}^2$, which rearranges to the claim.

Bucket damage. Each bucket epoch has $n_e f_e(x_e)^2 \leq \min(2^{\ell+1} \varepsilon_j^2, \tilde{\beta}) = \min(8 \cdot 2^\ell \varepsilon_{j+1}^2, \tilde{\beta})$, so the bucket's total damage is at most

$$N \cdot \min(8 \cdot 2^\ell \varepsilon_{j+1}^2, \tilde{\beta}) \leq d \left(\frac{R}{2^\ell \varepsilon_{j+1}^2} + 2 \right) \cdot 8 \cdot 2^\ell \varepsilon_{j+1}^2 \leq d(8R + 16\tilde{\beta}),$$

using $2^\ell \varepsilon_{j+1}^2 \leq n_e f_e(x_e)^2 \leq \tilde{\beta}$ for any bucket member in the last step. The value scales run from b down to at least $\alpha/2$ and the weight scales cover $1 \leq n_e \leq 2T$, so there are at most $\lceil \log_2(2b/\alpha) \rceil \cdot \lceil \log_2 4T \rceil$ buckets; summing the per-bucket damage and adding the floor contribution proves the lemma (constants consolidated). $\square$

Lemma D.12 (Eluder Bound Across All Epochs). *Suppose Assumptions 2.3, 2.4, and 3.1 hold, and condition on the high-probability events of Lemma D.4 (strengthened as in Appendix D.8) and Lemma D.6 for all epochs. With the channel classes of Section 2, $d_E = d_{\text{agg}}^W + d_{\text{agg}}^G$ evaluated at scale $1/(2T)$:*

$$\sum_e n_e \Delta_e^2 \leq \tilde{O}(d_E H \beta_T / c_{\text{KL}}) = \tilde{O}(d_E \tilde{\beta}_T). \quad (42)$$

Here and below, $\tilde{O}(\cdot)$ additionally absorbs constants depending on B_0 and c_{KL} , as reflected in the constant-dependence statement of Theorem 3.3.

Proof. We establish past consistency and per-epoch caps for each channel, apply the capped summability lemma channel by channel, and combine.

Past consistency, per channel. Since $\xi_e \in \mathcal{C}_{e-1}$, Lemma D.3 gives

$$\sum_{k < e} n_k \text{KL} \left(P_{\xi_e}^{\pi_k} \parallel P_{\xi_e}^{\pi_k} \right) \leq C\beta_e \leq C\beta_T.$$

(The concentration argument is unaffected by the data-dependent epoch lengths, because the martingale runs over episodes in temporal order, with each episode's policy adapted to the filtration.) By the two clauses of Assumption 3.1(e) and Cauchy–Schwarz over h (as in Definition D.10), writing F_{ξ_e} , $F_{\theta_e}^W$, $F_{\Phi_e}^G$ for the composed and channel aggregate errors of the optimistic model at past policies,

$$\sum_{k < e} n_k F_{\xi_e}(\pi_k)^2 \leq \frac{CH\beta_T}{c_{\text{KL}}} =: R, \quad \sum_{k < e} n_k F_{\theta_e}^W(\pi_k)^2 \leq R,$$

and, by the channel-triangle transfer (4) together with $(a+b)^2 \leq 2a^2 + 2b^2$,

$$\sum_{k < e} n_k F_{\Phi_e}^G(\pi_k)^2 \leq 2 \sum_{k < e} n_k F_{\xi_e}(\pi_k)^2 + 2 \sum_{k < e} n_k F_{\theta_e}^W(\pi_k)^2 \leq 4R.$$

Per-epoch cap, per channel. Lemma D.6 covers every data-collection episode of every epoch, terminated or complete; via the two clauses of Assumption 3.1(e) and Cauchy–Schwarz over h ,

$$n_e \Delta_e^2 \leq H\tilde{\beta}_e/c_{\text{KL}} \leq \tilde{O}(H\beta_T/c_{\text{KL}}) =: \tilde{\beta}, \quad n_e(\Delta_e^W)^2 \leq \tilde{\beta}, \quad n_e(\Delta_e^G)^2 \leq 4\tilde{\beta},$$

the last by the triangle transfer, and the middle estimates using $\tilde{\beta}_e = O(B_0\beta_T)$ with the B_0 -dependent factor absorbed.

Summability, per channel. The functions $F_{\theta_e}^W$ belong to $\mathcal{F}_{\text{agg}}^{W,\xi^*}$ and the $F_{\Phi_e}^G$ to $\mathcal{F}_{\text{agg}}^{G,\xi^*}$. The world-channel class takes values in $[0, 2|B|H]$, since each b -component of $W_h^\theta q$ carries mass at most one, so $\|W_h^\theta q\|_{1,\mathcal{V}} \leq |B|$ and each h -summand is at most $2|B|$. The adversary-channel class takes values in $[0, 2H]$, since $G_h^{\Phi,\pi} \circ W_h^{\theta^*} = J_h^{(\theta^*,\Phi),\pi}$ is an observable operator of the model $(\theta^*, \Phi) \in \Xi$, whose image of a normalized state has mass at most one. The ranges enter Lemma D.11 only through $\lceil \log_2(2b/\alpha) \rceil$, so the extra $\log |B|$ is absorbed into $\tilde{O}$. Apply Lemma D.11 to the world channel with $(R, \tilde{\beta})$ and to the adversary channel with $(4R, 4\tilde{\beta})$; the lemma's floor scale for the adversary application is 2α with $\alpha = \sqrt{\tilde{\beta}/(2T)}$, and in both applications the dimension is evaluated at a scale at least $\alpha/2 \geq 1/(2T)$. Noting that $\tilde{\beta} \geq cH/c_{\text{KL}} \geq 2/T$ (taking $c_{\text{KL}} \leq 1$ without loss of generality, since Assumption 3.1(e) only weakens as c_{KL} decreases) gives $\alpha/2 \geq 1/(2T)$, so that by scale monotonicity $\dim_E(\mathcal{F}_{\text{agg}}^W, \alpha/2) \leq d_{\text{agg}}^W$ and $\dim_E(\mathcal{F}_{\text{agg}}^G, \alpha/2) \leq d_{\text{agg}}^G$, we obtain

$$\sum_e n_e (\Delta_e^W)^2 \leq \tilde{O}(d_{\text{agg}}^W (R + \tilde{\beta})), \quad \sum_e n_e (\Delta_e^G)^2 \leq \tilde{O}(d_{\text{agg}}^G (R + \tilde{\beta})).$$

Combining. By $\Delta_e \leq \Delta_e^W + \Delta_e^G$ and $(a+b)^2 \leq 2a^2 + 2b^2$,

$$\sum_e n_e \Delta_e^2 \leq 2 \sum_e n_e (\Delta_e^W)^2 + 2 \sum_e n_e (\Delta_e^G)^2 \leq \tilde{O}((d_{\text{agg}}^W + d_{\text{agg}}^G) H\beta_T/c_{\text{KL}}),$$

which is (42). □

D.11. Applying Cauchy–Schwarz and Summing Over Epochs

Using the channel-sum Eluder dimension $d_E = d_{\text{agg}}^W + d_{\text{agg}}^G$, Lemma D.12 gives (absorbing constant and problem-dependent factors into C_1 , and a $\log H$ into the $\log^2 T$ factor, using $H \leq \text{poly}(T)$):

$$\sum_e n_e \Delta_e^2 \leq C_1 d_E \tilde{\beta}_T \log^2 T.$$

Applying Cauchy–Schwarz with $\sum_e n_e \leq T$:

$$\sum_e n_e \cdot \Delta_e \leq \sqrt{\left(\sum_e n_e \right) \cdot \left(\sum_e n_e \cdot \Delta_e^2 \right)} \leq O\left(\sqrt{C_1 d_E T \tilde{\beta}_T \log^2 T} \right).$$

The warm-up cost across all epochs is $(m-1)H S_T$, with the number of deployed policies S_T bounded in Lemma D.7. Combining with the factor H from $R_e \leq n_e H \Delta_e + (m-1)H$:

$$\begin{aligned} PR(T) &\leq H \sum_e n_e \Delta_e + (m-1)H S_T \\ &\leq C \left(H \sqrt{d_E \bar{\beta}_T T \log T} + mH S_T \right), \end{aligned}$$

which is the bound of Theorem 3.3. Lemma D.7 gives, unconditionally,

$$mH S_T \leq O(mH \log T) + \tilde{O} \left(\frac{mB_0}{1+mB_0} H \sqrt{d_E \bar{\beta}_T T} \right) \leq O(mH \log T) + \tilde{O}(H \sqrt{\bar{\beta}_T d_E T}),$$

so the warm-up term is dominated by the leading term plus $mH \log T$, with no condition relating m , B_0 , and T ; this is precisely what the mB_0 floor in β_e ensures. The total failure probability is at most 2δ : δ for the confidence events (Lemma D.4, strengthened as in Appendix D.8) and δ for the per-epoch events of Lemmas D.6 and D.7 ($\sum_e \delta_e \leq \delta$).

Remark D.13 (Sharper bound under aggregate KL conversion). If Assumption 3.1(e) is strengthened to the aggregate form

$$\text{KL}(P_{\xi^*}^{\pi} \| P_{\xi}^{\pi}) \geq c_{\text{agg}} \left(\sum_{h=1}^H \mathbb{E}[\|J_h^{\xi, \pi} q_{h-1}^{\xi^*} - J_h^{\xi^*, \pi} q_{h-1}^{\xi^*}\|_1] \right)^2,$$

then the past-consistency step yields $\sum_{k \leq e} n_k F_{\xi_e}(\pi_k)^2 \leq C \beta_T / c_{\text{agg}}$ directly (no Cauchy-Schwarz factor H), and the per-epoch cap of Lemma D.6 similarly loses its H factor; strengthening the world-channel clause in the same way transfers the improvement to both channels. The final theorem then holds with the sharper $\bar{\beta}_T = \beta_T$ in place of $H\beta_T$. This aggregate KL condition holds for tabular/finite classes with c_{agg} absorbing an H -dependent constant from the weak-revealing argument.

Remark D.14 (Stepwise alternative). One can instead run the summability argument on per-step channel classes, whose queries are $(\pi, h, \tau_{h-1}, o_h^A, a_h)$, converting back through $\Delta_e^2 \leq H \mathcal{E}_e^2$ (Definition D.10); this costs an extra $\sqrt{H}$ in the final regret bound. We do not pursue it. For the parametric classes of Proposition 2.6, the per-step and aggregate channel dimensions admit the same span-dimension upper bounds (Lemma C.1 and the proof of Proposition 2.6 in Appendix C), so the aggregate route is never worse.

Remark D.15 (Time accounting). Algorithm 1 executes $m-1$ warm-up episodes per epoch in addition to the data-collection episodes. If T denotes the number of data-collection episodes actually counted in the regret, then the total number of calendar episodes is $T + O(m S_T)$. The regret bound $PR(T)$ is stated in terms of the T data-collection episodes; warm-up episodes contribute the additional $O(mH S_T)$ term already included above (which the mB_0 floor reduces to $O(mH \log T)$ plus a dominated term). If one wishes to count T as the total calendar budget, one should reduce each epoch's data-collection length by the warm-up length accordingly; this changes constants but not the asymptotic rate.

E. Proof of Theorem 4.1 (Lower Bound)

We prove that any algorithm must incur $\Omega(\sqrt{dT})$ policy regret by embedding the standard d -armed stochastic bandit lower bound into a two-step POMG ($H = 2$) that fits within the POMG protocol.

Proof of Theorem 4.1.

Construction. Fix $d \leq T$ and $\Delta = c_0 \sqrt{d/T}$. The POMG has:

- Latent states $S = \{s_1, \dots, s_d\}$; initial state s_1 (the identity of the initial state is irrelevant since step 1 is only a routing step).
- Learner actions $A = \{a_1, \dots, a_d\}$; single adversary action $B = \{b_{\text{coop}}\}$ (adversary is trivially memoryless, $m = 1$, $L = 0$).
- Horizon $H = 2$.
- **Step $h = 1$:** All states emit the same observation o_0 . The learner chooses a_i ; the state transitions deterministically to s_i . Reward $r_1 = 0$.
- **Step $h = 2$:** In state s_i , the learner observes $o = (i, y)$, where $y \sim \text{Bernoulli}(\mu_i)$. The reward is $r_2(i, y) = y$.

Although all states share the same step-1 observation, the initial state is fixed and known to be s_1 , so there is no hidden-state ambiguity at step $h = 1$. The first coordinate i of the step-2 observation $o = (i, y)$ reveals the latent state at step $h = 2$; together, this makes the construction fully revealing with $\kappa = 1$ and a constant signal parameter independent of Δ . The reward is the second coordinate y , so the learner both observes the routed state and receives the corresponding Bernoulli reward sample.

Policy regret equals bandit regret. With $H = 2$, $m = 1$, and a fixed adversary, the expected policy regret of any learner algorithm against the best fixed policy $\pi^* = a_{i^*}$ satisfies:

$$\mathbb{E}[PR(T)] = \sum_{t=1}^T \left(\frac{1}{2} + \Delta - \mathbb{E}[r_2^{(t)}] \right) = \Delta \cdot \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{A_t \neq i^*\} \right], \quad (43)$$

which is exactly the expected cumulative regret of the learner against arm i^* in a d -armed Bernoulli bandit with gap Δ .

Policy class, ambient model class, and Eluder dimension. The learner's policy class for this instance is $\Pi = \{a_1, \dots, a_d\}$, the deterministic routing policies; the learner may still randomize its choice across episodes, and the bandit lower bound below applies to arbitrary algorithms. Define the ambient model class as $\mathcal{M} = \{\mu \in [1/2, 1/2 + \Delta]^d\}$, parameterizing the step-2 emission means. The adversary channel is trivial (a single constant response), so $d_{\text{agg}}^G = 0$ and $d_E = d_{\text{agg}}^W$. Since Π has d elements, any repeated query is dependent on its predecessors at every scale, giving $d_E \leq d$. Conversely, for $dT \geq C_{\text{lb}}$, with C_{lb} an absolute constant determined by c_0 , the d single-arm queries form a $1/(2T)$ -independent sequence. The witness for query a_j sets $\mu_j = 1/2 + \Delta$ and matches the reference elsewhere, so its aggregate error at a_j is exactly Δ (the step-1 operators coincide across the class, and the step-2 emission difference carries ℓ_1 mass $|\mu_j - \frac{1}{2}| = \Delta$ at the realized observation), while its error at every other query is zero; and $\Delta = c_0 \sqrt{d/T} > 1/(2T)$ exactly when $dT > 1/(4c_0^2)$. Hence $d_E = \Theta(d)$ for $dT \geq C_{\text{lb}} := \lfloor 1/(4c_0^2) \rfloor + 1$. The hard-instance prior uses the subclass where exactly one $\mu_{i^*} = 1/2 + \Delta$ and the rest equal $1/2$; this subclass is contained in the ambient class $\mathcal{M}$, and the ambient class is the model class for which $d_E = \Theta(d)$.

Minimax lower bound. By the standard d -armed stochastic bandit minimax lower bound (Lattimore & Szepesvári, 2020, §15), in its Bernoulli form (the change-of-measure step uses $\text{KL}(\text{Ber}(1/2) \parallel \text{Ber}(1/2 + \Delta)) \leq 4\Delta^2$ for $\Delta \leq 1/4$, and requires $d \geq 2$), for $\Delta = c_0 \sqrt{d/T}$ with $c_0 > 0$ sufficiently small, every algorithm satisfies

$$\max_{i^* \in [d]} \mathbb{E}_{i^*} \left[\Delta \sum_{t=1}^T \mathbf{1}\{A_t \neq i^*\} \right] \geq c\sqrt{dT}, \quad (44)$$

for a universal constant $c > 0$. Since the policy regret of the constructed POMG equals the bandit regret in (43), the same lower bound holds for policy regret.

Assumption verification.

- **Posterior-Lipschitz (Assumption 2.3):** The adversary is constant ($|B| = 1$), so $L = 0$.
- **Weak revealing (Assumption 2.4):** $\kappa = 1$; at step $h = 1$ the next observation $o = (i, y)$ has first coordinate i equal to the routed state index, so world-channel differences are detected with a constant signal parameter (independent of Δ); at the final step the condition is not imposed, per the window convention of Assumption 2.4, and the operative conversion is Assumption 3.1(e), verified directly in the next item. The Δ -dependence belongs to the reward/emission mean estimation problem, not to state revelation.
- **Statistical regularity (Assumption 3.1):** Bernoulli likelihoods are bounded, the model class $\mathcal{M}$ is compact and has finite covering number, predictive states are finite-dimensional probability vectors with unit ℓ_1 -norm, and both clauses of the KL-to-operator conversion hold exactly, via $\text{KL}(\text{Ber}(p) \parallel \text{Ber}(q)) \geq 2(p - q)^2$ (the adversary channel is trivial).

Since $\max_{i^*} \mathbb{E}[PR(T)] \geq c\sqrt{dT}$ and $d_E = \Theta(d)$, we have $\sup_{\text{instance} \in \mathcal{M}} \mathbb{E}[PR(T)] \geq c'\sqrt{d_E T}$ for a universal constant $c' > 0$, completing the proof. $\square$

F. Proof of Theorem 5.1 (Adaptive Horizon)

The epoch-based structure of Algorithm 1 naturally provides adaptation to unknown time horizons. The key insight is that each epoch is self-contained. Epoch e determines its behavior from its cap $T_e = 2^e$, the data collected so far, and the termination test, while building its confidence set from all historical data. The algorithm never needs to know the total horizon T in advance.

Proof of Theorem 5.1. The algorithm operates through epochs $e = 1, 2, \dots$, where epoch e has length at most $T_e = 2^e$ (the termination test may end it earlier). The important point is that the algorithm does not need to know T in advance. All choices made in epoch e depend only on the epoch index e and on the data collected up to the current episode.

At the start of epoch e , the algorithm performs the following steps. First, it computes the MLE $\hat{\xi}_{e-1}$ from the cumulative dataset $\mathcal{D}_{e-1}$ containing all trajectories from epochs $1, \dots, e-1$. Second, it forms the confidence set

$$\mathcal{C}_{e-1} = \left\{ \xi : L_{e-1}(\xi) \geq L_{e-1}(\hat{\xi}_{e-1}) - \beta_e \right\},$$

where

$$\delta_e = \frac{\delta}{2e^2}, \quad \epsilon_e = 2^{-\lceil \log_2(|\mathcal{D}_{e-1}|+2) \rceil}, \quad \beta_e = c(\log \mathcal{N}(\epsilon_e; \Xi) + \log(1/\delta_e) + mB_0).$$

Thus β_e depends only on the current epoch index and the data collected so far, and not on the eventual stopping time. Third, the algorithm selects an optimistic pair

$$(\pi_e, \xi_e) \in \underset{(\pi, \xi) \in \Pi \times \mathcal{C}_{e-1}}{\operatorname{argmax}} V^\pi(\xi).$$

It then executes π_e for at most $T_e = 2^e$ data-collection episodes, subject to the statistical termination test, after the $m-1$ warm-up episodes needed for the m -memory adversary to stabilize, and adds the resulting trajectories to the cumulative dataset.

The confidence analysis is also horizon-free. By construction, $\sum_{e=1}^{\infty} \delta_e \leq \delta$. Therefore, applying the likelihood concentration and confidence-set validity arguments of Lemma D.4 with failure probability δ_e in epoch e , and union bounding over all $e \geq 1$, gives a single high-probability event of probability at least $1 - \delta$ on which all epoch confidence sets are valid simultaneously. This event is defined without knowing how many epochs will eventually be run.

Now fix any final horizon T . Each epoch e runs for n_e data-collection episodes (ended by its cap, the termination test, or the horizon), and

$$\sum_e n_e \leq T,$$

with at most $\log_2(2T)$ scheduled completions and N_{term} terminations bounded as in Lemma D.7.

Define, as in Appendix D.2,

$$\begin{aligned} \beta_T &:= c(\log \mathcal{N}(\epsilon_T; \Xi) + 5 \log(2T) + \log(1/\delta) + mB_0) = O(\log \mathcal{N}(\epsilon_T; \Xi) + \log T + \log(1/\delta) + mB_0), \\ \epsilon_T &:= 1/(2T+4), \quad \bar{\beta}_T := H\beta_T, \end{aligned}$$

so that $\beta_e \leq \beta_T$ and $\beta_e(t) \leq \beta_T$ for every epoch and episode boundary reached by time T . On the simultaneous confidence event, the proof of Theorem 3.3 applies verbatim to this prefix of epochs, with the final epoch truncated at the horizon.

In particular, if Δ_e denotes the aggregate observable-operator prediction error in epoch e , the simulation lemma gives the epoch regret bound

$$R_e \leq n_e H \Delta_e + (m-1)H.$$

The capped summability argument gives

$$\sum_e n_e \Delta_e^2 \leq \tilde{O}(d_E \bar{\beta}_T).$$

Since $\sum_e n_e \leq T$, Cauchy–Schwarz yields

$$\sum_e n_e \Delta_e \leq \sqrt{\left(\sum_e n_e \right) \left(\sum_e n_e \Delta_e^2 \right)} \leq \tilde{O}\left(\sqrt{d_E T \bar{\beta}_T} \right).$$

Combining the data-collection regret with the $(m-1)H S_T$ warm-up cost (Lemma D.7; all quantities involved are prefix-stable), we obtain

$$PR(T) \leq \tilde{O}\left(H \sqrt{\bar{\beta}_T d_E T} + mH S_T \right). \quad (45)$$

This bound holds for every final horizon T simultaneously on the same high-probability event. The algorithm’s adaptation comes from three features: geometric epochs, epoch-wise confidence parameters defined without reference to the final horizon, and a union bound over all possible future epochs. The per-epoch cap and switch-count arguments (Lemmas D.6 and D.7) are likewise prefix-stable, since they treat each epoch separately; their per-epoch failure probabilities consume a second budget $\sum_{e \geq 1} \delta_e \leq \delta$, also defined without reference to a final horizon, so all statements hold simultaneously for every T with probability at least $1 - 2\delta$. Finally, using the warm-up consequence of Lemma D.7 we bound $(m-1)H S_T \leq \tilde{O}(mH \log T + H \sqrt{d_E \bar{\beta}_T T})$; substituting this into (45) gives the form stated in Theorem 5.1. $\square$

G. Proof of Theorem 5.2 (Fading Memory Adversaries)

When the adversary has fading memory, meaning its response depends on an exponentially weighted average of past policies rather than a fixed window, we can still apply our epoch-based algorithm with one straightforward modification: extend the warm-up phase at the beginning of each epoch to allow sufficient time for the adversary's internal state to stabilize. The proof shows that (1) we can truncate the adversary's infinite memory to a finite window with negligible error, and (2) the warm-up cost of stabilizing this truncated adversary remains modest because scheduled epochs are logarithmic in number and every statistical termination is priced by evidence against the deployed model (Lemma D.7).

Proof of Theorem 5.2. Consider an adversary whose internal state evolves according to the *normalized* recursion

$$z_t = \gamma z_{t-1} + (1 - \gamma)\phi(\pi^t), \quad (46)$$

where $\gamma \in (0, 1)$, $\|\phi(\pi)\|_2 \leq 1$ for all π , and $g^t = g(z_t)$ for an L_g -Lipschitz map g . Whenever $\|z_0\|_2 \leq 1$, we have $\|z_t\|_2 \leq 1$ for all t (the normalized recursion preserves the unit ball). Unrolling gives:

$$z_t = (1 - \gamma) \sum_{i=0}^{t-1} \gamma^i \phi(\pi^{t-i}) + \gamma^t z_0.$$

Truncation analysis. For truncation length $m_{\text{trunc}} \geq 1$, define

$$\tilde{z}_t^{(m_{\text{trunc}})} := (1 - \gamma) \sum_{i=0}^{m_{\text{trunc}}-1} \gamma^i \phi(\pi^{t-i}).$$

The approximation error is:

$$\|z_t - \tilde{z}_t^{(m_{\text{trunc}})}\|_2 \leq (1 - \gamma) \sum_{i=m_{\text{trunc}}}^{\infty} \gamma^i + \gamma^t \|z_0\|_2 \leq \gamma^{m_{\text{trunc}}} + \gamma^t.$$

For $t \geq m_{\text{trunc}}$, we have $\gamma^t \leq \gamma^{m_{\text{trunc}}}$, hence $\|z_t - \tilde{z}_t^{(m_{\text{trunc}})}\|_2 \leq 2\gamma^{m_{\text{trunc}}}$. Since the warm-up step runs $m_{\text{warmup}} - 1$ episodes, every data-collection episode is at least the m_{warmup} -th consecutive episode of the deployed policy with $m_{\text{warmup}} = m_{\text{trunc}}$, so this is the only regime used in the regret analysis.

Choosing

$$m_{\text{trunc}} = m_{\text{warmup}} = \left\lceil \frac{\log(2T)}{1 - \gamma} \right\rceil \quad (47)$$

ensures $\gamma^{m_{\text{trunc}}} \leq 1/(2T)$, giving $\|z_t - \tilde{z}_t\|_2 \leq 1/T$ for all data-collection steps. Since g is L_g -Lipschitz, the per-step response error is $O(L_g/T)$ in total variation; over an episode this perturbs the trajectory law by $O(HL_g/T)$ and hence the episode value (which lies in $[0, H]$) by $O(H^2L_g/T)$, for a total truncation contribution of $O(H^2L_g)$ over T episodes, which is absorbed into lower-order terms.

Stabilization under constant policy. The stationary state under constant policy π is $z_\infty = \phi(\pi)$ (the unique fixed point of $z = \gamma z + (1 - \gamma)\phi(\pi)$ is $z = \phi(\pi)$, and $\|\phi(\pi)\|_2 \leq 1$). After k steps of constant π from any z_0 with $\|z_0\|_2 \leq 1$:

$$\|z_k - z_\infty\|_2 = \gamma^k \|z_0 - z_\infty\|_2 \leq 2\gamma^k.$$

Choosing the warm-up length as in (47) gives

$$\|z_{m_{\text{warmup}}} - z_\infty\|_2 \leq 1/T,$$

and hence an $O(L_g/T)$ error in the adversary's response per step; as above, this costs $O(H^2L_g)$ in total value over T episodes, a lower-order term.

Robustness of the confidence analysis to residual non-stationarity. The concentration argument (Lemma D.3) is stated for data generated exactly under the stationary response. During data collection the adversary's response deviates from stationarity by at most $\delta_t^{\text{ns}} \leq 2L_g\gamma^{m_{\text{warmup}}} \leq L_g/T$ in total variation per step, so each episode's trajectory law deviates by at most $H\delta_t^{\text{ns}}$. Since log-likelihood increments are uniformly bounded by B_0 (Assumption 3.1(c)), replacing each episode's law by its stationary counterpart shifts every conditional mean in the martingale argument by at most $2B_0H\delta_t^{\text{ns}}$, and hence all cumulative quantities by at most $\sum_{t \leq T} 2B_0HL_g/T = O(B_0HL_g)$, a T -independent constant. Enlarging β_e and the termination threshold $\tau_e(t)$ by a sufficiently large absolute multiple of this constant restores every statement of Lemmas D.3, D.4, D.6, and D.7 (their martingale arguments shift by the same conditional-mean bound, and the multiple preserves the firing margins of Lemma D.7). This enlargement is the B_0HL_g term inside β_T^γ in the theorem statement; it enters the leading term through the radius, not as additive regret.

Regret decomposition. The total regret has three components:

Component 1 (Truncation): $O(H^2 L_g)$, independent of T and absorbed into lower-order terms; the residual non-stationarity is accounted inside the enlarged radius β_T^γ (robustness paragraph above), not as additive regret.

Component 2 (Warm-up): Each epoch requires $m_{\text{warmup}} = O(\log T/(1-\gamma))$ warm-up episodes. With S_T epochs started (Lemma D.7):

$$\text{Warm-up regret} = S_T \cdot m_{\text{warmup}} \cdot H = O\left(\frac{H S_T \log T}{1-\gamma}\right),$$

which is $O(H \log^2 T/(1-\gamma))$ when only scheduled epoch endings occur.

Component 3 (Exploration-exploitation): Once stabilized, the analysis is identical to Theorem 3.3 with the same d_E and the enlarged radius $\tilde{\beta}_T^\gamma = H \beta_T^\gamma$ (which carries the $B_0 m_{\text{warmup}}$ floor and the $B_0 H L_g$ robustness constant):

$$\text{Statistical regret} = \tilde{O}\left(H \sqrt{d_E \tilde{\beta}_T^\gamma T}\right).$$

Combining all three yields the result,

$$PR(T) \leq \tilde{O}\left(H \sqrt{\tilde{\beta}_T^\gamma d_E T}\right) + O\left(\frac{H S_T \log T}{1-\gamma}\right).$$

Domination of the warm-up term. The algorithm runs with memory parameter $m = m_{\text{warmup}}$, so the floor in β_e is $B_0 m_{\text{warmup}}$ and Lemma D.7 applies verbatim with this m . Every termination requires $n_e \text{KL}_e \geq \frac{3}{16} \beta_e(n) \geq \frac{3c}{32}(1 + B_0 m_{\text{warmup}})$, so $N_{\text{term}} \leq \frac{32}{3c(1+B_0 m_{\text{warmup}})} \sum_e n_e \text{KL}_e$, and since $m_{\text{warmup}} B_0/(1 + B_0 m_{\text{warmup}}) \leq 1$, the terminated-epoch warm-up cost obeys $m_{\text{warmup}} H N_{\text{term}} \leq \tilde{O}(H \sqrt{d_E \tilde{\beta}_T^\gamma T})$. Scheduled completions and the final epoch contribute at most $m_{\text{warmup}} H \cdot O(\log T) = O(H \log^2 T/(1-\gamma))$. Hence the total warm-up cost is unconditionally dominated by the leading statistical term at the radius $\tilde{\beta}_T^\gamma$, plus $O(H \log^2 T/(1-\gamma))$, as discussed after Theorem 5.2. Choosing the warm-up per epoch as $\lceil \log(2^{e+1})/(1-\gamma) \rceil$ instead removes the dependence on T with the same rate; we state the fixed- T version for simplicity. $\square$

The extreme cases illustrate the dependence on the effective memory.

Case 1: $\gamma = 0$ (memoryless adversary). The state becomes $z_t = \phi(\pi^t)$, and the warm-up may be set to zero (the stated algorithm still runs $\lceil \log(2T) \rceil - 1$ warm-up episodes per epoch; setting them to zero is valid at $\gamma = 0$ since the state carries no memory). With zero warm-up the bound reduces to

$$PR(T) = \tilde{O}\left(H \sqrt{\beta_T d_E T}\right),$$

matching the memoryless case up to logarithmic factors.

Case 2: $\gamma \rightarrow 1$ (long memory). The warm-up length grows as $m_{\text{warmup}} = O(\log T/(1-\gamma))$, and the total warm-up cost over the S_T epochs is

$$O\left(\frac{H S_T \log T}{1-\gamma}\right),$$

capturing the increasing cost of stabilizing an adversary with slowly decaying memory.

Case 3: Intermediate γ . For fixed $\gamma \in (0, 1)$, the additional fading-memory cost is $O(H S_T \log T/(1-\gamma))$; as we discuss above, it is $O(H \log^2 T/(1-\gamma))$ outright when no early terminations occur, and in general the $B_0 m_{\text{warmup}}$ floor makes it dominated by the leading statistical term $\tilde{O}(H \sqrt{\tilde{\beta}_T^\gamma d_E T})$ plus the scheduled residual $O(H \log^2 T/(1-\gamma))$.

Remark G.1 (On the $1/(1-\gamma)$ dependence). We expect the $1/(1-\gamma)$ dependence to be unavoidable, because the adversary's internal state converges to stationarity only at rate γ^t , so distinguishing adversaries whose stationary responses differ by ϵ should require on the order of $\log(1/\epsilon)/(1-\gamma)$ episodes of play. Turning this heuristic into a formal policy-regret lower bound, which must also account for the learner's ability to exploit transient responses, is left open.

H. Proof of Theorem 5.3 (Tabular Bounds)

For finite POMGs, we can instantiate the abstract quantities in Theorem 3.3, specifically, the Eluder dimension d_E and confidence parameter β , in terms of the concrete problem parameters: state space size $|S|$, action spaces $|A|$ and $|B|$, observation spaces $|O_A|$ and $|O_B|$, and so forth. This gives us an explicit regret bound that practitioners can evaluate numerically for specific problem instances.

Proof of Theorem 5.3. Consider a tabular POMG with $|S|$ states, $|A|$ learner actions, $|B|$ adversary actions, $|O_A|$ learner observations, $|O_B|$ adversary observations, horizon H , adversary memory m , and a linear adversary response function with dimension d_{adv} .

Parameterizing the World Model The world model consists of two components, transition dynamics and emission distributions. At each time step $h \in [H]$, we need to specify:

1. **Transition kernels** $T_h(s'|s, a, b)$: For each combination of current state s , learner action a , and adversary action b , we have a distribution over next states s' . This requires specifying $|S|$ probabilities for each of the $|S| \times |A| \times |B|$ combinations (with normalization removing one degree of freedom per distribution). The total number of transition parameters is thus $\Theta(|S|^2|A||B|H)$.
2. **Emission kernels** $E_h^A(o|s)$ and $E_h^B(o^B|s)$: For each state s , we have distributions over O_A and over O_B . This requires $|S|(|O_A| + |O_B|)$ parameters per time step, giving $\Theta(|S|(|O_A| + |O_B|)H)$ total emission parameters.

The world model thus carries $\Theta(|S|^2|A||B| + |S|(|O_A| + |O_B|)H)$ parameters; under the standard tabular scaling $|O_A| + |O_B| = O(|S||A||B|)$ the transition parameters dominate.

Bounding the World Model Eluder Dimension. In the finite-window representation invoked by Proposition 2.6 (the exact factorization of Theorem 2.5 lives on the full-history space), the observable operators act on the reduced augmented state space $\tilde{S}_{\text{red}}^+ = S \times O_A^{\kappa-1} \times O_B^{\kappa-1}$ of dimension $d_W = |S| \cdot |O_A|^{\kappa-1} \cdot |O_B|^{\kappa-1}$, which tracks the information needed from the last $\kappa - 1$ learner and adversary observations. The $|O_B|^{\kappa-1}$ factor is needed because the adversary aggregation operator $G_h^{\Phi, \pi}$ must be expressible without referring to the world parameter θ . The world model must also parameterize the adversary's emission kernel $E_h^B(o^B|s)$.

Following the span-dimension route of Proposition 2.6 (cf. the covering-dimension calculation of Liu et al. (2022a) for the single-agent analogue), and summing the per-step operator complexity over $h \in [H]$, the tabular world-channel contribution satisfies:

$$d_{\Theta}^{[\kappa]} = O(H \cdot |S|^2|A||B| \cdot |O_A|^{\kappa} \cdot |O_B|^{\kappa} \cdot \log(|S||A||B||O_A||O_B|HT)),$$

where the logarithm is the sign-witness scale factor at the evaluation scale $1/(2T)$. The factors O_A^{κ} and O_B^{κ} come from the κ -step observation windows used by the weak-revealing representation, while the factor H comes from summing over the H time-dependent operator classes.

Parameterizing and Bounding the Adversary Model. For a linear adversary response model, the response function takes the form:

$$g_h(b|\tau_B, \pi) = [\Phi^* w_h(\tau_B, \pi)]_b,$$

where $\Phi_h^* \in \mathbb{R}^{|B| \times d_{\text{adv}}}$ is a column-stochastic matrix (each column sums to 1 and has non-negative entries; one matrix per step, a shared matrix only tightening the bounds) and $w_h : \mathcal{T}_{h-1}^B \times O_B \times \Pi^m \rightarrow \mathbb{R}^{d_{\text{adv}}}$ is a feature function mapping adversary decision histories and recent learner policies to features. The parameter space for Φ^* has dimension $(|B| - 1) \cdot d_{\text{adv}}$ due to column-stochasticity (one degree of freedom per column is fixed by the normalization constraint).

For fixed π and feature map w_h , the adversary aggregation operator $G_h^{\Phi, \pi}$ is linear in Φ . The Eluder dimension of a bounded linear class is controlled by its parameter dimension up to logarithmic factors (Russo & Van Roy, 2013). Summing over $h \in [H]$ gives

$$d_{\Psi}^{[\kappa]} = O(H \cdot d_{\text{adv}} \cdot |B|).$$

This reflects the fact that identifying Φ_h^* requires observing its effect on $O(d_{\text{adv}} \cdot |B|)$ linearly independent feature vectors.

Joint Eluder dimension. The channel-sum Eluder dimension of Section 2 is the sum of the world-channel and adversary-channel contributions:

$$d_E^{\text{tab}} = d_{\Theta}^{[\kappa]} + d_{\Psi}^{[\kappa]} = \tilde{O}(H(|S|^2|A||B||O_A|^{\kappa}|O_B|^{\kappa} + d_{\text{adv}}|B|)).$$

This agrees with Proposition 2.6. The factor H reflects the sum of the per-step operator dimensions over $h \in [H]$. Typically, the world-channel term dominates unless the adversary response model has very large feature dimension.

Confidence radius. The confidence radius β_T^{tab} must be chosen so that the true parameter remains in the confidence set with high probability. This requires β_T^{tab} to scale with the logarithm of the covering number of the joint model class.

For tabular models, the parameters are the probabilities in the transition and emission kernels, together with the adversary-response parameters. We discretize the corresponding probability simplices to precision ϵ , chosen small enough that the

discretization error is absorbed into the confidence radius. Taking $\epsilon = 1/(2T + 4)$, the horizon-level scale ϵ_T matching the dyadic schedule of Algorithm 1, suffices for the stated bound. The world-model covering number satisfies

$$\log \mathcal{N}(\epsilon; \Theta) = O(H(|S|^2|A||B| + |S||O_A| + |S||O_B|) \log(1/\epsilon)) = \tilde{O}(H(|S|^2|A||B| + |S|(|O_A| + |O_B|))),$$

where the emission terms, $|S||O_A|$ and $|S||O_B|$, are retained; under the standard tabular scaling $|O_A| + |O_B| = O(|S||A||B|)$ they are dominated by the transition term. Similarly, for the adversary response class,

$$\log \mathcal{N}(\epsilon; \Psi) = O(Hd_{\text{adv}}|B| \log(1/\epsilon)) = \tilde{O}(Hd_{\text{adv}}|B|).$$

Combining these covering terms with the union bound over all epochs and the failure probability δ gives

$$\beta_T^{\text{tab}} = \tilde{O}(H(|S|^2|A||B| + |S|(|O_A| + |O_B|) + d_{\text{adv}}|B|) + \log(1/\delta) + mB_0).$$

Substituting this tabular confidence radius into the general effective radius $\bar{\beta}_T = H\beta_T$ gives

$$\bar{\beta}_T^{\text{tab}} = \tilde{O}(H^2(|S|^2|A||B| + |S|(|O_A| + |O_B|) + d_{\text{adv}}|B|) + HmB_0),$$

which is the quantity used in Theorem 5.3, with the additive $H \log(1/\delta)$ from $H\beta_T^{\text{tab}}$ absorbed into the $\tilde{O}$; under the standard tabular scaling $|O_A| + |O_B| = O(|S||A||B|)$ the emission term is absorbed and the familiar two-term form is recovered.

Applying the Main Theorem. Substituting d_E^{tab} and $\bar{\beta}_T^{\text{tab}}$ into Theorem 3.3 gives

$$\begin{aligned} PR(T) &\leq \tilde{O}\left(H\sqrt{\bar{\beta}_T^{\text{tab}} d_E^{\text{tab}} T} + mH \log T\right) \\ &= \tilde{O}\left(H\left(H^2(|S|^2|A||B| + |S|(|O_A| + |O_B|) + d_{\text{adv}}|B|) + HmB_0\right)^{1/2}\right. \\ &\quad \left.\cdot \left(H(|S|^2|A||B||O_A|^\kappa|O_B|^\kappa + d_{\text{adv}}|B|)\right)^{1/2}\sqrt{T} + mH \log T\right). \end{aligned}$$

Understanding the scaling. The tabular bound reflects several sources of difficulty. The state and action spaces enter through both the transition parameterization and the aggregate Eluder dimension. In particular, the transition model contributes factors of $|S|^2|A||B|$, while the adversary-response class contributes $d_{\text{adv}}|B|$, and the emission kernels contribute the additive $|S|(|O_A| + |O_B|)$ term in the confidence radius, absorbed under the standard tabular scaling. The observation-window factors $|O_A|^\kappa|O_B|^\kappa$ appear in d_E^{tab} and capture the cost of partial observability. Larger κ means that longer observation windows are needed to reveal the predictive state, increasing the effective dimension.

The horizon appears in several places. The outer factor H comes from the simulation lemma, the factor H inside d_E^{tab} comes from summing the time-dependent operator dimensions over $h \in [H]$, and the effective confidence radius $\bar{\beta}_T^{\text{tab}}$ also carries horizon dependence. The adversary memory m appears through the warm-up cost $(m-1)H S_T$, which the mB_0 floor in β_e makes unconditionally dominated by the leading term plus $O(mH \log T)$ (Lemma D.7), and through the additive mB_0 term in β_T^{tab} . Thus longer-memory adversaries require longer stabilization periods, but m does not increase the Eluder dimension, and its effect on the statistical term is only the additive mB_0 floor in the confidence radius.

Computational implications. The theorem is primarily a statistical guarantee. The optimistic planning step requires solving

$$\max_{(\pi, \xi) \in \Pi \times \mathcal{C}_{e-1}} V^\pi(\xi),$$

which can be computationally expensive in tabular POMGs. In the worst case, the history-dependent policy class has size exponential in H , and planning in a known finite-horizon POMG is generally intractable in H . The advantage of the epoch structure is that this optimistic planning problem is solved only once per deployed policy, S_T times in total (Lemma D.7), rather than after every episode; the per-episode work of the termination test is a single MLE re-fit, not a new planning problem. Computationally efficient implementations for special tabular or structured subclasses are an important direction beyond the statistical scope of this theorem.

Comparison with POMDP results. It is instructive to compare our POMG bound with the corresponding single-agent POMDP scaling. When the adversary component is absent, equivalently when the adversary has a single trivial action and no strategic response, the model reduces to a POMDP. Existing weakly revealing POMDP analyses give regret bounds of the form $\tilde{O}(\sqrt{d_{\text{POMDP}}T})$, with

$$d_{\text{POMDP}} = \tilde{O}(H|S|^2|A||O_A|^\kappa),$$

under the same convention of summing the operator dimension over time steps. Our POMG bound adds:

- an additional factor of $|B|$ in the world-model Eluder dimension, since adversary actions affect transitions;
- an adversary-observation factor $|O_B|^\kappa$ in the predictive representation;
- the adversary model dimension $d_\Psi^{[\kappa]} = \tilde{O}(H d_{\text{adv}} |B|)$;
- the warm-up cost $(m - 1)H S_T$ from adversary stabilization, dominated by the leading term plus $O(mH \log T)$.

Thus the POMG bound reduces to the familiar weakly revealing POMDP scaling when the adversary component is absent, while separating the additional costs due to adversary actions, adversary observations, response complexity, and memory. $\square$

I. Empirical Validation

We report simulations that exercise the paper’s three main predictions that

1. data-dependent termination is necessary (Experiment 1),
2. regret and the number of deployed policies scale as the theory says they should (Experiment 2), and
3. the full algorithm behaves as intended against an adaptive adversary whose environment statistics come from real data (Experiment 3).

The code, a self-contained Python notebook, and the exact seeds are available at <https://github.com/ramanarora/pomg-policy-regret>; the full suite runs in a few CPU minutes. Following standard practice, the schedule constants are set to practical values, with $c = 1$ and termination threshold $2\beta_e(t)$ in Experiments 1 and 2; Experiment 3 uses the variant stated in Section I.3. The theoretical constants are pessimistic and affect none of the qualitative predictions.

Metric. Policy regret is the quantity defined in Section 2.2: $PR(T) = \sum_{t \leq T} [V^{\pi^*}(R_\infty(\pi^*)) - v_t]$, where v_t is the expected reward of the policy played in episode t under the adversary’s current response and π^* is the best fixed policy under its own stationary response. In Experiments 1 and 2 the environment is the bandit-embedded POMG family of Appendix E, whose value function is available in closed form, so both terms are computed exactly rather than estimated from realized rewards. Figure 3 plots the per-episode average $PR(T)/T$; all other figures plot cumulative quantities, as their axis labels indicate. Every curve averages independent seeds (20 in Experiments 1 and 2, 10 in Experiment 3) and shaded bands show two standard errors of the mean, i.e., $\pm 2\hat{\sigma}/\sqrt{n_{\text{seeds}}}$ with $\hat{\sigma}$ the across-seed standard deviation, approximately a 95% confidence band for the mean curve.

Exact implementation of the algorithm. For a product Bernoulli class the optimistic MLE step of Algorithm 1 is computable in closed form, so Experiments 1 and 2 run the algorithm itself rather than an approximation. The MLE is the clipped empirical mean of each arm. The optimistic value of arm i is its KL upper confidence value, the largest q in the class with $n_i \text{kl}(\hat{\mu}_i, q) \leq \beta_e$, computed by bisection; an arm with no data attains the top of the class. Ties among unplayed arms are broken in favor of fresh arms, which is exactly the valid tie-breaking under which Proposition D.5 operates. The termination test also follows Algorithm 1. The epoch records the deployed arm’s promised optimistic value μ_{dep} at selection time, and after every data-collection episode it recomputes the running radius $\beta_e(t)$ and ends the epoch as soon as the arm’s own data certify refutation, $n_{\text{arm}} \text{kl}(\hat{\mu}_{\text{arm}}, \mu_{\text{dep}}) > 2\beta_e(t)$.

I.1. Experiment 1: Necessity of the Termination Test

Setting. This experiment runs the exact construction behind Proposition D.5. There are $d = \lceil \log_2 T \rceil + 1$ arms. One arm has mean $3/4$ and all others have mean $1/2$. The model class is the widened product family $[1/8, 7/8]^d$. The horizon ranges over $T \in \{2^8, \dots, 2^{15}\}$, so the instance itself grows with T . We compare two algorithms that differ in exactly one line: the epoch-based optimistic MLE with the termination test, and the identical algorithm with the test removed, so that every epoch runs to its scheduled cap 2^e .

Why the instance is hard. The widened class is what drives the failure mode. An arm that has never been played can optimistically be a $7/8$ arm, which is better than anything already observed, so an optimistic learner always prefers a fresh arm. Without the test, each such preference commits an entire cap- 2^e epoch to a bad arm. With $d = \lceil \log_2 T \rceil + 1$ arms, an unplayed arm exists at the start of every epoch, so every episode is spent on an arm of true mean $1/2$ while the best arm has mean $3/4$. The per-episode regret is therefore pinned at the gap $1/4$, which is exactly the proposition’s bound $\mathbb{E}[PR(T)] \geq T/4$.

Findings. Figure 3 shows the per-episode average $PR(T)/T$. The fixed schedule sits at 0.250 across all horizons, matching the proposition’s $T/4$ exactly. The identical algorithm with the termination test refutes each bad commitment within a few episodes and its average regret falls to 0.100 by $T = 2^{15}$. The blue curve is not monotone because the instance changes with T , as d grows with the horizon.

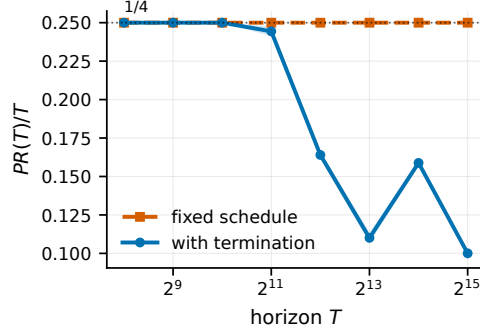


Figure 3. Experiment 1: necessity of the termination test on the hard instance of Proposition D.5, with $d = \lceil \log_2 T \rceil + 1$ arms and 20 seeds. The fixed schedule stays pinned at average regret $1/4$; the termination test drives the average down.

I.2. Experiment 2: Scaling in the Horizon, the Dimension, and the Memory

Setting. All four panels of Figure 4 use the bandit-embedded family with the full algorithm. In panels (a) and (b) there are $d = 8$ arms and the horizon sweeps $T \in \{2^8, \dots, 2^{17}\}$. The best arm’s gap is set to $0.6\sqrt{d/T}$, shrinking with the horizon. The reason is that a fixed gap would eventually let the algorithm identify the best arm, after which regret grows only logarithmically; shrinking the gap like $\sqrt{d/T}$ keeps every horizon in the worst-case regime that the $\sqrt{T}$ theory describes. Panel (c) fixes $T = 2^{14}$ and sweeps the number of arms $d \in \{2, 4, 8, 16, 32\}$ with the same gap rule. Panels (a)–(c) use log-log axes, on which a power law $y \propto x^k$ is a straight line of slope k ; the fitted slopes are least-squares fits in these coordinates. Panel (d) fixes $d = 8$, $T = 2^{14}$, and a gap of 0.15, and sweeps the adversary memory $m \in \{1, 2, 4, 8, 16\}$. After every policy switch the next $m - 1$ episodes are warm-up and are excluded from the dataset, exactly as in Algorithm 1. During these episodes the simulated adversary is still re-stabilizing, and the instance models that transient as a fixed reduction of the learner’s reward by 0.3; the value 0.3 is a design parameter of the simulation (twice the gap), chosen so that warm-up episodes carry a visible cost and the warm-up component of the regret can be measured.

Horizon scaling (panel a). We plot cumulative policy regret against the horizon on log-log axes, where a straight line of slope $1/2$ corresponds to $\sqrt{T}$ growth. The least-squares slope over the full range is 0.53, in close agreement with Theorem 3.3.

Number of deployed policies (panel b). This panel counts the policies S_T the algorithm deploys. Two reference curves frame the measurement. The lower one is the scheduled count $\log_2(2T)$, which would be exact if every epoch ran to its cap. The upper one grows at the $\sqrt{T}$ scale and is the allowance of Lemma D.7, which prices every early termination in statistical evidence. The measured count exceeds $\log_2(2T)$ once $T \geq 2^{11}$, because on these shrinking-gap instances the test keeps firing, and it grows with fitted slope 0.35, well inside the allowance. Both regimes described by the lemma are visible at once: scheduled switching at small horizons, evidence-priced termination at large ones.

Dimension scaling (panel c). Cumulative regret grows with the number of arms at fitted slope 0.78. The two natural reference rates are $\sqrt{d}$, the dependence of the lower bound in Theorem 4.1, and d , the dependence the upper bound implies for this class, whose covering radius satisfies $\beta_T = \Theta(d)$. The measurement falls between the two. The gap between the reference rates reflects the known looseness of covering-based radii in the dimension, not a defect of either bound.

Warm-up cost (panel d). Total policy regret is flat as the memory m grows, while the warm-up component, the regret incurred during warm-up episodes and tallied separately, grows linearly in m and remains lower order, about 14% of the total at $m = 16$. This is the unconditional-domination behavior that the mB_0 floor guarantees through Lemma D.7.

I.3. Experiment 3: An Adaptive Adversary on Real Data

Task. The learner repeatedly classifies handwritten digits against an opponent that controls which digits it sees. In every episode the adversary chooses a distribution over the ten digit classes, a class is drawn from it, an image of that class is observed through a deliberately coarse channel, and the learner’s currently deployed decision rule predicts the class. The learner is rewarded when the prediction is correct. Its choice is which of $K = 6$ fixed decision rules to deploy, and its goal is to compete with the best single rule under the class distribution that this rule itself would eventually induce, which is exactly policy regret against the stationary response.

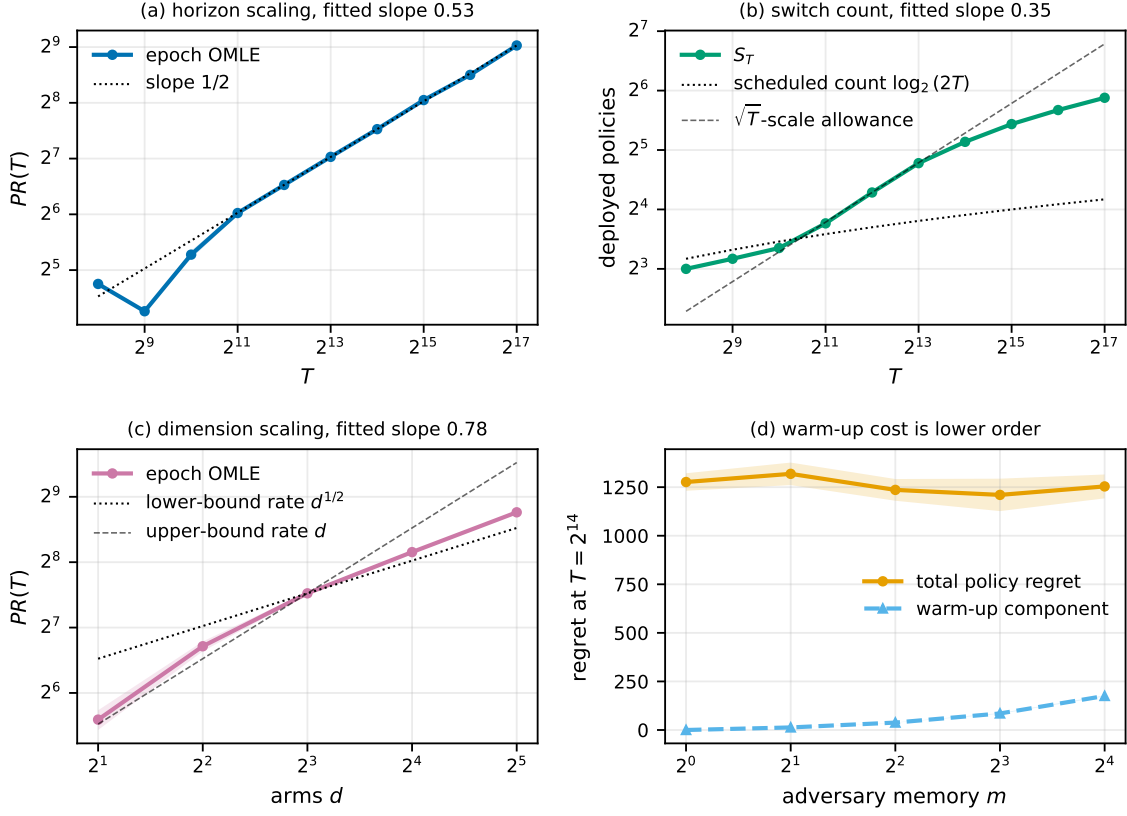


Figure 4. Experiment 2: scaling behavior of the epoch-based optimistic MLE on the bandit-embedded family (20 seeds; bands are two standard errors). (a) Cumulative policy regret against the horizon on minimax-regime instances with gap $0.6\sqrt{d/T}$; the fitted slope 0.53 matches the $\sqrt{T}$ theory. (b) Deployed-policy count S_T between the scheduled count $\log_2(2T)$ and the $\sqrt{T}$ -scale allowance of Lemma D.7. (c) Cumulative policy regret against the number of arms at $T = 2^{14}$; the fitted slope 0.78 lies between the reference rates $\sqrt{d}$ and d . (d) Total policy regret and its warm-up component against the adversary memory m at $T = 2^{14}$.

Data and observation channel. Latent states are the ten digit classes of the scikit-learn handwritten digits corpus (1,797 images of 8×8 pixels). The observation is deliberately coarse. Each image is projected onto the first two principal components of the corpus, each of the two coordinates is split into four bins at its empirical quartiles, and the learner observes only the index of the resulting grid cell, an integer in $\{1, \dots, 16\}$. The learner never sees the image itself. The emission $P(o | s)$ is the empirical distribution of grid cells within class s . Partial observability therefore comes from the genuine confusion structure of the corpus, overlapping classes in the projection, rather than from injected noise. Panel (a) of Figure 5 shows the projected data and the grid.

Decision rules. Each rule is a fixed lookup table $h_k : \{1, \dots, 16\} \rightarrow \{0, \dots, 9\}$ from observed cells to predicted classes; on observing cell o , the deployed rule predicts $h_k(o)$. The tables are built once, before learning begins, and never change. Rule k is fit on its own random half of the corpus by assigning to each cell the majority class among the training images falling in it, except that with probability 0.15 ($k - 1$) the cell's assignment is replaced by a uniformly random class (a cell with no training images also receives a random class). Rule 1 is thus a plain majority-vote prototype classifier and later rules are progressively corrupted copies, so the six rules have genuinely different accuracy profiles across classes, with mean accuracies from 0.57 down to about 0.24. Panel (b) of Figure 5 shows the full accuracy matrix $\text{acc}(k, s)$, the probability that rule k correctly classifies an image of class s drawn through the channel.

The adversary. A memory-4 adversary watches the learner's last four deployed rules and shifts the class distribution toward the classes those rules misclassify most, through a smooth softmax tilt, $\rho(s) \propto \exp(\lambda \overline{\text{err}}(s))$ with $\lambda = 2$, where $\overline{\text{err}}(s)$ is the average error of the four recent rules on class s . The tilt is a smooth function of the recent policies, an instance of the posterior-Lipschitz reaction rules of Section 2.3. Whenever the four recent rules are not identical the adversary is in transit between tilts, and the learner additionally pays a re-stabilization penalty of 0.15; this is the transient that warm-up exists to absorb.

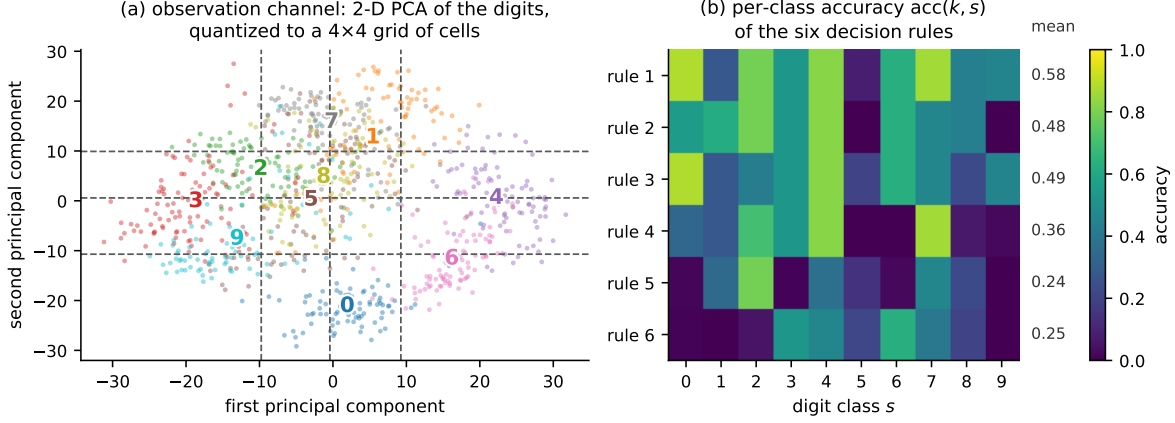


Figure 5. The setup of Experiment 3. (a) The observation channel: the digits corpus projected onto its first two principal components, with points colored and labeled by class and the dashed lines showing the 4×4 quantization grid; the learner observes only the index of the cell containing the projected image, so overlapping classes are genuinely confusable. (b) The per-class accuracy matrix of the six decision rules (row means at right); the adversary tilts the class distribution toward the columns where the learner’s recent rules are weakest.

Metric. Regret is accumulated in expectation rather than from realized rewards. Let v^* be the best rule’s value under its own stationary tilt. In episode t the deployed rule earns its expected accuracy under the adversary’s current class distribution, reduced by the 0.15 penalty whenever the recent window is not constant, and the episode’s regret is v^* minus this quantity.

Algorithms. Four algorithms are compared over $T = 2^{14}$ episodes:

- **Termination:** the epoch-based optimistic MLE of Algorithm 1. Epochs are capped at 2^e ; the optimistic value of each rule is its KL upper confidence value computed from realized rewards; the deployed rule is abandoned by a one-sided refutation test that fires, after at least four stable episodes, when $n \text{kl}(\hat{\mu}, \mu_{\text{dep}}) > 2\beta_e$ with $\hat{\mu}$ the rule’s in-epoch mean reward and μ_{dep} its promised optimistic value.
- **Fixed:** the identical epoch algorithm with the test disabled, so every epoch runs to its cap.
- **ϵ -greedy:** plays the rule with the best empirical mean and explores a uniformly random rule with probability $\epsilon = 0.1$ in every episode. It has no epoch structure, so it switches constantly.
- **Uniform:** plays a uniformly random rule in every episode, as a floor.

The practical schedule for this experiment uses $c = 0.5$ with a unit covering term and no warm-up discard; the fixed baseline uses the identical radius, so the comparison isolates the termination rule.

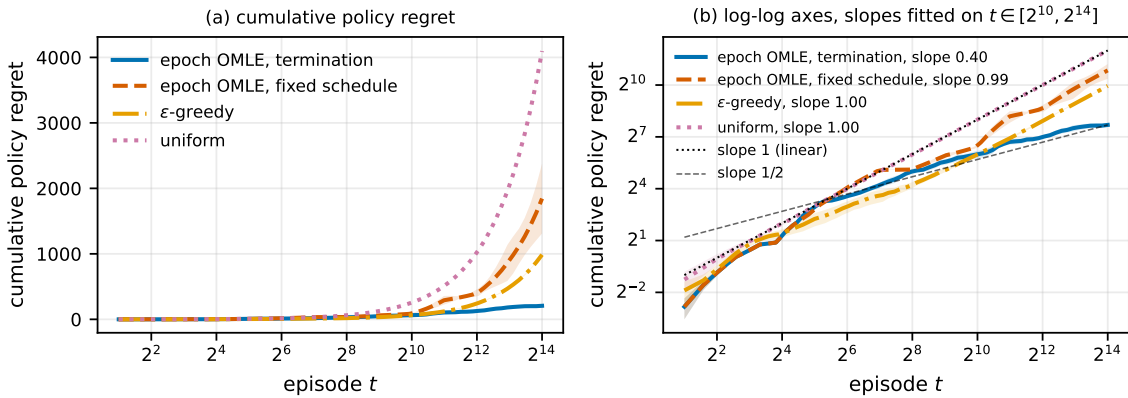


Figure 6. Experiment 3: policy regret on the digits POMG, with a memory-4 softmax-tilt adversary and re-stabilization transients (10 seeds; bands are two standard errors of the mean). (a) Cumulative policy regret of the four algorithms. (b) The same curves on log-log axes, with least-squares slopes fitted over $t \in [2^{10}, 2^{14}]$; the three baselines grow linearly (slope ≈ 1), while the epoch algorithm with termination grows at fitted slope 0.40 and flattens as it settles on the best rule.

Findings. Panel (a) of Figure 6 shows cumulative policy regret. The epoch algorithm with termination ends at about 207, against 1844 for the fixed schedule, 987 for ϵ -greedy, and 4097 for uniform play. The unit of policy regret here is one expected correct classification, so the numbers have a concrete reading. Over the 16,384 episodes, the termination algorithm

forgoes about 207 expected correct predictions relative to always playing the best fixed rule under its own stationary tilt, about 1.3% of the horizon; the fixed schedule forgoes about 11%, ϵ -greedy about 6%, and uniform play about 25%. The ordering matches the theory. The fixed schedule overcommits to poorly fitting rules for entire epochs. The ϵ -greedy learner switches in almost every episode, so the adversary never settles and the learner pays the transient penalty nearly continuously. The epoch structure with termination stabilizes the opponent, and still abandons refuted commitments quickly.

Trend. Panel (b) of Figure 6 repeats the comparison on log-log axes, where a power law is a straight line whose slope is the exponent, with least-squares slopes fitted over the last four octaves $t \in [2^{10}, 2^{14}]$. The three baselines grow at slope ≈ 1 , that is, linearly, because each keeps paying a per-episode cost that never decays: refuted commitments for the fixed schedule, transient penalties for ϵ -greedy, and both for uniform play. The termination algorithm’s fitted slope over the same window is 0.40, below the $1/2$ of the $\sqrt{T}$ theory, and its curve visibly flattens once the algorithm settles on the best rule.

Scope. Two limitations are inherent. Strategic-response data does not exist in logged form, so the adversary is necessarily simulated even when the environment statistics come from real data. And the experiments instantiate the algorithm exactly only where the optimistic MLE is computable in closed form; scaling the oracle steps to rich model classes is the computational open problem discussed in Section 5.

J. Technical Comparison with Concurrent Work

In this appendix, we provide a detailed technical analysis of the concurrent work by Sang & Nguyen-Tang (2025) on policy regret minimization in POMGs. We identify fundamental issues in their problem formulation and analysis. Most critically, we show that their flat-batching analysis allows the historical transport term to grow polynomially with T , which obstructs a direct $\tilde{O}(\sqrt{T})$ guarantee via the Eluder/Cauchy–Schwarz route. We explain how our work resolves these issues.

J.1. Issue 1: Circularity in the Posterior-Lipschitz Condition

The posterior-Lipschitz condition in Sang & Nguyen-Tang (2025) is intended to identify adversary responses that vary smoothly with the learner’s policy. In their formulation, for policies π, ν and adversary history τ_B , the condition takes the form

$$\|g(\cdot \mid \tau_B, \pi) - g(\cdot \mid \tau_B, \nu)\|_1 \leq L \max_h \|S_{\tau_B}(\pi_h) - S_{\tau_B}(\nu_h)\|_1, \quad (48)$$

where the posterior-predictive learner policy is defined by

$$S_{\tau_B}(\pi_h) := \mathbb{E}_{\tau_A \sim P^{\pi, g, \theta}(\cdot \mid \tau_B)} [\pi_h(\cdot \mid \tau_A)]. \quad (49)$$

The difficulty is that the right-hand side of (48) is computed using the same adversary response g that the condition is trying to constrain. Thus the metric with respect to which g is required to be Lipschitz is itself induced by g . This makes the adversary class circular because, before one can check whether g is admissible, one must already use g to define the posterior-predictive quantities appearing in the admissibility condition.

This circularity is not merely cosmetic. The posterior-predictive quantity $S_{\tau_B}(\pi_h)$ is used to compare the adversary responses induced by different learner policies, and this comparison is then invoked in the Lipschitz transfer and constraint-propagation steps of the analysis. If the comparison metric itself depends on the candidate adversary response, the resulting regularity condition does not define a fixed model class over which one can form confidence sets and perform uniform concentration.

Our formulation removes this dependence by computing posterior-predictive learner policies under a fixed reference measure μ^{ref} . Specifically, we use

$$S_{\tau_B, h}^{\text{ref}, \theta}(\pi) := \mathbb{E}_{\tau_A \sim P^{\pi, \mu^{\text{ref}}, \theta}(\cdot \mid \tau_B)} [\pi_h(\cdot \mid \tau_A)], \quad (50)$$

and state posterior-Lipschitzness with $S_{\tau_B, h}^{\text{ref}, \theta}$ in place of S_{τ_B} . The reference measure μ^{ref} is fixed independently of the adversary response being constrained, so the right-hand side of the Lipschitz condition is well-defined before choosing g . The adversary class is therefore specified non-circularly, while still measuring how the learner’s policy appears from the adversary’s information state.

J.2. Issue 2: A $T^{3/4}$ -Scale Obstruction from the Batched Structure

Beyond the circularity issue, the flat-batching structure in Sang & Nguyen-Tang (2025) allows the historical transport term to grow polynomially with T . The point is not merely that many batches are used, but that the posterior-Lipschitz transport term accumulates over previously deployed policies. When this term is carried through the same weighted Eluder/Cauchy–Schwarz route used to convert prediction errors into regret, it creates a $T^{3/4}$ -scale contribution.

Their algorithm uses a flat batching schedule with

$$K = \sqrt{d_E T}, \quad n = \frac{T}{K} = \sqrt{\frac{T}{d_E}}, \quad (51)$$

where K is the number of batches, n is the number of data-collection episodes per batch, and we write d_E for the corresponding complexity parameter of their analysis.

A key quantity in their transport analysis is the historical accumulation term

$$\Gamma_{j-1} := \sum_{k=1}^{j-1} \sum_{h=1}^H \mathbb{E} \left[\left\| q_h^{\text{mid},*}(\pi_k) \right\|_1^2 \right]. \quad (52)$$

This is the analogue of the transport term in our analysis. It measures how much historical predictive mass must be carried when an adversary-channel error is transported from the historical policies π_k to the current batch policy. In their Lemma 10, this term appears in the adversary-channel constraint as

$$\sum_{k=1}^{j-1} \sum_{h=1}^H \mathbb{E} \left[\left\| \left[G_h^{\Phi_j, \pi} - G_h^{\Phi^*, \pi} \right] q_h^{\text{mid},*} \right\|_1^2 \right] \leq C\beta + C'\Delta_\sigma(\pi, \nu_j)^2 \Gamma_{j-1}. \quad (53)$$

Since $q_h^{\text{mid},*}$ is a predictive state, its ℓ_1 -norm is $O(1)$. Therefore

$$\Gamma_j = \sum_{k=1}^j \sum_{h=1}^H \mathbb{E} \left[\left\| q_h^{\text{mid},*}(\pi_k) \right\|_1^2 \right] = O(jH). \quad (54)$$

At the final batch, using (51),

$$\Gamma_K = O(KH) = O\left(H\sqrt{d_E T}\right). \quad (55)$$

Thus the historical transport term grows as $\sqrt{T}$, rather than polylogarithmically.

In the final assembly of their analysis, this transport contribution is absorbed into a constant described as depending on “the covering radius used to select a nearest historical policy”; no argument is given that optimistically selected policies form such a net, so we track the term explicitly below.

We next spell out how this growth affects the regret conversion. For batch j , define the per-episode adversary-channel prediction error

$$A_j^2 := \sum_{h=1}^H \mathbb{E} \left[\left\| \left[G_h^{\Phi_j, \pi_j} - G_h^{\Phi^*, \pi_j} \right] q_h^{\text{mid},*} \right\|_1^2 \right]. \quad (56)$$

Equivalently, the batch-weighted squared error is nA_j^2 . This matches the episode-weighted quantities used in the regret analysis, since batch j contributes n data-collection episodes under the same deployed policy π_j .

The adversary-channel part of the value error in one episode of batch j is bounded by $H\sqrt{H} A_j$, where the factor H is the simulation-lemma horizon factor and the $\sqrt{H}$ is the Cauchy–Schwarz conversion over the H steps, since A_j is the root of a stepwise *squared* sum. Hence the adversary-channel contribution to regret satisfies

$$\text{Regret}_G \leq H^{3/2} \sum_{j=1}^K nA_j. \quad (57)$$

Applying Cauchy–Schwarz in the weighted form gives

$$\sum_{j=1}^K nA_j \leq \sqrt{\sum_{j=1}^K n} \sqrt{\sum_{j=1}^K nA_j^2} = \sqrt{T} \sqrt{\sum_{j=1}^K nA_j^2}, \quad (58)$$

where we used $Kn = T$.

Thus the relevant quantity for regret is the weighted square-error sum $\sum_j nA_j^2$. Granting their weighted Eluder summability step (which, as Proposition D.5 shows, itself requires a per-batch cap that a fixed batch length does not provide) and carrying the transport term in (53) through it gives a bound of the form

$$\sum_{j=1}^K nA_j^2 \leq \tilde{O}(d_E(\beta + \Gamma_K)). \quad (59)$$

Substituting (55) into (59), and ignoring the lower-order confidence term β , yields

$$\sum_{j=1}^K n A_j^2 \leq \tilde{O}\left(d_E \cdot H \sqrt{d_E T}\right). \quad (60)$$

Combining (57), (58), and (60), we obtain

$$\text{Regret}_G \leq H^{3/2} \sqrt{T} \sqrt{\tilde{O}\left(d_E \cdot H \sqrt{d_E T}\right)} \quad (61)$$

$$= \tilde{O}\left(H^2 d_E^{3/4} T^{3/4}\right). \quad (62)$$

Therefore, once the historical transport term Γ_K is retained in the weighted square-error bound, the final Cauchy–Schwarz step produces a $T^{3/4}$ -scale contribution.

To summarize, the calculation above isolates the obstruction caused by flat batching. With $K = \sqrt{d_E T}$ batches, the transport term satisfies $\Gamma_K = O(H \sqrt{d_E T})$ by (55). Substituting this growth into the weighted Eluder bound (59) and then applying the regret conversion (58) gives the $T^{3/4}$ -scale contribution in (62). Thus the difficulty is not a logarithmic artifact. The flat-batching structure allows the historical transport term to grow polynomially with T , which obstructs a direct $\tilde{O}(\sqrt{T})$ guarantee by this analysis route.

J.3. Our Solution: One-Policy-Per-Episode Achieves $\tilde{O}(\sqrt{T})$

We now summarize how our algorithmic design avoids the two difficulties discussed above. The first issue is handled at the modeling level by defining the posterior-predictive learner policy with respect to the fixed reference measure μ^{ref} , as in (50). This makes the posterior-Lipschitz condition a non-circular condition on the adversary response class. The second issue is handled algorithmically by using epochs rather than a flat batch schedule.

Our algorithm caps epochs at geometrically increasing lengths $T_e = 2^e$ and, crucially, terminates an epoch early as soon as the collected data refute the deployed optimistic model. At the beginning of each epoch, the learner forms a confidence set from all previously collected data and selects one optimistic policy π_e , which it plays until the cap or the termination test ends the epoch. Concretely, after every data-collection episode the learner recomputes the running MLE and terminates the epoch as soon as the deployed model’s log-likelihood falls more than $\tau_e(t) = 2\beta_e(t) + C_0 B_0 \log(2t(t+1)/\delta_e)$ below it, so a refuted commitment is abandoned within the episodes needed to certify the refutation. Two consequences follow. First, under the joint KL-to-operator conversion (Assumption 3.1(e)) our analysis requires no policy-transport term at all. Past consistency is established directly at the data-collection policies, the termination test caps every epoch’s contribution at $\tilde{O}(H\beta_T)$ (Lemma D.6), and the Eluder summability argument (Lemma D.12) yields

$$\sum_e n_e \Delta_e^2 \leq \tilde{O}(d_E \bar{\beta}_T), \quad \bar{\beta}_T := H\beta_T. \quad (63)$$

Applying Cauchy–Schwarz with $\sum_e n_e \leq T$,

$$\sum_e n_e \Delta_e \leq \sqrt{\left(\sum_e n_e\right) \left(\sum_e n_e \Delta_e^2\right)} \leq \tilde{O}\left(\sqrt{d_E T \bar{\beta}_T}\right). \quad (64)$$

The simulation lemma contributes the outer factor H , and the warm-up episodes contribute $(m-1)H S_T$, which the mB_0 floor in β_e makes unconditionally dominated by the leading term plus $O(mH \log T)$ (Lemma D.7). Therefore,

$$PR(T) \leq \tilde{O}\left(H \sqrt{d_E T \bar{\beta}_T} + mH \log T\right). \quad (65)$$

Second, even from the transport viewpoint relevant to the flat-batching route (Remark D.8), our *scheduled* switches number only $O(\log T)$, so any transport accounting over them stays logarithmic; the data-driven terminations are not free in that accounting, but our proof never needs that accounting, since the per-epoch cap replaces transport entirely. This is the substantive difference from a flat schedule with $K = \sqrt{d_E T}$ batches, where every one of the K switches is structural and the transport term necessarily accumulates to $\Gamma_K = O(H \sqrt{d_E T})$.

The essential difference is that a flat schedule with polynomially many deployed policies both forgoes the per-epoch cap (each batch runs its full length regardless of what the data say) and, on the transport route, lets the historical term grow polynomially in T , as shown in Appendix J.2.

In summary, our reference posterior-predictive formulation removes the circularity in the posterior-Lipschitz condition, and our epoch design, with geometric caps and statistical termination, supplies the per-epoch cap that makes infrequently updated optimistic planning sound while keeping policy switches rare. These changes are what allow the optimistic MLE analysis to recover the $\tilde{O}(\sqrt{T})$ policy-regret rate.